PAGSANJAN FALLS ECO-TOURISM: A WEB BASED SYSTEM FOR THE MUNICIPALITY OF LAGUNA

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In Partial Fulfillment of the Requirements for the Degree
Bachelor of Science in Information Technology
Specialized In Service Management Program

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ABSTRACT

Manual and uncoordinated tourism operations at Pagsanjan Falls have resulted in inefficiencies in reservation management, boat tour scheduling, fee collection, and access to reliable tourism information, often leading to booking conflicts, service delays, and inconvenience for both tourists and service providers. To address these challenges, this study developed a web- based ecotourism management system for the Municipality of Pagsanjan, Laguna, integrating online reservations, boat tour scheduling, payment and fee management, and tourism data reporting under the supervision of the Municipal Tourism Office. The system aims to improve operational efficiency, enhance the overall tourist experience, and promote organized and sustainable ecotourism by strengthening coordination among landing areas, hotels, resorts, restaurants, and the United Boatmen Association of Pagsanjan (UBAP). System development followed the Agile Scrum methodology and was evaluated using ISO 25010 software quality standards and the Technology Acceptance Model (TAM). The ISO 25010 outcomes show that the system has met the necessary software quality requirements with the general composite mean of 3.17 (SD= + -16), which is considered to be agreeable, hence, reflecting an acceptable functional, usability, reliability, security and safety. The TAM assessment indicates that user acceptance is high with a composite mean of 3.69 (SD= +0.04), which translates to excellent, and strong results in Perceived Usefulness (3.74), Perceived Ease of Use (3.56), Attitude Towards Usage (3.73), and Behavioral Intention to Use (3.73), which confirms that the users feel that the system is advantageous, user-friendly, and will continue to use it. Comprehensively, the results indicate that the created web-based system helps to improve coordination between the Tourism Office, landing grounds, accommodation facilities, and boatmen, facilitate the adoption of municipal policies, and provide a more systematic and more convenient ecotourism service to the visitors of Pagsanjan Falls.

Keywords: *Ecotourism, web-based system, tourism management, reservation system, ISO 25010, Technology Acceptance Model (TAM), Pagsanjan Falls*

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DEFINITION OF TERMS

Agile Scrum	A project management framework emphasizes iterative development by breaking down projects into smaller, manageable sprints, typically lasting two to four weeks. This approach allows for the incremental completion of critical features and the production of deliverable systems
System Architecture	A setting that determines whether the system should send an immediate notification warning to the farmer when critical environmental thresholds are breached.
Conceptual Framework	An analytical tool used in research defines essential concepts, variables, and relationships within a study. It provides an organized strategy for understanding and analyzing a research problem.
Flowchart	A graphical representation of a process, system, or algorithm. It employs symbols and arrows to depict the actions, decisions, and flow of a certain activity or operation, making complicated operations simpler to comprehend.
Php and Laravel Framework	Laravel is an open-source PHP web application framework that follows the Model- View-Controller (MVC) architecture. It is designed to make web development more enjoyable and efficient by providing an expressive and elegant syntax. Laravel is a powerful and versatile PHP framework that simplifies web development by providing a rich set of features and tools. Its elegant syntax, robust architecture, and extensive ecosystem make it a popular choice among developers for building modern web applications.

Technology Acceptance Model	The Technology Acceptance Model serves as a theoretical framework to comprehend and forecast the adoption of technology by users. It evaluates aspects such as perceived usefulness and ease of use that influence users' intentions to adopt and use a technological system.
Business Plan	A business plan is a comprehensive document that outlines a company's goals, strategies, and financial projections. It serves as a strategic roadmap detailing how a business intends to achieve its objectives, including descriptions of its products or services, target market, and competitive landscape. Additionally, it acts as a blueprint for new ventures seeking capital or for established businesses aligning their growth objectives.
Marketing Plan	A strategic document designed to guide the promotion of Pagsanjan Falls as an ecotourism destination through a web-based system. It outlines the target tourists, promotional strategies, pricing schemes, and communication channels to attract visitors while supporting sustainable tourism.
SWOT Analysis	A strategic planning tool used to identify and evaluate the Strengths, Weaknesses, Opportunities, and Threats of a business, project, or system. It helps in understanding internal capabilities and external factors to support informed decision-making, improve strategies, and achieve organizational or project objectives effectively
Report and Analytics	Report and Analytics refers to the system component that collects, processes, and presents data such as tourist arrivals, revenues, and feedback to support decision-making and system monitoring.

Reservation and Scheduling	A system component that allows tourists to book services and select preferred schedules while coordinating availability with landing areas and boat tour providers.
Boat Ride Scheduling	Boat Ride Scheduling refers to the process of assigning specific dates and time slots for tourists to participate in boat tours, ensuring organized operations and coordination with service providers.
Tourist Feedback	Tourist Feedback refers to the comments, ratings, and suggestions provided by tourists through the system, used to evaluate service quality and improve ecotourism operations.
System Analytics Dashboard	A visual interface within the system that displays key metrics, such as the number of bookings, peak visiting hours, and user demographics. This helps administrators make data-driven decisions to improve tour operations and support sustainable tourism management.
Marketing Plan	A strategic document designed to guide the promotion of Pagsanjan Falls as an ecotourism destination through a web-based system. It outlines the target tourists, promotional strategies, pricing schemes, and communication channels to attract visitors while supporting sustainable tourism.
Web-Based System	A web-based system is an application that operates through a web browser and does not require installation on a local device. It allows users to access services such as reservation, scheduling, and payment using the internet.

CHAPTER I

INTRODUCTION

Project Context

Ecotourism, which has appeared in academic literature since the late 1980s, is a special form of nature-based tourism that maintains the well-being of the local community while protecting the environment and provides tourists with a satisfying nature experience and enjoyment (Ceballos-Lascuráin, 1996; Higgins, 1996; Orams, 1995). With years of research and development, ecotourism has risen to be a subject of investigation in the field of tourism research (Weaver & Lawton, 2007). In 2002, the United Nations declared it the International Year of Ecotourism (IYE), and the professional Journal of Ecotourism was established in the same year. With the progress and maturity of ecotourism as an academic research field, countless scholars have put forward standards and definitions for ecotourism (Sirakaya et al., 1999; Wight, 1993).

The main objectives of ecotourism emphasize long-term sustainable development (Whitelaw et al., 2014), including the conservation of natural resources, the generation of economic income, education, local participation, and the promotion of social benefits such as regional economic development and infrastructure (Ardoin et al., 2015; Coria & Calfucura, 2012; Krüger, 2005; Oladeji et al., 2021; Ross & Wall, 1999; Valdivieso et al., 2015). It can also boost rural economies and alleviate poverty in developing countries (Snyman, 2017; Zhong & Liu, 2017). With unrestricted increasing attention to the ecological environment and the improvement of environmental awareness, ecotourism is becoming ever more prevalent, and the demand for tourism is

increasing year by year (CREST, 2019).This increase, however, leads to a number of environmental, social, and economic challenges in the development of ecotourism. For example, due to the low public awareness of ecotourism, the increase in tourists has brought a series of negative impacts on the local ecological environment, culture and economy, including disrespect for local culture and environmental protection, as well as more infrastructure construction and economic burden to meet the needs of tourists (Ahmad et al., 2018; Chiu et al., 2014; Shasha et al., 2020; Xu et al., 2020). Such challenges and contradictions are urgent problems to be tackled by the sustainable development of ecotourism. Especially against the backdrop of the current pandemic, tourism has experienced a severe blow, but climate change and other environmental issues have not improved (CREST, 2020). In this context, facing these challenges and difficulties, it is essential to re- examine the future development path of ecotourism, to explore how government agencies can formulate appropriate management policies while preserving the environment and natural resources to support sustainable tourism development. Accordingly, it is necessary to consult literature in the field of ecotourism to understand the research progress and fundamental research issues, to identify challenges, suitable methods, and future research directions of ecotourism.

Tourism is a major industry and source of employment in the Municipality of Pagsanjan, wherein the majority of the residents rely on tourist arrivals for business opportunities and jobs in hotels and recreational places (Russel Ching, March 12, 2016). During our research, we did not hear from the people we interviewed about the cultural impact of tourism; we were not able to identify anything on this aspect. However, what we can say is that. The Bangkero River

Festival is an opportunity to reintroduce the lost traditions of the town. The brochure of the festival, the water parade, will highlight the town's history, people's legend, and arts. And the Bangkero festival is the town's cultural practice. Since the cleanliness of the river is an important condition for tourism to thrive in Pagsanjan, boatmen have become more conscious of their environment. Local people think that the boating activities themselves do not make much of a negative impact on the environment. The development of Pagsanjan tourism is part of an overall Philippine tourism, which seeks to intensify tourism promotion and development, to maximize the generation of foreign exchange from tourism and the spreading of its benefits to the different regions in the country (Ching, 2016).

The global ecotourism market size was valued at USD 216.49 billion in 2023 and is projected to grow from USD 260.76 billion in 2024 to USD 759.93 billion by 2032, exhibiting a CAGR of 14.31% during the forecast period. Europe dominated the ecotourism market with a market share of 38.87% in 2023. The United Nations World Tourism Organization (UNWTO) defines ecotourism as all forms of nature-based tourism in which the key aim of the tourist is to observe and appreciate nature and the traditional cultures that predominate the natural areas. This type of tourism helps in attracting a large number of national and international travelers, which preserves the integrity of the natural environment. It is an eco-friendly tourism that is marketed as responsible travel due to its travel destination, where the cultural heritage, flora, and fauna are the primary attractions for tourists. The sudden onset of the COVID-19 pandemic badly affected the global economy. COVID-19 was considered a public health emergency that transformed lives and businesses and brought significant

changes to the ecotourism industry. The pandemic had a strong negative impact on the manufacturing, logistics, travel, and tourism sectors. The complete closure of national and international borders due to the pandemic was the major reason behind the decline in tourists worldwide. The outbreak of the virus forced governments of various countries to impose lockdown measures to reduce its spread across the world, which further slowed the market progress. Rising Trend of Online Booking by Tourists to be a Major Factor Boosting Market Growth.

Furthermore, the frequency of online booking has increased due to easy and quick flight & hotel bookings. Ecotourism, an alternative to mass tourism, is poised to reign supreme in the Philippines' travel industry's transition to the new normal post-COVID19. According to experts at the Ecotourism Philippines webinar. Organized by Masungi Georeserve, Visita, Eco Explorations, and Andrew Tan Center for Tourism of the Asian Institute of Management, the webinar was held to discuss the country's tourism recovery post-COVID. John Roberts, Minor Hotels group director of sustainability and conservation, said that ecotourism already has a head start with its outdoor-centric activities, offering itineraries in safer, low-risk environments. As such, he reckoned that ecotourism would rebound faster than mass tourism. "Hotels and other operations are going to have to look at what we do in ecotourism – how we manage to make money with fewer guests, greater space, more personalised service, and smaller groups," Roberts pointed out. "For ecotourism, the new normal is actually the old normal, and that is, ecotourism is run in a very different way by and large from many of the other mass tourism products," said Tony Charters, board member of the Asian Ecotourism Network and vice-chair of

Global Ecotourism Network. "In some ways, ecotourism is the leader in what the new normal is," Charters said, adding that this sector is "adopting practices that are probably much safer from environmental and health points of view than many other forms of tourism. He further elaborated: "The experiential side of ecotourism is very important, and often, you will find that many different ecotourism activities like walking treks and canoe treks are carried out in quite remote areas where the risk (of virus infection) is very low." Boboi Costas, who organised the Bojo Aloguinsan Ecotourism Association in Cebu.

Developed the Aloguinsan River Eco-cultural Tour, underscoring the need to look at product development in the new normal. He suggested that the ecotourism market should be segmented into different product types, for example, health and wellness, culinary, and bird watching, with the addition of more private activities and groups. He added that ecotourism providers should establish contacts with local health practitioners. On the other hand, Roberts advised industry players to start learning and training in ecotourism as it's poised to be the most popular vacation type in the new normal. He added that they should also step up efforts to woo the domestic market, which will recover ahead of the international market, noting that Minor Hotels is strongly eyeing the domestic market in Bangkok once the Golden Triangle opens up in July.

Furthermore, Roberts also suggested that ecotourism providers in Southeast Asia share inspirational stories created around their destinations on social media, retrain the digital community, and start differentiating themselves from mass tourism by building on the region's trademark friendliness and hospitality. Roberts cited the case of him and his community hosting Facebook live sessions for Minor Hotels' Anantara Golden Triangle Elephant Camp,

connecting with their audience and telling them about the camp's work so they remain top of mind, and raising awareness of their donation drive. For destinations, Charters advised pushing for quality and sustainability, and marketing their merits, which, in the Philippines' case, is friendliness. With global discussions about Boracay's rehabilitation, Charters said the key is to keep the project at the forefront, emphasizing that the Philippines is sustainably doing things.

During the webinar, the panelists also Debunked ecotourism operations as very small-scale and expensive, claiming that it can be carried out on a large scale if done correctly, and that the element of scale only comes into relevance when sustainability or the experience is threatened. Large-scale ecotourism spots included a $10 billion. industry, which generates a $10 billion (US$6.9 billion) annual contribution to Australia's economy; Queensland's Skyrail Rainforest Cableway, a long cableway that can carry hundreds of thousands of people yearly to the top of the rainforest; and English Bay Resort on the world heritage site Fraser Bay. (Ocampo, 2020).

Project Objectives

The main objective of this study is to develop a web-based online services platform system for the Ecotourism of Pagsanjan Falls of Laguna, to promote the majestic Ecotourism of Pagsanjan, and to preserve the tradition of Bangkero. Specifically, the project aims to:

1. Develop an information management system for tourists and access relevant ecotourism information efficiently.

2. Create a portal for tourists, such as a landing area, resorts, restaurants, and a hotel.

3. Conduct a user acceptance evaluation using the Technology Acceptance Model (TAM) to determine perceived usefulness, ease of use, and user intent to adopt the new technology.

Project Purpose

The purpose of this project closely corresponds to the United Nations' Sustainable Development Goals (SDGs), supporting the sustainable development of ecotourism in Pagsanjan Falls by enhancing tourist check-in management and boat tour bookings through a web-based system. This system ensures compliance with the Municipality of Pagsanjan's regulations, requiring tourists to check in at accredited resorts before availing of boat tours. By implementing this structured process, both resorts and boatmen benefit from fair and organized income opportunities, aligning with Sustainable Development Goal 8: Decent Work and Economic Growth to promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all. Additionally, the system optimizes tour scheduling and booking efficiency, integrating digital solutions in tourism management to support Sustainable Development Goal 9: Industry, Innovation, and Infrastructure. By promoting regulated and responsible tourism, the project also contributes to Sustainable Development Goal 11: Sustainable Cities and Communities, preventing overcrowding and unauthorized boat rentals.

Landing Area and UBAP (United Boatmen Association of Pagsanjan). Accredited landing areas, such as resorts, restaurants, and hotels, and UBAP

will benefit from the system since it will provide a regulated and systematic reservation process. This system will allow resorts to increase their guest occupancy rates, while the boatmen will receive a fair number of clients, thereby ending unauthorized and direct transactions. With the improvement of the whole management of tourism services, this system benefits their economic growth and stability.

Tourists. Tourists traveling to Pagsanjan Falls will be rewarded with a more convenient and hassle-free tour booking experience. Under the scheme, they can check into an accredited landing area, schedule their boat ride tours, and get official tour confirmations, thereby ensuring safety and compliance with local regulations. The systematic process provides a more organized and enjoyable tourist experience. Visitors can view available schedules, select preferred landing areas, and confirm reservations in advance, reducing the uncertainty and waiting time often experienced in traditional walk-in bookings.

Municipal Tourism Office of Pagsanjan. The Municipality of Pagsanjan Tourism Office will benefit from strengthened enforcement of ecotourism policies and enhanced monitoring of tourist operations. The existing system will ensure that all tours are conducted through legitimate channels, thereby eliminating illegal operations, but at the same time ensuring responsible tourism.

Future Researchers. This research can be used as an effective guide for future researchers who study tourism management systems, ecotourism policy, and technology-based solutions to tourism. They can study the efficiency of a

web based check-in and tour management system in improving tourist registration, booking efficiency, and regulatory compliance.

SCOPE AND LIMITATION OF THE STUDY

This project aims develop a web-based tour management and check-in system that will be utilized in managing and streamlining the tourist registration process of Pagsanjan Falls. The system will ensure that all tourists comply with the tourism policy of the Municipality of Pagsanjan, which requires tourists to check in at accredited resorts before taking a boat tour. The system includes the following modules:

Account Management Module

In the account management module, registration is required for tourists, resort staff (Landing Area), UBAP, and the Tourism Office (Admin) to access the full features of the system. While general information about resorts and boat tour services is publicly viewable, users must register and log in with their credentials to use the system's core functions. First, tourists must create an account to request a reservation, pay a down payment, and avail themselves of a boat ride. Once the resort confirms the check-in, the tourist can proceed with the boat tour reservation. Second, the landing area must also register and log in to verify tourist check-ins, confirm boat ride requests scheduling to UBAP, and coordinate with UBAP. Lastly, the Tourism Office registers as an Admin user to manage the entire system, generate tourism and income reports, enforce policy compliance, and support sustainable tourism operations in Pagsanjan. This process ensures that all users are properly identified and authorized, which helps maintain system security and organized transactions.

Payment and Fees Management Module

The Payment and Fees Management Module enables a safe and transparent process of collection of all fees needed, including landing area, hotel, restaurant, and resort fees, boat tour fees, environmental fees, and system maintenance fees. Tourists can see a detailed breakdown of their charges before they confirm any transaction. Payment can be made online through the system or in cash upon arrival. For reserving a slot, tourists are requested to pay a down payment of 50%, which can be done during the reservation. GCash or bank transfers can be made to the respective account of the selected resort, hotel, restaurant, or designated landing site, with each receiving its respective share of the payment. A fixed subscription fee of ₱100 will also be automatically charged and paid upon confirmation of the reservation of tourists for the Tourism Office of Pagsanjan (Admin). The subscription fee will apply when the system is under maintenance and troubleshooting.

Landing area, hotel, restaurant and resort staff will be tasked with verification of payments. At the same time, the collection of government-imposed fees and the smooth and transparent operation of the platform are the tasks of the Tourism Office.

Data Visualization Module

The report provides the Tourism Office (Admin) with essential information needed to monitor and analyze tourism patterns at Pagsanjan Falls. These are Tourist Arrival Trends, which monitor the number of visitors over time, thus enabling tourism forecasting and policy updates. Revenue & Fees Collection Reports enable effective financial management by reporting resort

income, boat tour fees, and environmental fees. Tourist Demographics Reports analyze visitor information, including gender, age group, and travel type, thus providing insights into future tourism expansion. Lastly, the Feedback Analysis module collects and processes visitor comments to help service enhancement and a better visitor experience through data-driven decision-making. This capability supports strategic planning and helps the local government make informed decisions regarding tourism management, infrastructure improvements, and resource allocation.

CHAPTER II

THEORETICAL FRAMEWORK OF THE PROJECT

Review of Related Literature, Studies, and Systems

This chapter presents the theoretical foundation and review of related literature, studies, and existing systems that support the development of a web-based online services platform for ecotourism in Pagsanjan Falls, Laguna. Moreover, this chapter also includes a review of systems that have been implemented to enhance ecotourism experiences through digital platforms. The purpose of this chapter is to provide insights into the conceptual and practical frameworks guiding the study. It identifies the gap in the current ecotourism service delivery in the area and highlights the importance of technological integration to improve accessibility, sustainability, and user engagement in ecotourism management.

Web-based Systems and Ecotourism Management

Ecotourism is a type of tourism that is one way of traveling to nature and culture with responsibility, without disturbing or destroying natural resources and the environment, but with the main goal of seeing, learn, and enjoying the scenery, forest, wild animals, and also the culture that appears in that location. The national park is just one of the ecotourism sites that tourists pay attention to due to high competition in the tourist industry, and a high number of TV commercials and other media. Over the last 10 years of visitor statistics in the national park, there has been a decline in tourists and less interest in ecotourism. Based on the above problems, the researcher raises the issue of creating a web-based decision support system for ecotourism planning by choosing the national park in Chiangmai province, where the majority of

national parks are situated. Additionally, national parks in Chiangmai province have biological diversity and natural diversity appropriate for promotion as an ecotourism destination. This study illustrates the concept and two models of a web-based decision support system for ecotourism planning to enable the user to select. The system integrates digital tools that assist users in evaluating different tourism options, helping them make informed decisions regarding travel planning and ecotourism activities. The outcome of the study showed that users were generally satisfied with the system because it improved accessibility to tourism information and simplified the decision-making process (Tritilanunt et al., 2021).

Technology, especially the Internet and digital services, has greatly transformed tourism by digitizing services and empowering travelers to access information independently. This shift affects the entire tourism chain from businesses to consumers, leading to the rise of digital tourism. The study highlights how social media, digital marketing, and information systems influence travel behavior, with users now relying more on smart devices and peer-shared content than traditional agencies. Ultimately, digital tools help travelers save time and money by enabling easier trip planning and decision-making.

Impact of Online Booking Systems in Ecotourism

Tourism has now emerged as a significant contributor to various sustainability matters. Moreover, online booking sites have developed to become a highly utilized method of accommodation booking, which can prove to be significant in promoting sustainability. In addition, the promotion of

sustainable tourism (ST) and accommodation services on an international scale has now become imperative. Even as the number of sustainability- certified lodging options has been growing, few efforts have been made to market ST via online booking websites. This research seeks to determine if the idea of sustainability has become part of online booking websites. In order for the purpose to be served, a representative sample of websites was discovered and subsequently examined both in terms of their content and through testing. The findings identify that this idea has not been properly incorporated in such websites and that the pandemic of COVID-19 has altered the focus towards health and safety concerns from sustainability. Yet, the possibility of capitalizing on the pandemic to increase ST does exist (Arzoumanidis, Petti, & Raggi, 2022).

The tourism sector has also significantly changed with the emergence of digital technologies, and especially online booking systems. Online e- booking systems have revolutionized the manner in which consumers organize and reserve their travel arrangements, providing ease, cost-saving, and a means to tap into real-time information using mobile phones and other digital devices. Not only do these systems streamline day-to-day operating and administrative functions for companies, but they also give useful feedback by analyzing customer information to customize services and anticipate trends. Redonovic et.al. (2024) discuss the pivot e-booking plays in the digital age of tourism, using cluster analysis to categorize nations by their implementation and performance of e-booking services. The report identifies the increasing worldwide dependence on e-booking websites and their ability to improve the customer experience and streamline business processes. Overall, this literature

underscores the importance of online e-booking systems as essential tools in modern tourism, supporting digital transformation, operational effectiveness, and improved service delivery.

Digital Transformation in Tourism Services

Digital transformation does not relate only to the use of cutting-edge technology offer innovative experiences. Indeed, digital transformation is about improving certain processes in creating and delivering customer experiences. The focus should be on making the customer's journey as exceptional as possible. In fact, nowhere is the pursuit of generating the best customer experience more important than in the tourism and hospitality industry, as highlighted by many, if not all, of our interviewees. Digitization saves time and effort. Digital tools streamline operations, reducing repetitive tasks and freeing up employees to provide more personalized, human-centric service. Automation of operations ensures quality and generates better results (Cheng et.al. 2023).

In this study, Darkel et.al. (2024) examine the impact of digital transformation in tourism on travel experiences among tourists using a systematic literature review. The key emphasis is on online booking platforms, implementation of virtual and augmented reality technology, and implementation of the Internet of Things (IoT) by tourist destinations. A systematic literature review was employed to research global trends and the case in Indonesia against the backdrop of digitalization in tourism. The analysis results indicate that these aspects have a positive influence on tourists' travel experience through easier access, more immersive and engaging experiences,

and more integrated facilities. Still, challenges and barriers persist in the form of improved regulation and deeper integration of technology into tourist infrastructure. This study emphasizes how digital technologies improve travel experiences while highlighting the infrastructure and regulatory issues that must be resolved for successful application.

According to the study of Marx et.al. (2021), they point out that the adoption and use of new information and communication technology solutions are influenced by individual, environmental, and government factors. Whether the digital transformation is being utilized to its full potential by tourists and tourism service providers depends less on access and ownership but more on keeping abreast with the development and its speed through continuous development of one's digital skills and how they align, making use of available online resources in order to promote continuing professional development (CPD).

User Experience in Ecotourism Apps

Smart tourism through the extensive adoption of ICTs has progressed from traditional tourism and ecotourism to encompass concepts such as the smart destination, the smart experience, and smart business. The smart destination represents the integration of advanced ICTs and innovations into the physical infrastructure and other facilities. For example, it facilitates the visitor’s interaction with the location, and it increases the quality of tourist experiences at the destination, as well as improving decision-making for stakeholders. It is necessary to have a close understanding of how digital technologies are used by travelers in order to comprehend the challenges of

local ecotourism destinations in leveraging new technology to make their communities known to potential tourists. Trends identified from academic literature are that tourists are increasingly employing digital technologies as their information sources during the process of choosing ecotourism destinations, and the trend is being fueled by the global emergence of personal devices, including mobile phones. Therefore, the potential of destinations to maximize the potential competitive advantages delivered by digital technologies necessitates that they harmonize the technologies that they use with those used by their prospective consumers (Ayalalert, 2022).

Sustainable ecotourism goes beyond minimizing harm and aims to harmonize the desires of the current and future generations. It is centered on optimizing the usage of resources, promoting local economic growth, and advocating for equitable social and economic resource distribution. This involves establishing long-term job opportunities, offering public facilities to residents, and taking measures to alleviate resource deficiency. But there must be greater regard for environmental concerns and proof of how such objectives can be met in ecotourism that is sustainable (Zhang & Deng, 2024).

Web-Based Eco-Tour Guide Systems

A web-based tour package system. The purpose of this paper is to introduce a review of a web-based system for the ecotourism industry in Bangladesh. This paper examines and discusses the software development cycle and the software development tools in the development cycle. The design influence of the application is what the paper explores and tries to determine the elements influenced by the application. By using the created web

application, users can access and sign up for posted tours and pay for the tour using an electronic payment system. There is information for each posted tour, and users can see comments written by previous users. With the web application developed, users can see the location of tourist places and get an idea of the distance from their current location to the tourist places. There will be tour guides to take the travelers through all the tourist places under a tour package. This application will assist the development of the tourism sector by assisting the users with improved accessibility to cost-effective alternatives (Khan et.al. 2021).

The tourism industry is one of the top-performing industries for the economy of India. But there are no proper resources to gather detailed information regarding the precious spots of India. The tourists even get into trouble seeking the elementary requirements, which may be hard to manage. India is a vast nation extending 3,214 km (1,997 mi) from north to south and 2,933 km (1,822 mi) from east to west. It is a task in itself to manage the tourism industry of a country as big as ours. India, being a developing nation, finds it is not easy to compete with developed nations in order to bring tourists here. Tourists get cheated at every step during their visit, which sort of angers the tourists. In the long run, this would have an impact on the tourism industry of our nation as a whole, which could ultimately hit the economy. Thus, we have created a "Tourist Guide System" through which full guidance of tourism in India would be available at a single place. Details such as information about interesting tourist spots, best and most economical stay places, transport facilities, and price details regarding all things would be available at the click of

a button. We will also offer various kinds of tourism packages (e.g., Beach, Adventure to the user (Thombare et.al. 2023).

Tourist Information System

In the study of Hulat and Freitas (2024), tourism Information Systems (TIS) refer to electronic platforms that offer insightful data and knowledge for efficiently running and marketing tourist destinations, attractions, and services. TIS combines several technologies, such as Geographic Information Systems (GIS), databases, and mobile apps, to upgrade the experience of traveling by making available real-time information and location-based suggestions. Through the application of TIS, tourism stakeholders can enhance operational efficiency, augment visitor satisfaction, and enable sustainable tourism growth.

Service Dominant Logic

Rebecca (2023) stated that service-Dominant (S-D) Logic is a theory of marketing that looks at all economic activity as service-based instead of goods-based. It stresses value co-creation in which customers play active roles as opposed to consumers being mere buyers. In contrast with the common goods-dominant logic, S-D logic places greater emphasis on intangible assets, cooperation, and practical knowledge. Firms are viewed as nodes in a network of contributors to a shared value, transforming the marketing function from selling to interacting with customers as co-creators.

Tourism and ICT in the Philippines

LGUs in the Philippines have transformed the tourism industry through their imaginative digital efforts. Perhaps the most remarkable initiative has been the creation of online platforms that offer detailed information on tourist sites,

accommodations, and activities. The beneficial effect of these efforts has been industry-changing, allowing tourists to easily plan their journey and find necessary information at their fingertips. For instance, Baguio City introduced an online site named 'Baguio Visita,' which gives travelers detailed information regarding the tourist attractions, hotels, and activities of the city. The website's online reservation system has made it that much more convenient for tourists to plan their travel and reserve a stay in the city. This effort, among others, has helped a lot in making LGUs' digitalization of the tourism sector a huge success.to plan their travel and reserve a stay in the city. This effort, among others, has helped a lot in making LGUs' digitalization of the tourism sector a huge success. This program goes on to describe how LGU-developed web platforms serve as centralized, dependable, and user-friendly sources of tourism information that greatly lower travel uncertainty and increase accessibility to necessary services.

These platforms improve operational efficiency and expedite the entire travel planning process by combining online reservation and scheduling tools with extensive destination details. The availability of digital booking platforms allows travelers to rely on reputable and official sources of information while minimizing manual transactions and administrative workload. LGUs can monitor visitor data, handle reservations in an organized manner, control the flow of tourists, and more successfully implement local tourism policy thanks to these digital initiatives. Consequently, digital tourism platforms support enhanced destination management, better service delivery, increased visitor satisfaction and the sustained success of tourism digitalization initiatives in the Philippines**.**

As a result of these digital initiatives, the Philippine Department of Tourism (DOT) reported a record-breaking 8.26 million international tourist arrivals in 2019, representing a 15.24% increase compared to the previous year. LGU digital platforms like Baguio Visita have significantly driven this growth, allowing tourists to access information more efficiently and promoting the country as a top destination for travelers worldwide (Gumban, R., 2023). This further shows that the use of digital technologies in destination management not only empower destination marketing and increases the visibility of local attractions on the global tourism map, presenting tourists with more information, but also allows making decisions in a data-driven manner, planning and allocating resources to specific projects in the most efficient way, and coordinate the work of the tourism stakeholders.

Local government units and service providers can offer superior control over the flow of visitors, spread precise and timely information, optimize visitor booking and schedule planning, and provide increasingly responsive customer service through the digital media. Such innovations decrease the inefficiencies in the functioning, decrease the overcrowding, and create safer and more organized traveling experiences. In the end, embrace of such technologies makes the total experience of local and international tourists much more comfortable, hassle-free, and fulfilling as it becomes possible to make their travel experience quicker, more structured, and provide access to the most vital services and information easily. The system will enable visitors to spend more time on their tourism activities and less time on administrative issues by minimizing the routine inconveniences of making a reservation manually and processing their check-in and check-out in the hotel.

Table 1. Matrix of Related Literatures and Studies

AURTHOR	YEAR	TITLE	FINDINGS
TandafatuN. K., Ermilinda, L., & Darkel, Y. B. M.	2024	Digital transformation in tourism: Exploring the impact of technology on travel experiences.	The analysis results indicate that these aspects have a positive influence on tourists' travel experience through easier access, more immersive and engaging experiences, and more integrated facilities. Still, challenges and barriers persist in the form of improved regulation and deeper integration of technology into tourist infrastructure.
Chai-Arayale rt, S.	2024	Exploring the nexus of smart technologies and sustainable ecotourism.	Sustainable ecotourism goes beyond minimizing harm and aims to harmonize the desires of the current and future generations. It is centered on optimizing the usage of resources, promoting local economic growth, and advocating for equitable social and economic resource distribution.
Hulat & Freitas	2024	Tourism Information systems.	Through the application of TIS, tourism stakeholders can enhance operation efficiency, augment visitor satisfaction, and enable sustainable tourism growth.
Marikyan, D.& Papagiannidi s, S.	2024	Technology Acceptance Model.	The use of the model for testing usability in IS has facilitated the assessment of users' motivation to embrace a variety of technologies an activity that had not occurred previously as a consequence of the unavailability of validated subjective measures.
Chen,Inversini & Kelly	2023	Digital Transformation: A Competitive Edge in Tourism & Hospitality.	Digitization saves time and effort. Digital tools streamline operations, reducing repetitive tasks and freeing up employees to provide more personalized, human-centric service. Automation of operations ensures quality and generates better results.
Thombare, Chinmay and Parate, Ritik and Rakhunde, Vaibhav N.,	2023	Web-Based Tourist Travel Guide System.	Thus, we have created a "Tourist Guide System" through which full guidance of tourism in India would be available at a single place. Details such as information about interesting tourist spots, best and most economical stay places, transport facilities, and price details regarding all things would be available at the click of a button.

Synthesis and Relevance of the Study

The related literature and studies serve as a valuable foundation for the researchers to clearly establish the scope and flow of this research. Several systems have been developed to enhance tourism services through digitized platforms, particularly in destinations where visitor management and environmental conservation are critical. Implementing an online check-in, reservation, and ecotourism management system has shown numerous advantages in previous research. Studies emphasized that a centralized and web-based platform improves the overall experience for tourists by streamlining reservation processes, reducing long queues, and making trip planning more accessible and convenient.

Web-based booking systems have been widely applied in the tourism and travel industry to manage visitor flow, provide real-time updates, and gather data for better service delivery. These digital platforms are also instrumental in reducing physical contact and minimizing paper usage, making them environmentally friendly and aligned with sustainable tourism goals. This form of digitization has allowed other ecotourism sites to maintain the balance between promoting tourism and preserving their natural resources. Relevant studies have further highlighted the impact of digital platforms in supporting ecotourism efforts. Features such as visitor analytics, scheduling systems, and feedback collection allow stakeholders to make informed decisions that protect the environment while boosting local tourism. These platforms are beneficial not only to tourists but also to local government units, tour operators, and conservation groups who aim to uphold both economic growth and environmental sustainability.

This study aligns with those findings by proposing a comprehensive digital solution tailored to the unique needs of Pagsanjan Falls. It demonstrates how such a system can enhance tourism services, improve ecological management, and contribute to a more organized, sustainable, and contribute to a more organized, sustainable and modern ecotourism experience. providing a centralized web-based platform, the system allows tourists, service providers, and administrators to interact more efficiently while maintaining accurate records of tourism activities.

Conceptual Framework

Figure 1 illustrates the conceptual framework of the web-based online services platform system whose area of focus is the ecotourism of Pagsanjan Falls in Laguna. It specifies the fundamental modules, features and the development methodology of the system. It consists of major modules, such as administration, UBAP, resort management, reservation, tourist services, and data visualization, which cooperately support the overall work functioning of the platform ensuring that there is a smooth coordination among all the stakeholders involved in tourism operations. The framework also reveals that the system was built with the use of Agile Scrum methodology, thus, allowing the system to enhance on a continuous basis by planning, developing, reviewing and receiving feedback. Also, it lists the used technologies, Visual Studio Code, Reactjs, MySQL, Node.js, and PHP, that help to create a responsive and functional system. In short, the conceptual framework can be used to explain how the different modules, tools and processes are linked to one another to form an organized and effective web based platform to manage tourism services in Pagsanjan falls.

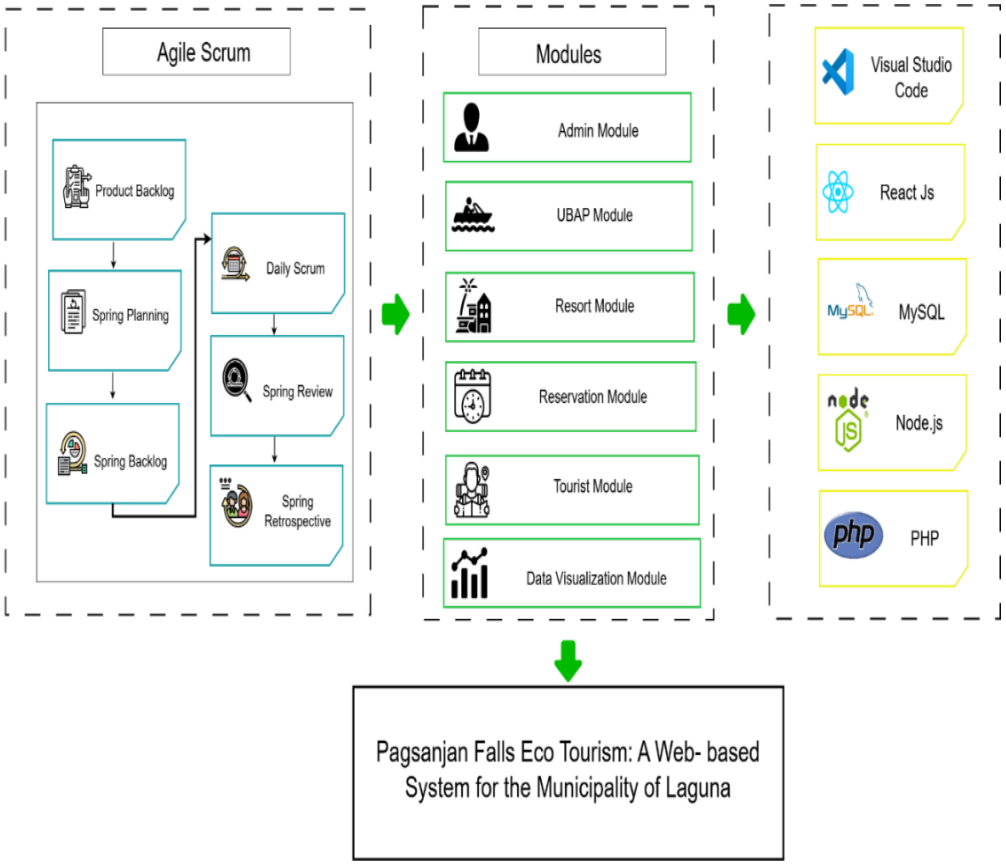


Figure 1. Concept of the proposed Web-based System for the Municipality of Laguna

CHAPTER III

METHODOLOGY

The analysis of this chapter focuses on evaluating the web-based reservation system development methods systematically. A structured approach defined the study process which combined project design with development processes and testing and evaluation procedures for assuring system functionality and usability effectiveness.

Project Design

The developers applied architectural techniques to produce the system architecture that establishes how end-user needs relate to each other.

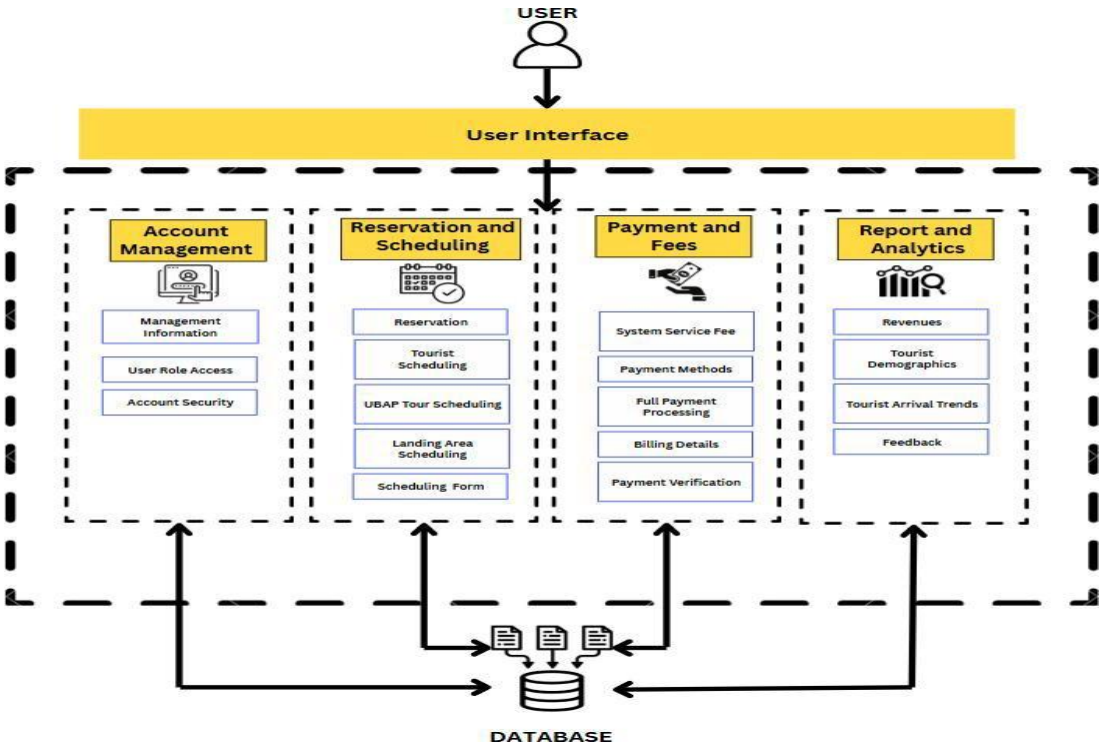


Figure 2. System Architecture

Figure 2 shows the system's architecture. An illustration presents the system layout by showing all modules available in the platform. The Account

Management Module handles user profiles, roles, and security, making sure only registered users can access the system. The Reservation and Scheduling Module allows tourists to book resorts and boat tours, while the system helps manage tour schedules, landing areas, and coordination with the United Boatmen Association of Pagsanjan (UBAP). The Payment and Fees Module includes all payment processes such as system service fees, online or manual payments, and payment verification, ensuring all transactions are smooth and secure. Lastly, the Report and Analytics Module collects important data such as tourist arrivals, revenues, and feedback, which helps the admin make better decisions. All modules are connected to a central database to store and manage information efficiently and support the operations of the entire system.

Project Development

To efficiently design and build the web-based booking system, the developers used the Agile Software Development Life Cycle (SDLC) technique. A Scrum methodology of Agile development divided project work into separate phases for the purpose of flexible user-responsive development. The development method used continuous user interactions for continuous improvement during every phase of development.

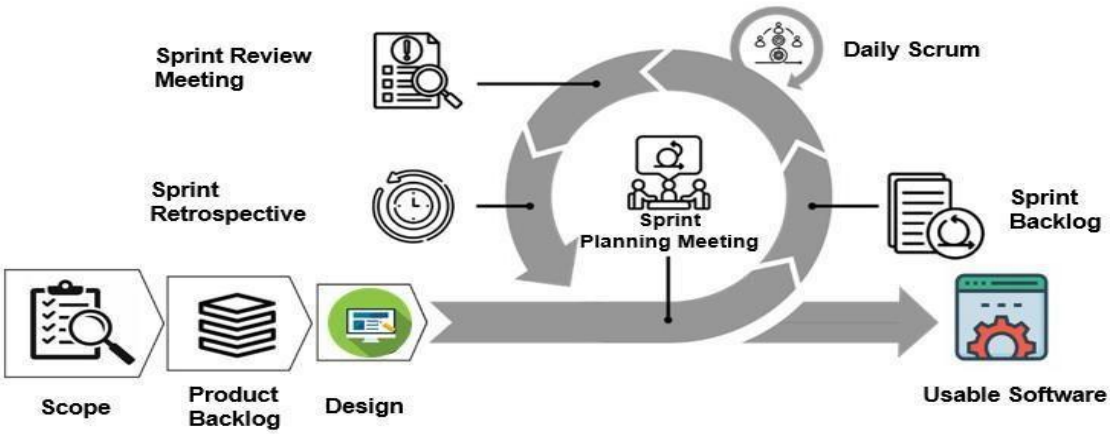


Figure 3. Agile Scrum Methodology

Figure 3 presents Project management framework Agile Scrum which enables development through continuous stages with periods between two weeks and four weeks. A typical cycle within Agile Scrum spans two to four weeks and groups most important features as priorities to achieve fast functionality alongside usability. Fast design changes become possible through user input in this method which helps reduce development risks while improving system quality. Through this approach, collaboration between developers, boatmen, resort partners, and tourists was enhanced. Constant assessments made sure that the system stayed true to its main goal which resulted in an advanced web- based booking system. Agile Scrum places a strong emphasis on active cooperation amongst all stakeholders, including developers, boatmen, resort partners, and visitors, creating an atmosphere of openness, responsibility, and constant communication.

Frequent sprint planning meetings, daily stand-up meetings, sprint reviews, and retrospectives offer organized chances to track advancement, spot problems early, and improve system elements. This approach is especially well-suited for dynamic situations like ecotourism management because it facilitates flexibility in response to evolving technology, environmental factors, and changing requirements. The final web-based booking system will be effective, dependable, and user-friendly thanks to the integration of stakeholder participation and ongoing assessment. It is in line with Pagsanjan Falls' sustainable management objectives, improving tourist happiness, ecological monitoring, and tourism services while offering a contemporary and structured approach to ecotourism. This flexible and participatory approach will make the system relevant in the long-term, whereby there will be continuous

improvements to meet the emerging needs and improve the overall quality, efficiency, and sustainability of tourism operations.

PRODUCT BACKLOG

Table 2. Product Backlog for Tourism Management Office of Pagsanjan (Host)

Priority	Features	Estimated hr/s per day
1	As a user, I want to register and log in to the system.	1hr
2	As a user, I want to manage the user account management system.	3hrs
3	As a user, I want to manage the reservation and scheduling of the user.	2hrs
4	As a user, I want to manage payments and transactions.	2hrs
5	As a user, I want to manage all data reporting and analytics.	3hrs
6	As a user, I want to manage the content management of the system.	2hrs

Table 2 outlines the Product Backlog for host using the system. This backlog represents the tasks required to be completed during development. The developers put the things that they want to see in the system as an Admin. For the initial step of the project, the product backlog is the method of the first phase which is needed. It collected the client data and list of procedures for creating the system. Principally, it contains the user or the administrator's procedure of working and acting in the system.

It needs to be a generalized task and include the probable performance of the project. It helps assure effective progress system performance, directs developers in creating functions, and aligns the system with user and administrative demands.

Table 3. Product Backlog for Landing Area, Hotel, Restaurant and Resort

Priority	Features	Estimated hr/s per day
1	As a user, I want to register and log in to the system.	1hr
2	As a user, I want to add available and accredited landing area, hotel, restaurant, and resort.	3hrs
3	As a user, I want to manage the reservation.	2hrs
4	As a user, I want to manage payments and transactions.	2hrs
5	As a user, I want to manage data reporting and analytics	3hrs
6	As a user, I want to manage all the items for landing area, hotel, restaurant and resort	2hrs

Table 3 outlines the Product Backlog for host using the system. This backlog represents the tasks required to be completed during development. The developers put the things that they want to see in the system as Landing Area, Hotel, Restaurant and Resort Management. The development project includes sufficient time allocation for all features to guarantee efficient execution. It guarantees that every system component complies with user and administrator requirements in addition to assisting developers in setting priorities and managing time effectively.

Table 4. Product Backlog for Tourists (Client)

Priority	Features	Estimated hr/s per day
1	As a user, I want to register and log in to the system.	1hr
2	As a user, I want to make a reservation to available landing area, hotel, restaurant, and resort.	3hrs
3	As a user, I want to avail and schedule for a boat ride tour.	2hrs
4	As a user, I want to provide feedback and rate my experience.	2hrs

Table 4 represents the Product Backlog for tourists, listing the essential functionalities needed in the system. After signing in, every system user has

unique module or project to work on. The developers will prioritize these tasks to enhance user experience. The login system grants users permission to perform unique activities like manage reservation and boat tour scheduling and rating submission. The implementation time for each feature as estimated by developers amounts to daily hours. Users experience smooth navigation through the application with the help of an item monitoring dashboard that tracks system progress.

Managing reservations, setting up boat tours, providing ratings, and gaining access to customized modules after logging in are all included. Developers can concentrate on creating features that directly improve usability and navigation by giving these activities top priority.

SPRINT PLANNING

Sprint planning is focused about prioritizing and doing the development work required to develop a user-friendly and functional web-based ecotourism portal for Pagsanjan Falls landing area, hotel, restaurant and resort. By establishing definite goals and following a systematic planning process, the sprint backlog is created to allow the team to develop high-quality features and system documentation in line with the project objectives and user requirements. Structured approach to sprint planning ensures that every development cycle produces high-quality, functional, and user-friendly components of the web-based ecotourism system, paving the way for a reliable, efficient, and integrated platform that meets the needs of administrators, service providers, and tourists

Use Case Diagram

Figure 4 illustrates the Use Case Diagram, which consists of 4 main actors: Tourists, UBAP, Landing Area management, and Admin. The diagram

demonstrates how each system user interacts with the platform at different levels. The Use Case Diagram presented in Figure 4 shows that the four major actors are Tourists, UBAP, Landing Area Management and Admin. The diagram will explain the relationship of every system user and the platform at various functional levels. Through the system, tourists use it to reserve services, get comprehensive information and make payments. The UBAP and Landing Area Management is in charge of the work that involves reservations, booking confirmation, and the control of daily operations. The Admin remains the general master of the system in the administration of the users, data repositories, and analytical reports.

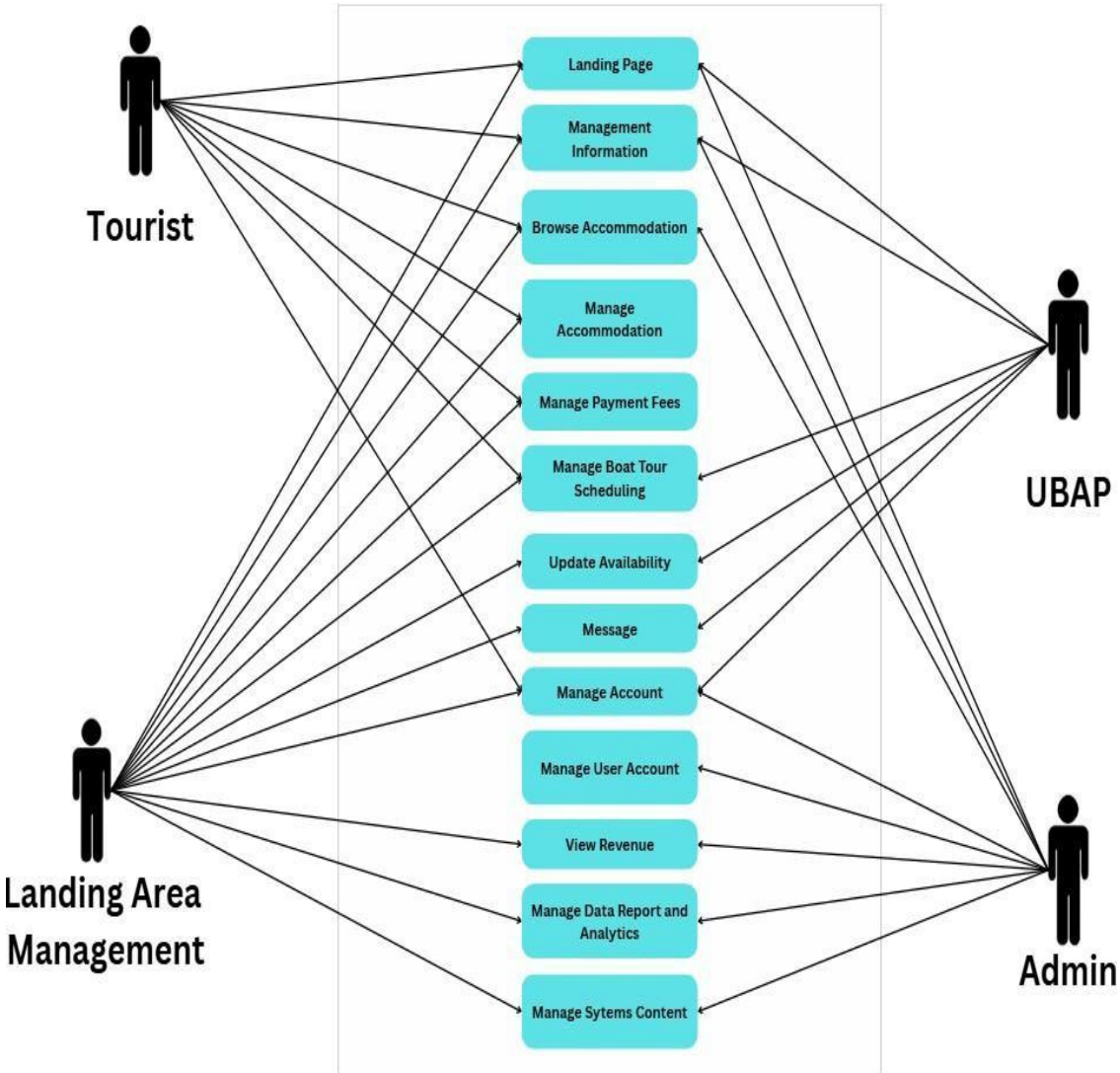


Figure 4. Use Case Diagram

System Flowchart

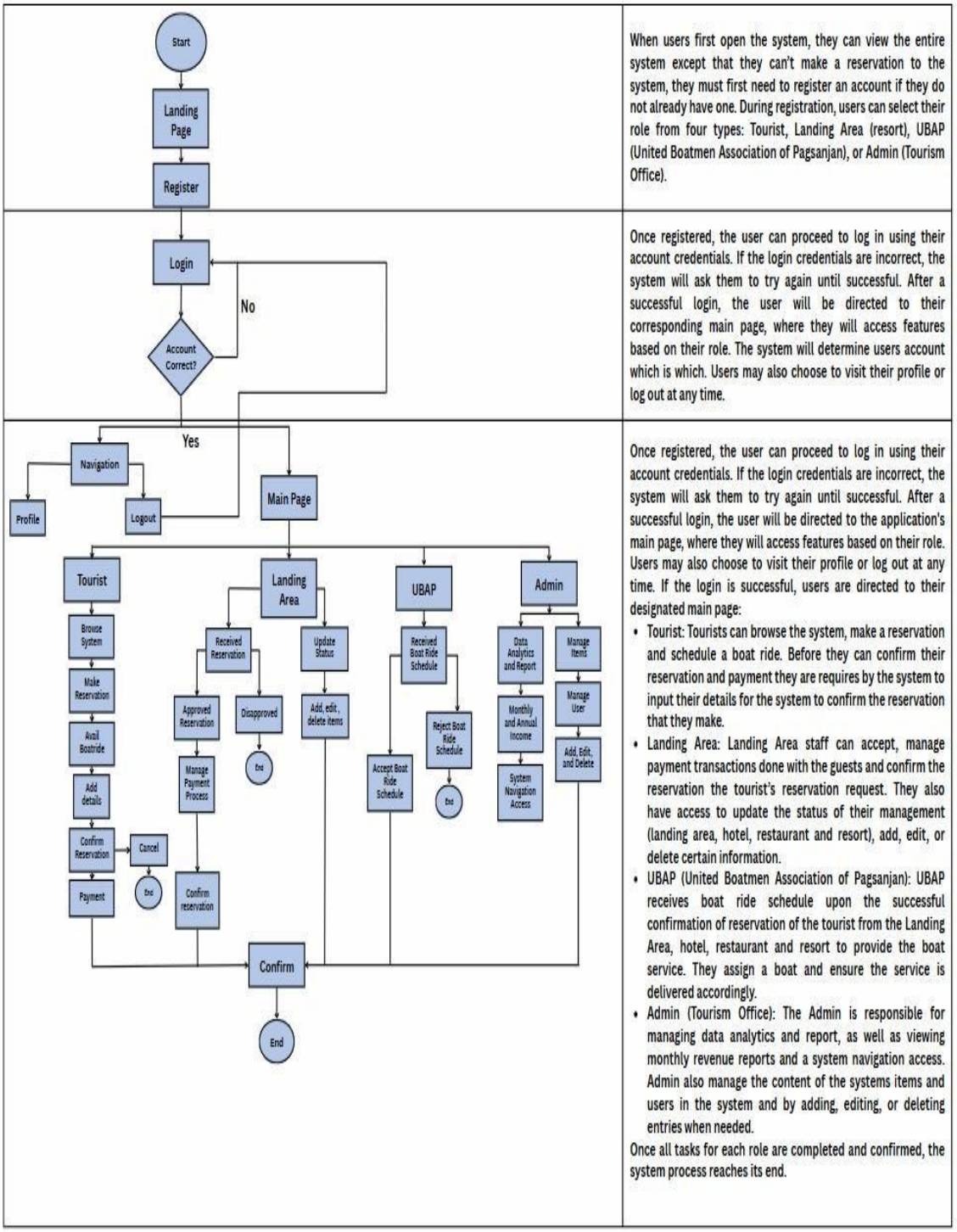


Figure 5. System Flowchart

Figure 5 illustrates the system's overall flow, beginning with user registration where individuals choose their roles as Tourist, Landing Area (resort staff), UBAP (United Boatmen Association of Pagsanjan), or Admin

(Municipal Tourism Office of Pagsanjan). After registering, users log in using their credentials; if incorrect, the system prompts for re-entry until successful. Once logged in, users are directed to their respective main pages. Tourists can make a reservation, avail and schedule a boat ride, and pay a 50% down payment to confirm their booking. Landing Area staff manage and accept tourist reservations, update statuses, and maintain records. UBAP receives notifications upon successful reservation of tourists and provides a boat ride schedule service accordingly to the requests date of tourists. The Admin is responsible for managing data analytics and report, as well as viewing monthly revenue reports and a system navigation access.

Materials

The development phase of the web-based booking system utilized particular software along with hardware tools. The tools proved necessary to maintaining system efficiency and functionality but also for ensuring reliable performance, smooth operation, and successful implementation of all features throughout the development phase.

Software

The following software was utilized during the development of the web-based booking system. PHP and the Laravel Framework were employed to handle the structure, functionality, and scalability of the backend features. These technologies were used to process reservations and payments, manage business logic, and control user administration, ensuring secure and efficient system operations. MySQL Database served as the primary database management system, responsible for securely storing and organizing user

information, reservation details, and transaction records, thereby supporting data integrity, reliability, and efficient retrieval throughout the system.

Table 5. Hardware Specifications

Hardware	Specification
RAM	8GB
Storage	SSD 237GB (Recommended: 256, 512GB SSD or higher)
CPU.	AMD Ryzen™ 3
Operating System.	Windows 11

Table 5 shows hardware specification that the programmer will use as the basic equipment requirements for the project to be started. The system should have at least 8GB of RAM but 16GB RAM will provide enhanced multitasking capabilities. The necessary CPU for this project should be a quad-core model starting from an AMD Ryzen™ 3 or equivalent because of its programming complexity. Windows 11 Operating System will be heavily recommended for the latest and better version of windows. The computer requires a minimum SSD storage capacity of 256GB but 512GB SSD provides the fastest data access together with better storage performance. Furthermore, creating this web-based system uses two or more laptop.

BUSINESS PLAN

In this part of the research this will take in business plan action that will take in Pagsanjan Ecotourism Web-Based Platform and will initiate under of Laguna's Pagsanjan Municipality Tourism Office. It is a move to modernize and improve the local tourism experience by incorporating an online reservation and boat ride schedule system, and easy booking for the landing point, hotels, restaurants, and resorts. It is a strategic initiative to cater to the increasing demand for online services from local and foreign tourists.

Situation Analysis

Currently, the Pagsanjan tourism operations are mostly manual. There is no existing online-based system to accommodate the increasing number of tourist bookings and coordination needs thereof. In the absence of an integrated system, the operations are often causes delays, there is confusion or miscommunication, and inconvenience to customers. The use of a centralized online system would not only improve tourist satisfaction but also more efficient operations in all the tourism service outlets.

Key Assumptions

The creation of the Pagsanjan Ecotourism Web-Based Platform is based on several underlying assumptions. It is presumed that internet connectivity and the utilization of smartphones by tourists will further rise, thus promoting the mass use of the online platform. Partner facilities like the landing site, hotels, restaurants, and resorts are presumed to actively collaborate by updating their data within the system on a daily basis. The Pagsanjan Municipal Tourism Office is also presumed to monitor, maintain, and update the platform on a daily basis to make it reliable and efficient. Lastly, it is presumed that the use of the system will greatly improve the overall tourist experience, thus contributing to a rise in tourism revenue for the municipality.

Major Events

Some significant milestones have already been achieved in the development of the Pagsanjan Ecotourism Web-Based Platform. Conceptualization of the system has already been successfully done, providing a solid foundation for its functionalities and objectives. Initial consultations with key tourism stakeholders in Pagsanjan have already been done to make sure

that the platform suits the needs and desires of the Pagsanjan local tourism community. Moreover, the assignment of the system management function to the Pagsanjan Tourism Office has already been done, thereby assuring proper monitoring and long term sustainability of the platform.

Why We Are Undertaking This Activity

The project seeks to improve the overall tourist experience in Pagsanjan, increase operational efficiency, and increase the competitiveness of the municipality as a tourist destination. Since tourism is one of the biggest economic drivers in Pagsanjan, modernization is needed to sustain global standards and expectations.

How the Project Will Established

The platform that will build for the Municipal Tourism Office of Pagsanjan, Laguna was proposed by the students of LSPU-LB Campus for their Capstone Project for the Pagsanjan Tourism Office in an effort to upgrade tourism facilities and comply with the plans for the development of the municipality ecotourism.

How the Project will Financed

The business will be financed through the efforts of the students from Laguna State Polytechnic University - Los Baños Campus (LSPU-LB Campus), who will develop the entire platform. They will also provide funding from their own resources if any development costs is needed.

Past Problems and Resolutions

Traditionally, manual booking procedures tended to cause overbooking as well as conflicts over bookings, leading to inefficiencies in operations and customer dissatisfaction. In efforts to address this problem, the new web-based

system seeks to centralize booking information and computerize schedule management, thus minimizing errors significantly and improving coordination. Another problem identified was the lack of adequate information sharing across the United Boatmen Association of Pagsanjan (UBAP) landing area management, resorts, and restaurants, which tended to lead to miscommunications most of the time. The system is intended to solve this issue through the provision of up-to-date messages, which are available to all partner establishments, thus promoting smooth teamwork and better service delivery.

Likelihood of Similar Problems and Preventive Measures

Technical downtime and lack of proper system upgrades pose the greatest risks. To prevent this, regular system maintenance cycles, staff training, and a special technical support team will be undertaken. Periodic system audits and customer grievances will also help to identify issues and resolve them in advance.

EXECUTIVE SUMMARY

This business plan offers the web-based reservation and boat ride scheduling platform system design and implementation to assist the ecotourism operations in Pagsanjan, Laguna. The system will assist the management and efficiency of tourist services in the landing area, and in the hotels, restaurants, and resorts that make up the Pagsanjan Falls experience. Today, the business runs without a digital system, and bookings, scheduling, and customer management are manual. The project seeks to automate the processes to make reservations easier, encourage responsible tourism, and improve overall the experience of local and foreign tourists. The system will enable online booking of accommodations, restaurant bookings, and guided boat tours for

tourists, thus reducing long queues and miscommunication, while at the same time encouraging environmental conservation and compliance with local policies.

MARKETING PLAN

Target Market

The primary market includes local and international tourists who are interested in nature-based experiences, specifically the iconic boat ride to Pagsanjan Falls. These tourists often seek convenience, safety, and authenticity when planning their travel.

Market Needs

There is a growing demand for digital convenience in travel, such as online bookings, itinerary planning, and cashless transactions. Tourists also look for environmentally responsible and well-organized tourism experiences.

Marketing Objectives

The marketing goals are to increase awareness of the new online platform among tourists, encourage them to book tours, accommodations, and restaurant tables online, and promote the Municipality of Pagsanjan's commitment to sustainable and organized tourism.

Promotional Strategies

The promotion strategies are to initiate a Website Launch Campaign to promote the platform via the official tourism portal and social media channels. Second, introduce Social Media Marketing via Facebook, Instagram, and TikTok for promoting attractions, user reviews, and booking guides. Third, establish Tourism Partnerships through association with local restaurants, hotels, and travel agencies for promoting the platform and introduce In-Town

Promotions via posting of posters and electronic information boards at Pagsanjan's tourist spots, resorts, terminals, and checkpoints.

Pricing Strategy

The platform will charge a minimal convenience fee for online transactions to support system maintenance, while the core services (accommodations, meals, tours) will remain at competitive local rates. Discounts may be offered in Pagsanjan Festival, summer promo and etc.

Customer Relationship Management

The following will be the Customer Relation Ship Management. The email confirmations for bookings, customer feedback surveys to improve services and 247 online support managed by the Tourism Office. By that travelers receive prompt email confirmations for their reservations, these features facilitate efficient customer relationship management and enable travelers to stay up to date on their bookings. The Tourism Office uses feedback questionnaires to get important information to improve services over time and quickly resolve any problems.

SWOT ANALYSIS (Strength, Weaknesses, Opportunities and Threats)

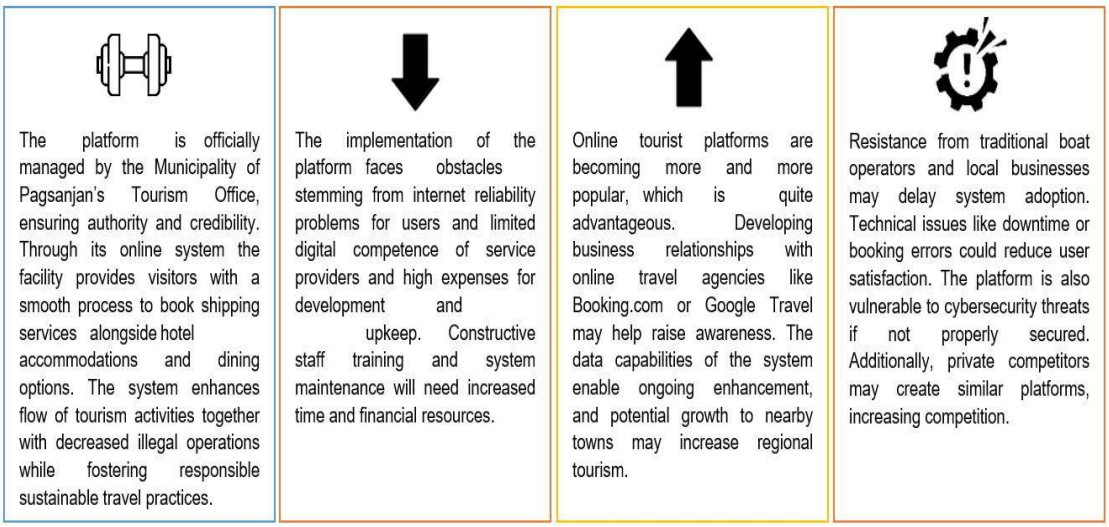


Figure 6. SWOT Analysis

Strengths

The proposed platform is to be managed as a government-backed system, which shall be directly controlled by the Municipality of Pagsanjan's Tourism Office, thus ensuring its credibility and official authority. It provides enhanced convenience to tourists with online reservation and boat tour booking, highly improving the overall experience of travelers. By integrating services across the landing area, hotels, restaurants, and resorts, the platform is an inclusive solution for travelers. It promotes improved organization of the tourism flow by helping to neutralize illegal or unregulated boat tour operations by simplifying booking and scheduling processes. In addition, the system promotes sustainable tourism practices by encouraging environmentally friendly processes and responsible tourist behavior.

Weaknesses

Despite the potential of the platform, there are several weak links that necessitate consideration. There is limited digital infrastructure, as some of the local service providers may not be well aware of or capable of managing an online system. Moreover, internet connectivity may be a significant roadblock. As some tourists and locals may have limited access, which may adversely affect their ability to utilize the platform to the best advantage. Furthermore, the development and upkeep cost of the platform along with the training of employees will necessitate significant monetary investment and time.

Opportunities

There are many opportunities that can be leveraged to enhance the success of the platform. The growing trend of online tourism presents a tremendous advantage, as more and more tourists utilize online platforms for

planning and booking trips. There is potential to form strategic partnerships with leading travel platforms like Booking.com, Agoda, or Google Travel, which can bring in more traffic and visibility. In addition, the ability of the system to gather and analyze data will enable continuous service improvement and the development of better tourism strategies. In the long run, the platform also has the potential to expand to surrounding towns, thus contributing to regional tourism development.

Threats

Resistance from traditional boat operators or other local business organizations may hinder the implementation of the system. System malfunctions, such as site downtime or reservation errors, may adversely impact the satisfaction of the users. In addition, the platform is susceptible to cybersecurity because doing business online makes it vulnerable if not protected properly. Finally, private companies may duplicate the same services, leading to competition that can divert traffic from the municipal system.

HUMAN RESOURCES PLAN

Organizational Structure

The Tourism Office of the Municipality of Pagsanjan will monitor the web-based platform and will be charged with the responsibility of ensuring that the system runs its overall activities as well as sustaining its relationships with local tourism stakeholders. The office will be in charge of implementation, monitoring and continuous improvement of the platform in order to see to it that it serves the tourism activities and services in the municipality effectively. They will also track the usage of the system, create reports and analyze tourism information that will be used in planning and decision-making so as to improve the

management of tourism in Pagsanjan. Under the right supervision and alignment with the stakeholders, the platform will become an effective and structured electronic tool that amplifies the level of transparency, the flow of tourism services and the increased quality of experience of visitors who come to explore Pagsanjan falls and other attractions within the area.

Key Roles and Responsibilities

Table 6. Key Roles and Responsibilities

Position	Responsibilities
Tourism Officer (Project Head, System Administrator, and Partnership Coordinator)	Oversees the platform operations, manage partnerships with local businesses (landing area hotels, resorts, and restaurant), updates system content, and ensures alignment with tourist goals.
Customer Service Representatives (Landin Area, Hotel, Restaurant, Resort Staff)	Provide support to tourists by assisting with reservations, answering queries, and managing booking confirmations through their respective establishments.
IT Support Staff	Responsible for maintaining the technical functionality of the platform, addressing the system issues, and ensuring cybersecurity and data protection.

Recruitment Plan

There is no necessity for additional managerial hiring as the Tourism Officer shall be tasked with various tasks. Customer service activities shall be centralized with the existing staff from the involved hotels, restaurants, resorts, and landing area (arrival area). IT services may be undertaken by municipal IT personnel, a retained professional, or outsourced to a local computer vendor. The system can function efficiently without the need for extra administrative recruits by centralizing cost-effectiveness and streamlined operations.

Training and Development

Training will be organized for Tourism Office staff in system administration and partnership management; for establishment representatives, in handling platform reservations and communication with tourists; and for IT staff, in system maintenance and cybersecurity practices. Training may also be organized in collaboration with tourism or technology development agencies to provide quality and effectiveness in learning.

Salary and Compensation

The Tourism Office staff that will handle the entire system shall be compensated in accordance with the pay grade system of the local government. Partner establishments (landing area, hotel, restaurant, and resort) shall handle and manage the compensation of their own customer care representatives, whose task is to coordinate with the tourists via the platform. The IT support services may be compensated either through contracts of service or on-call payment arrangements depending on the agreed terms with the Municipal Tourism Office of Pagsanjan, Laguna.

All system employees will be fairly compensated in accordance with their duties thanks to this compensation scheme. Employees in the tourism office, representatives of partner establishments, and IT support workers get compensation commensurate with their tasks, encouraging responsibility, productivity, and efficient coordination. Contracts or on-call payments are examples of flexible arrangements for IT services that enable the system to function efficiently while adjusting to changing workload demands. This compensation scheme ensures fair and role-appropriate payment for all personnel involved in the system.

Human Resource Needs Summary

Table 7. Human Resources Need Summary

Department	Number of Staff	Status
Management & Admin (Tourism Office	1	None
Customer Service (landing area, hotel restaurant and resort	11	None
IT & Technical Support	3	Existing

KEY MANAGEMENT CONTROLS

Internal Controls

In this part of the Internal controls, Pagsanjan Ecotourism Web-Based Platform shall be ready to implement sound internal controls to make sure all transactions are consistent with the business goals of having well-organized and sustainable tourism. All procedures in reserving, boat ride schedules, and customer service operations will be managed by the landing area, hotel, restaurant, and resort and will be observed by Municipal Tourism Office as system administrator to guarantee quality service and effective management that such an arrangement makes the roles and responsibilities of each stakeholder in the system very clear. Regular system audits shall be conducted to determine any error in bookings, delay in service, or customer complaints. Monitoring mechanisms are built into the platform to monitor system uptime, booking rate fulfillment, and customer complaints, thereby allowing for constant improvement and consistency. This aspect assists in enhancing the co-ordination among tourism stakeholders as well as helping to enhance long-term sustainability and effectiveness of tourism operations in Pagsanjan Falls and it also makes all the transactions and operations in the system transparent and accountable.

Administrative and Accounting Controls

Administrative controls will include use of a central management dashboard which will be available only to authorized staff members of the Tourism Office. All financial transactions, including bookings, payments, and cancellations, will be automatically recorded to determine transparency and facilitate audit processes. The platform will also provide monthly and yearly reports relating to revenues, tourist volumes, geographical information, and operational processes, thus facilitating planning and decision-making processes. In accounting controls, financial operations will be recorded with utmost caution through a combined system for maintaining accurate and real-time management of revenues and expenses. Periodical reconciliation of the accounts will be done to determine discrepancies at the earliest. Independent annual auditing of finances by a third-party accounting firm will also be organized to verify financial integrity as well as enable investors and stakeholders to have clear records.

PRODUCTION PLAN

Production Process

The main product will be an online web-based reservation and boat ride booking site that can boost and simplify the tourism experience within Pagsanjan. This site will allow tourists to easily book their stays in areas like landing fields, hotels and resorts. Moreover, users have an opportunity to reserve restaurant table, book boat ride to Pagsanjan Falls organized by the selected accommodations, as well as, get full access to the tourism-related information including safety rules, activity schedule, and local attractions. The system will make planning easier to the tourists, enhance efficiency in

operations of the service providers and help in making ecotourism a safer and more orderly experience by incorporating these features in one digital platform.

Steps in the Production Process

The production process of the web-based platform will entail several major steps to ensure its successful development and implementation. First, system development will be initiated through consultation and coordination with IT professionals in designing and building the website. Second, partner integration will be conducted, where coordination with the Municipal Office of Pagsanjan, UBAP (United Boatmen Association of Pagsanjan), hotels, restaurants, resorts, and the landing area will ensure that their availability, rates, and services are integrated into the platform properly. Once the system is integrated, usability and technical testing will be conducted to determine and rectify any problems before the official public launch. In day-to-day operations, customer service representatives from each partner establishment will take care of and coordinate bookings while taking care of tourists as necessary. Finally, system maintenance will be taken care of by IT support staff, who will be responsible for updates, bug fixing, and data security preservation.

Tools, Materials, and Equipment

Table 8 shows the tools, materials, and equipment need the web-based tourist platform to be properly implemented and run. In order to support system development, data storage, online reservation processing, and communication between visitors, administrators, and partner establishments, these resources are crucial. By facilitating seamless system operation, safe information management, and ongoing online services, the aforementioned components guarantee the system's effectiveness, dependability, and accessibility to

efficiently handle updates and transactions pertaining to tourism while offering visitors a practical platform for accessing tourism services within the municipality.

Table 8. Tools, Materials, and Equipment

	Item	Purpose
1	Website Domain & Hosting	To publish and maintain the online platform
2	Computers	Used by staff to access and manage reservations
3	Internet Connection	Required for daily platform operations
4	Website CMS or Booking System Software	Backend system for managing content and bookings
5	Communication Tools (e.g., email, chat)	For communicating with tourists and partner establishments

Location and Facilities

The platform's operations will be managed at the Tourism Office of the Municipality of Pagsanjan, while customer service support will be extended through partner landing area, hotels, resorts, and restaurants and lastly the UBAP office for boat ride coordination.

Quality Control Measures

To provide consistent quality and reliability of the web-based system, a number of quality control steps will be taken. System performance checks at regular intervals and gathering of user feedback will be conducted to detect and resolve potential problems early. Pre-scheduled updates and routine bug testing will be conducted by either the IT support staff of the Municipal Tourism Office of Pagsanjan or local and outsourced IT experts, as the case may be. Regular coordination meetings with partner establishment will be conducted to

ensure current accuracy of service information and resolve operational issues. Tourist feedback forms and ratings will also be gathered to measure user satisfaction and improve service quality on a regular basis.

Suppliers and Partners

While the platform itself doesn't require physical suppliers, the key partners are local Municipal Tourism Office IT services providers for the development and maintenance of the system. The landing area, hotels, restaurants, resorts and UBAP for service coordination and Local internet service providers to make sure of reliable connection. They also assist in managing bookings, coordinating tourist services, and addressing user inquiries. Providers maintain connectivity, enabling efficient, org and accessible tourism services for all users.

RISK ANALYSIS

Table 9 lists some of the risks description, impact of the risk and the mitigation strategy that might affect the implementation and operation of the web-based travel platform, including technical issues, cybersecurity threats, challenges in digital literacy, power or internet-based issues, resistance to adoption, seasonality of demand, natural disasters, reputational risks. It outlines the possible impact of each risk and provides the recommended reduction methods to prevent or reduce disruptions. These risks are critically evaluated so that the system is still able to move on even where difficulties arise. Achievement of effective planning and preventive measures aid the reduction of adverse consequences on users and stakeholders. Moreover, it is noteworthy to monitor the system constantly and update it regularly to ensure stability, security, and the overall performance of the system.

Table 9. Type of Risks, Description, Impact, and Mitigation Strategy

Type of Risks	Description	Impact	Mitigation Strategy
Technical Issues	System crashes, bugs, or downtime may affect bookings and user experience.	High	Regular system maintenance for trouble shooting.
Cyber Security Risks	Unauthorized access, data breaches, or hacking attempts.	High	Implement strong data encryption, firewalls, and user authentication systems.
Low Digital Literacy	Some staff or tourists may struggle with using the platform.	Medium	Conduct training sessions for staff and provide a user-friendly interface with help guides and FAQs for users.
Power/Internet Outages	Interruptions in connectivity may disrupt real-time bookings.	Medium	Coordinate with local ISPs, use pocket wifi or modem and mobile internet as a backup during outages.
Resistance to Adoption	Some partner establishments may prefer traditional booking methods.	Medium	Conduct awareness campaigns and show the benefits of the platform. Offer onboarding assistance and continuous support.
Seasonal Demand Fluctuations	Tourist numbers vary, especially during off-season.	Low	Implement flexible promotions and dynamic pricing. Use data analytics to forecast trends and adjust operations accordingly.
Natural Disasters	Flooding, typhoons, or earthquakes may affect the tourism flow and infrastructure.	High	Develop a disaster response plan and allow for booking cancellations or re-scheduling through the system.
Reputation Risks	Negative reviews or poor customer service.	High	Provide high-quality responsive customer service and allow partners to address issues quickly. Monitor reviews and implement a feedback resolution.

FINANCIAL PLAN

This financial plan gives a thorough summary of the Pagsanjan tourism platform's predicted revenues, operating costs, startup costs, and break-even analysis. It ensures that resources are deployed effectively by highlighting the financial requirements required to launch and sustain the system. The plan assists in determining the platform's potential profitability and financial sustainability. It also helps stakeholders make well-informed decisions, efficiently control expenses, and contribute to the platform's long-term success.

Table 10. Existing Financial Expenses

Sources	Expenses
Bangkero	₱700.00
Municipal	₱250.00
UBAP	₱300.00
Landing Area	₱100.00
Total:	₱1,350.00

Table 10 show the existing financial contributions from key sources, showing the total expenses incurred to support the platform’s operations. It highlights the cost distribution among stakeholders and the overall funding required. This breakdown provides a clear understanding of the financial responsibilities and helps in planning future budgets to ensure the platform’s sustainability and continued development. It highlights the cost distribution among stakeholders and the overall funding required.

Monthly Operational Expenses (System)

Table 11 summarize the platform’s projected monthly revenue from user subscriptions and booking fees, providing an overview of expected income to support its ongoing operations and financial sustainability assists in determining the financial feasibility of the system.

Table 11. Monthly Operational Expenses

Source	Assumption	Monthly Income
Subscription/Registration Fee	11 x 199	₱2,189.00
Booking Fee	100 x 300 person	₱30,000.00
Total Monthly Income		**₱32,189.00**

EXIT STRATEGY

The Pagsanjan Ecotourism Web-Based Platform shall position itself in a sustainable way under the Pagsanjan Municipality Tourism Office. The main purpose is to expand tourism operations, advance local business, and improve the tourist experience. Since this is to be accomplished by the government, the platform shall not be subject to private sale. The primary goal of the platform is to serve the Municipality of Pagsanjan by enhancing tourism management and encouraging sustainable practice. However, if the Municipality decides to transfer or expand operations through a future public-private partnership, a number of exit opportunities will be available. To reach the expansion target, the business will concentrate on spreading its services by incorporating additional ecotourism sites in Laguna and using mobile applications for easier access.

An exit through handover is also anticipated; in the event that the platform is self-sustaining and successful in meeting its goals, management control can be transferred to a private local IT service firm, while maintaining periodic municipal supervision for long-term upkeep. In terms of financial return, since the platform is managed publicly, immediate financial returns are not the concern; instead, any revenue generated will be re-invested in the tourism industry to drive further development. In the event of future public- private partnerships, investor exit opportunities.

Conclusion and Recommendation

The Pagsanjan Ecotourism Web-Based Platform is a strategic way of enhancing the tourism experience within the municipality. By offering a centralized boat ride scheduling and reservation platform, the system optimizes operations, enhances tourist satisfaction, and creates local economic activity.

Recommendation

To address current gaps in tourist handling and boat ride management, immediate development and implementation of the platform should proceed. All stakeholders, including personnel from the landing area, hotel, restaurant, and resort, must be properly trained in using the system. The platform should be regularly updated and maintained to ensure accurate information and to adjust for seasonal demand. A targeted marketing campaign must be launched to introduce the system to both local and international tourists.

ANNEXURES

In these, it will show the annexure of the research and the following are the supporting documents which will be attached. Financial Statements and Projections, including estimated breakdown of the initial costs, monthly cost of operations, revenue projections, to be furnished by the Municipality of Pagsanjan, Letters of Intent of the clients, and the contracts of the suppliers, to be procured from the participating resorts, restaurants, and boat operators. These documents also support the credibility and completeness of the proposed system.

The supporting documents can be the evidence of the financial feasibility of the project, its operational preparedness, and the commitment of the stakeholders to the project. They offer more specific cost estimates, estimated revenues and

formalities with partners and clients, which would guarantee transparency, feasibility and adequate coordination in successful implementation of the proposed system. They serve as a guide for tracking expenses, projecting potential revenues, and ensuring that the project remains financially sustainable throughout its lifecycle. The documented commitments from partners and clients also strengthen accountability and coordination, reducing the risk of delays, miscommunication, or operational conflicts.

Table 12. Sprint Backlog (Admin)

Priority	Implementation	Task	Estimated hours per day
1	As a user, I want to register and log in to the system.	Design and develop the registration and login page.	3
2	As a user, I want to manage users, UBAP, landing area, hotel, restaurant and resort management.	Design and develop an interface for the reservation platform for tourists, UBAP, landing area, hotel, restaurant, and resort.	3
3	As a user, I want to view all the Data Analytic and Reports.	Integrate a report system registration.	2
4	As a user, I want to manage all the content of the system.	Design and develop a content Management user page.	3

Table 12 provides the exact tasks the researchers intend to complete during the upcoming sprint. It acts as a detailed guide, clearly indicating what needs to be achieved within the timeframe specified, and assists the team in efficiently organizing their workload. The product backlog, particularly the host (admin) section, contains a prioritized list of features, allowing researchers to select which work should be tackled first and which can be put off until later. Following this systematic strategy allows the researchers to focus on high-

priority items while ensuring that less important tasks are not ignored.

Additionally, using the sprint and product backlogs offers a methodical manner to keep track of completed activities, assess progress, and make required adjustments to ensure that project objectives and deadlines are met. While the product backlog provides a long-term view of all the features and tasks that must be executed, the sprint backlog enables researchers to concentrate on short-term goals within a given time window. Prioritize high-impact projects, and they disrupt the timetable by using this dual strategy.

Table 13. Sprint Backlog Landing Area, Hotel, Restaurant and Resort

Priority	Implementation	Task	Estimated hours /day
1	As a user, I want to register and login to the web-based platform.	Design and develop the registration and login page.	3
2	As a user, I want to manage landing area, hotel restaurant or resort for tourists.	Design and develop an interface for a reservation platform.	3
3	As a user, I want to manage the reservation made by tourists.	Develop reservation management with date schedule platform.	2
4	As a user, I want to manage payments process done by client.	Integrate a payment gateway into the web-based system.	3

Table 13 provides an overview of the key features planned for implementation in the upcoming sprint backlog. It presents all the tasks that need completion before the start of the next sprint. The priority lists consist of user-requested items, which are determined based on the needs of the users. Researchers can efficiently allocate resources, evaluate progress, and

concentrate on high-value tasks first by adhering to this prioritized list. This methodical approach encourages accountability, helps avoid delays, and improves team member collaboration. Overall, the project planned features and prioritized activities, which supports the accomplishment of the project's overall objectives and the effective completion of each sprint. This methodical approach supports the effective fulfillment of both sprint objectives and overall project goals by promoting accountability, improving teamwork, and offering a clear structure for finishing tasks on time and also minimizes errors, facilitates collaboration, and ensures that critical objectives are accomplished.

Table 14. Sprint Backlog (Tourist)

Priority	Implementation	Task	Estimated hours /day
1	As a user, I want to register and login to the web-based platform.	Design and develop the registration and login page.	3
2	As a user, I want to view available landing area, hotel restaurant or resort.	Design and develop an interface for a reservation and accommodation platform.	3
3	As a user, I want to set a reservation and schedule a boat ride.	Develop a booking form with a calendar interface to choose dates.	2
4	As a user, I want to make payment in online for hassle less free.	Integrate an online payment gateway into the web-based platform.	3

Table 14 gives a general idea of the features, which will be implemented in the next sprint backlog. It enumerates all activities that have to be done before the next sprint, and the priorities are decided by the needs of the users and requirements of the system. Some of the features are user registration and login to access it securely, an interactive landing page where the user can see the services available, a booking form which has a calendar to make a reservation and an online payment gateway that will allow the user to make

transactions conveniently. Every job is allotted an estimated number of hours a day to allow effective scheduling and development on time. This is a systematic plan that can be used to make sure that the necessary functionalities are provided in a systematic and efficient manner.

Daily Scrum. The developers scheduled a 20-minute meeting each day to meet the sprint target and decide whether any adjustments to the sprint or the product backlog were required.

Sprint Execution. The developers meet the client/user (tourist, landing area management, UBAP and Admin) to discuss what they can provide and how they'll do it in order to fulfill their sprint objective. This allows developers/programmers to estimate the amount of work ahead of you and positions your team for success

Sprint Review Meeting. The developers assemble a meeting per week to go over the work that has been accomplished and see if any modifications are required. The developers/programmers will showcase the accomplished work, whether there are missing features or if they want to add them for the system to be fully completed.

Sprint Retrospective. Each week, developers completed 2-3 priorities in terms of people, equipment, procedure, and relationships. The developers focus on the user as an admin or store proprietor, with 50% completed throughout the primary month of development. The developers made an improvement chart to focus on how they intended to create it. The team also had frequent reviews at the completion of each sprint to measure their achievements, to identify challenges and also to pinpoint the areas that needed more improvement for the system.

Software Testing

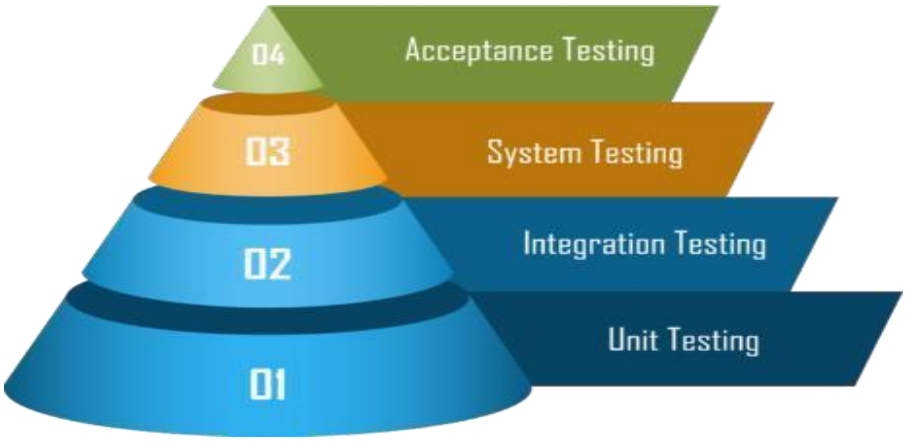


Figure 7. Levels of Testing

In these part of the study, it will show the Figure 7 that shows 4 the phases of software testing which is the acceptance, system, integration and unit testing that can be applied to any software unit or component. The purpose is to verify that every software component works as expected. The figure outlines the phases of software testing, where individual software units and components undergo examination. The goal is to make sure that every software component works as intended.

Unit Testing

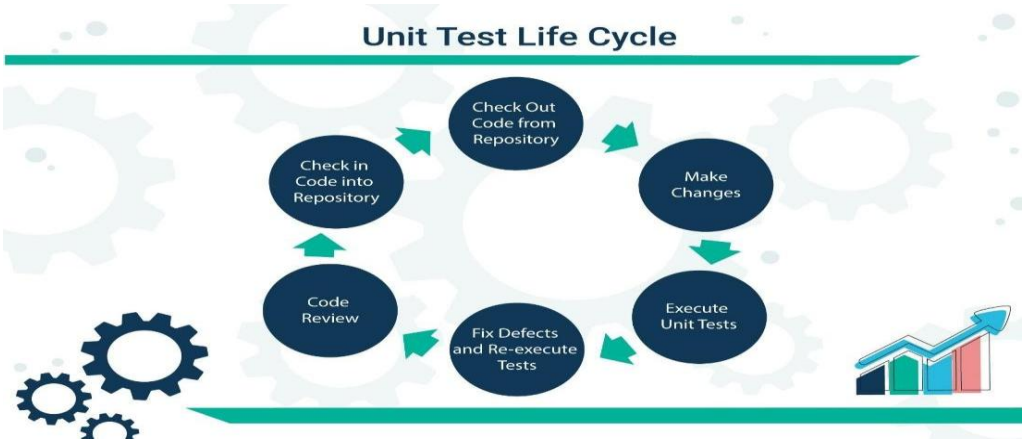


Figure 8. Unit Testing

Figure 8. In Unit Testing, every component or module of the system is tested thoroughly. Through this process, the system is ensured to be developed as per the scheduled research schemes and specifications designed by the

researchers. Moreover, the component is analyzed carefully, with special focus on its primary functionality and performance. The integrity and reliability of the system are analyzed carefully through this intense scrutiny, allowing for the detection and rectification of any potential issues or inefficiencies in the software development process.

Integration Testing

Figure 9 illustrates the Integration Testing, in which researchers confirmed the operation of the module as defined in the system design in the testing phase. This involves all parts of the system to identify faults and errors so that they work in harmony as intended. This strict testing procedure ensures that all modules work in harmony with one another, exposing any hidden bugs or integration issues. Thus, the performance and dependability of the system are significantly enhanced before final deployment. It enables the researchers to identify the inconsistencies, compatibility issues, and communication errors that otherwise can be not observed when testing the modules individually.

System Testing

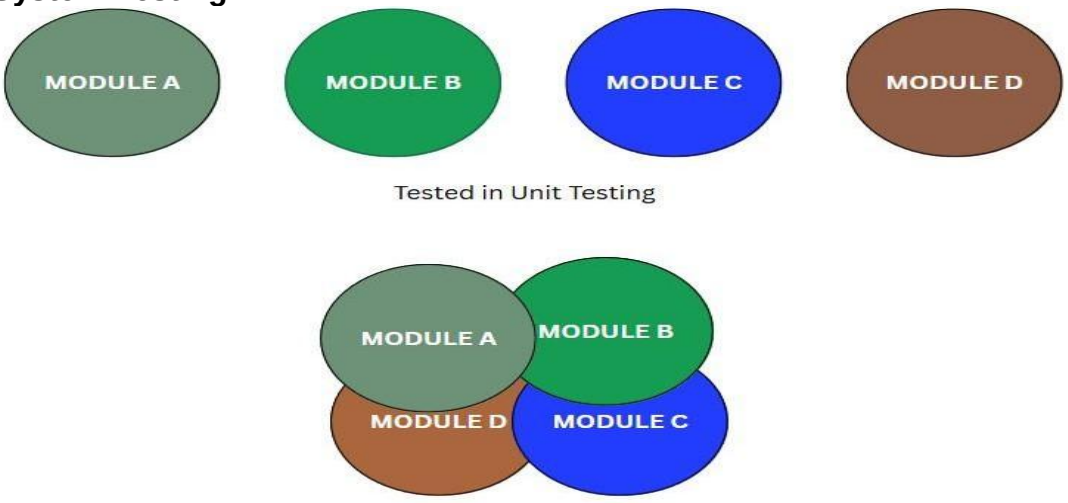


Figure 9. System Testing

System Testing was a comprehensive audit of the whole system to ensure it was performing well. We aimed to validate the integration and

execution functionality of all modules and components. Test cases entailed ensuring the responsiveness of key features like booking and payment processing, testing the robustness of data transmission between the server and users, and ensuring the overall performance and stability of the entire system.

Acceptance Testing

Acceptance testing is the final validation stage, designed to ensure that the project meets the expectations, requirements, and practical needs of its users. During this phase, researchers rigorously examine every aspect of the system, including its design, functionality, usability, and performance, to confirm that it aligns with the intended objectives and satisfies user requirements. Formal testing verifies whether the system complies with the established acceptance criteria, allowing users and stakeholders to make informed decisions regarding its approval. This stage not only confirms that the system operates as planned but also helps identify potential areas for improvement, which can enhance reliability, usability, and overall user satisfaction. Observing how actual users interact with the system provides valuable insights into their behaviors, preferences, and challenges, which can guide future refinements and updates. This evaluation ensures that the platform is intuitive, responsive, and capable of supporting the operational needs of administrators, service providers, and tourists alike. The last stage is acceptance testing where it is identified whether the system meets the needs of the user and the project requirement. During this stage, the system is tested on the basis of its design, functionality, usability and performance to make sure that the system works as per the expectations and meets the previously set standards. It also helps to define ways on how it can be improved and ensure that the system is sound,

user-friendly and suitable to administrators, service providers, and tourists.



Figure 10. User Acceptance Testing

EVALUATION PROCEDURE

The evaluation of the system was conducted using the ISO 25010 Software Quality Model and the Technology Acceptance Model (TAM). Theses standard and models were employed to guide the assessment, which focuses on the system's performance and user experience. This evaluation aims to provide a systematic approach for measuring how well the system meets its intended objectives and fulfills user requirements. By applying these established frameworks, the researchers ensured a comprehensive, reliable, and structured evaluation of both the technical quality and overall user acceptability of the system. The ISO 25010 model assessed key quality attributes such as functionality, reliability, security, and maintainability, while TAM measured users' perceived usefulness and ease of use.

Through this combined approach, detect areas for improvement, and validate whether the system is practical, efficient, and suitable for real-world

implementation. It also helps in understanding user behavior, preferences, and challenges when interacting with the system, allowing for targeted adjustments that enhance overall satisfaction. This ongoing assessment ensures that the system evolves in response to technological advancements and user expectations, ultimately supporting its long-term effectiveness, reliability, and successful integration into the intended operational environment. Can guide developers and stakeholders in making necessary refinements to further improve the system's efficiency, usability, and functionality. Continuous evaluation and feedback from users remain essential in ensuring that the system adapts to evolving technological trends and the growing demands of the tourism sector. The ISO 25010 has highlighted the important areas such as functionality, reliability, security, and maintainability, whereas the TAM determined the usefulness and easy usage of the system by users.

ISO 25010 Software Quality Model



Figure 11. Quality Characteristic of ISO 25010

IT experts tested the developed system with the ISO25010 software

quality standard to analyze system quality concerning the technical aspect and overall performance. The ISO model 25010 is also popular in the context of software evaluation because of the fact that it evaluates not only the functional but also the non-functional features such as internal structure and external behavior of the system.

These criteria are Functional Suitability, Performance Efficiency, Compatibility, Interaction Capability, Reliability, Security, Flexibility, Maintainability and Safety. The Likert Scale was based on four response options that included Strongly Agree (SA), Agree (A), Disagree (D), and Strongly Disagree (SD). The collected responses were then analyzed to measure the system's acceptability and identify areas for improvement and enhancement.

Technology Acceptance Model (TAM)

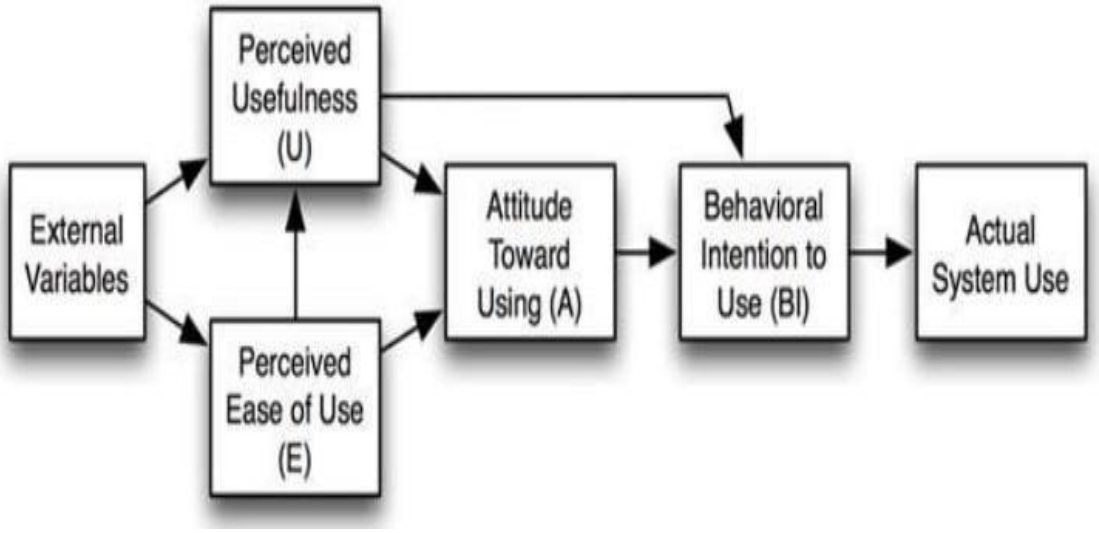


Figure 12. Technology Acceptance Model (TAM)

The four categories of TAM such as Perceived of Usefulness, Perceived Ease of Use, attitude Towards Use, and behavioral Intention to use will be evaluated by the 50 respondents. Each item on the respondent's evaluation has a 4-point Likert scale score, which is used to determine the results.

Sampling Design

Purposive sampling is a research methodology whereby the analyst will choose individuals from the immediate environment to participate in the study based on their own judgment. This type of non-probability sampling was utilized because it allows data to be collected. This study had a total of 50 respondents. It consists of four (4) Staff of Municipal Tourism Office of Pagsanjan Laguna, six (6) Landing Area Management Staff, (10) ten members of UBAP (United Boatmen Association of Pagsanjan) and thirty (30) tourists to test the technology acceptability of the made system.

Population of the Study

Table 15. Respondents of the Research Project (ISO25010)

CATEGORY OF RESPONDENTS	SAMPLE
IT expert	3
Total	3

Three (3) IT specialists have been chosen to evaluate the system in accordance with the ISO 25010 quality criteria. They will examine the system's secure processing of gathered data and identify any problems such errors, defects, and system failures. These specialists have demonstrated, examined, and evaluated the viability of creating a web-based system in addition to assessing the platform's general functionality and performance. Through evaluating elements like reliability, security, maintainability as well as functionality, these experts can offer crucial information on areas that may be weak and those that need to be developed. Their assessment can also be viewed as a reference point on the quality of the system as it aids the

developers to make the necessary refinements, improve the performance and make the platform to be more secure and efficient. Their analysis can be used as the guiding principle by developers to improve the performance, security, and efficiency of a system.

Table 16. Respondents of the Research Project (TAM)

CATEGORY OF RESPONDENTS	SAMPLE
Tourism Office Staff of Pagsanjan	4
Landing Area Management Staff	6
UBAP	10
Tourist	30
Total	**50**

To evaluate the system's user acceptance based on the Technology Acceptance Model (TAM), fifty (50) respondents composed of Municipal Tourism Office administrators, Accommodations, UBAP, and tourists were selected to participate in the study. These respondents were given access to the web-based system and were asked to explore and use its features related to ecotourism information, online booking, and tourism service management.

Data Collection Instrument

Data were gathered using questionnaires and surveys, which played an important role in the outcome of the project. For clarity purposes, participants will be given a questionnaire by the researchers to conduct surveys in a bid to obtain important data. The data form the basis of the web- based system. Conducted interviews were used as the primary data collection tool in this study; the researcher came up with the interview questions with a high degree of focus on developing a strategy with end- users' specific needs matching their desires. In analyzing the various aspects of the web-based system, questions were developed in an attempt to comprehend the Design and operation of the system

and ensure that the end-users' needs were adequately addressed in an attempt to analyze the intervention. The questionnaires will be collected and given to tourists, UBAP, Landing area and Municipal Tourism Staff of Pagsanjan Laguna in an attempt to measure their acceptance of the web-based application. The questionnaires will be created based on the Technology.

Statistical Treatment

The developers used population sampling, a type of purposive sampling approach where the entire collection of target respondents was gathered and analyzed to test and measure the previous characteristics and overall performance of the application to be developed. The Stratified Sampling Technique was also applied, which categorizes the population into a series of groups based on the responder type. The developers also used Likert Scale techniques to measure the application features. The measurement consisted of a series of questions that could be answered with one of four available responses or impressions which were sorted in the following order: four (4) – Strongly Agree, three (3) – Agree, two (2) – Disagree, and one (1) – Strongly Disagree.

The findings made using the technique is a significant source of determining the strengths and weaknesses of the application. It also helps the researchers to know the effectiveness of the system in satisfying the needs of the users and whether the features of the application work as expected. It provides a clear basis of how to evaluate the feedback of users and support the data-based decisions. The formula is as follows:

Highest Score – Lowest Score	4 – 1 = 3
Highest Score	3 / 4 = 0.7

Table 17. Frequentist Individual Item Reliability Statistics

Item	If item dropped Cronbach's α
PI1	0.955
PI2	0.955
PI3	0.956
PI4	0.955
IA1	0.955
IA2	0.959
IA4	0.955
SI1	0.955
SI2	0.955
SI3	0.957
SI4	0.956
PU1	0.954
PU2	0.955
PU3	0.955
PU4	0.954
PEOU1	0.953
PEOU2	0.955
PEOU3	0.956
PEOU4	0.957
IA3	0.955

Table 17 shows the Frequentist Individual Item Reliability Statistics with Cronbach's Alpha when Item dropped to determine the reliability of the survey questionnaire on the analysis of the Pagsanjan Ecotourism Web-Based Platform. The questionnaire questions in this study were founded on the constructs of the Technology Acceptance Model (TAM), such as Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Behavioral Intention (PI), System Interaction (SI), and Information Awareness (IA). The values presented in the table are between 0.953 and 0.959 which are very high values of reliability as values above 0.90 show there is high internal consistency.

Table 18. 5 Point Likert Scale for ISO 25010

RATING	MEAN RANGE	INTERPRETATION
4	3.25 - 4.00	Strongly Agree
3	2.50 – 3.24	Agree
2	1.75 – 2.49	Disagree
1	1.00 – 1.74	Strongly Disagree

Table 18 presents the 5-point Likert scale for ISO 25010, which serves as the basis to interpret respondents' ratings. A mean score of 3.25–4.00 indicates Strongly Agree, showing excellent system performance and high user satisfaction. Scores of 2.50–3.24 indicate Agree, reflecting good performance with minor areas for improvement. Mean scores of 1.75–2.49 represent Disagree, suggesting features that need enhancement, while 1.00–1.74 corresponds to Strongly Disagree, indicating poor performance requiring major improvements.

Table 19. 4 Point Likert Scale for TAM

RATING	MEAN RANGE	INTERPRETATION
4	3.25 - 4.00	Strongly Agree
3	2.50 – 3.24	Agree
2	1.75 – 2.49	Disagree
1	1.00 – 1.74	Strongly Disagree

Table 19 show the Verbal Interpretation Scale for TAM. The respondents will evaluate the system using this rating to interpret the user acceptance level and new technologies. The evaluation consisted of four responses, such as the following: four (4) – Strongly Agree, three (3) – Agree, two (2) – Disagree, and one (1) – Strongly Disagree.

The composite mean was calculated from the respondents' feedback regarding their perception of the technology's usefulness, ease of use, attitude, and behavior. This evaluation not only reflects the immediate reactions of the users but also provides insights into their overall experience.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the results and discussion derived from the methods and procedures stated in the previous chapter. The results are organized and discussed in relation to the research objectives and insights drawn from the collected data.

1. Presentation of the System

Information management for tourists and access relevant ecotourism information efficiently.

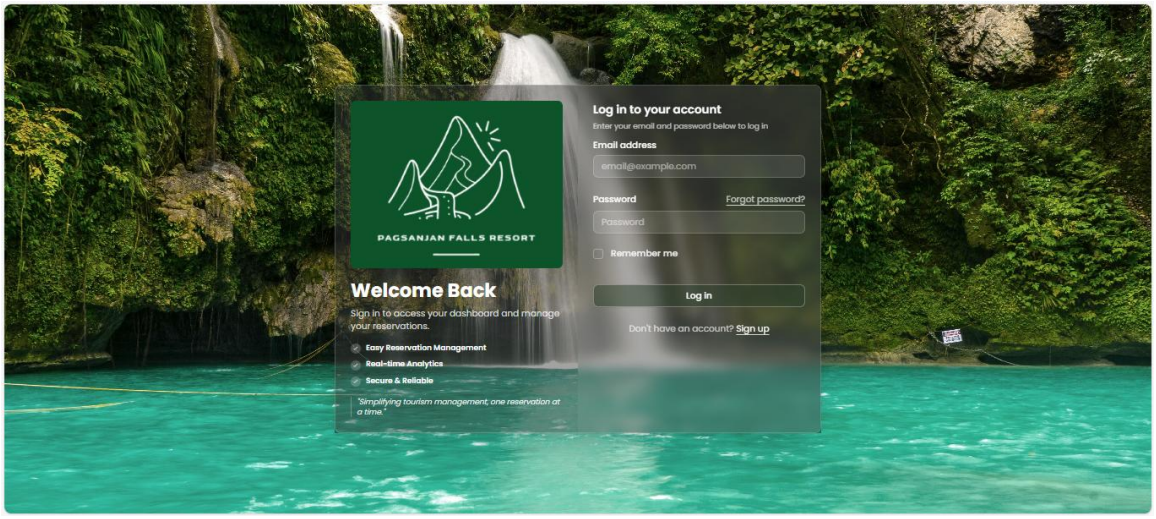


Figure 13. Login/Signup page

User Login Page will enable registered users to access the Pagsanjan Ecotourism Web-Based Platform with their email address and password. It provides an easy and straightforward log-in form with the following features like the options of Remember me and Forgot password to help the user with ease of use and account recovery. Ensures that user accounts remain secure while providing a personalized experience, as registered users can access their bookings, itinerary, and preferences seamlessly.

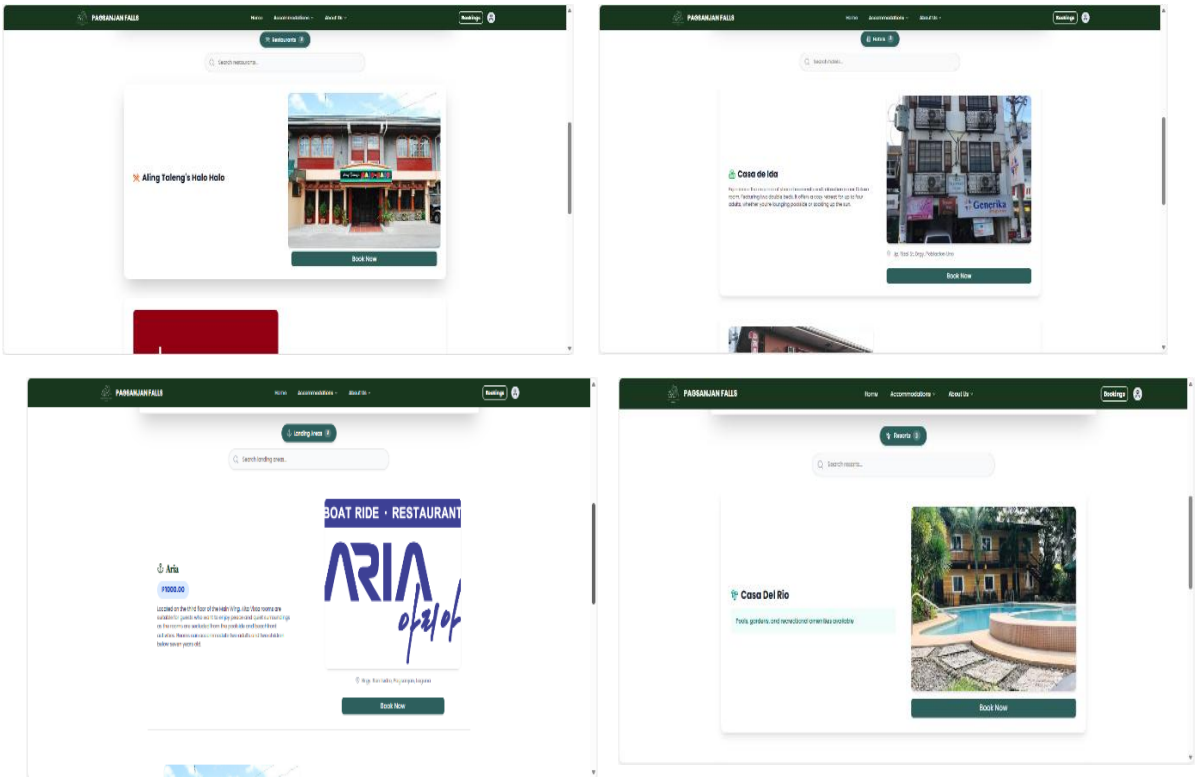


Figure 14, 15, 16, 17. Landing Area, Hotel, Resort and Restaurant Accommodation

Figures 14, 15, 16, 17. These four pictures depict the accommodation part that is displayed in the Pagsanjan web-based booking system, designed to help guests find the right accommodation. The pages list some of the different types of accommodation, such as landing areas, restaurants, resorts, and hotels, with clear photos and relevant information related to the services offered and prices. The available options can be easily viewed by the users to evaluate what each of the establishments can provide. The system provides a seamless booking experience where guests can check availability in real-time and make reservations instantly. The platform also includes detailed descriptions of facilities, nearby attractions, and special offers, allowing users to make informed decisions. This feature not only improves convenience for tourists but

also assists the accommodation providers in managing bookings more efficiently and increasing visibility to potential guests.

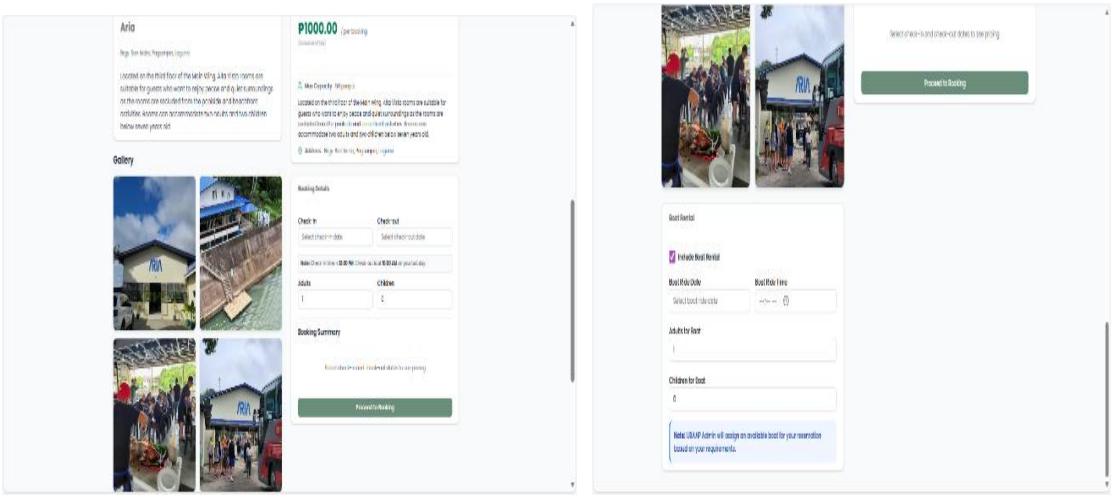


Figure 18. Booking Details for Accommodation

Figure 18 includes the booking information of accommodation, including its location, description, per-booking price, and number of guests. It will also provide a photo gallery to help tourists visualize the premises with clear booking fields to check-in and check-out dates, and the number of adults and children. It also gives a summary of the booking to direct the details to users before their reservation is confirmed, and thus makes it easier to do and easier to understand for the tourist using the system.

The website supports the introduction of new services, like boat tours to the famous Pagsanjan Falls or rapid-shooting trips, by enabling the user to choose the preferred time and the number of guests. This platform not only supports the visitors but also supports the accommodation providers and tour operators to manage the reservations efficiently. The system streamlines the administration by enabling information and booking to be centralized in a single location, thus reducing the number of administrative tasks done, double booking of rooms, and giving a clear track record of all transactions made. This

combination of accommodation and activity booking under a single platform would bring customers a seamless experience, and tourism in Pagsanjan will be promoted, which will make the services both more convenient and satisfying and more efficient, in general, to the users and service providers.

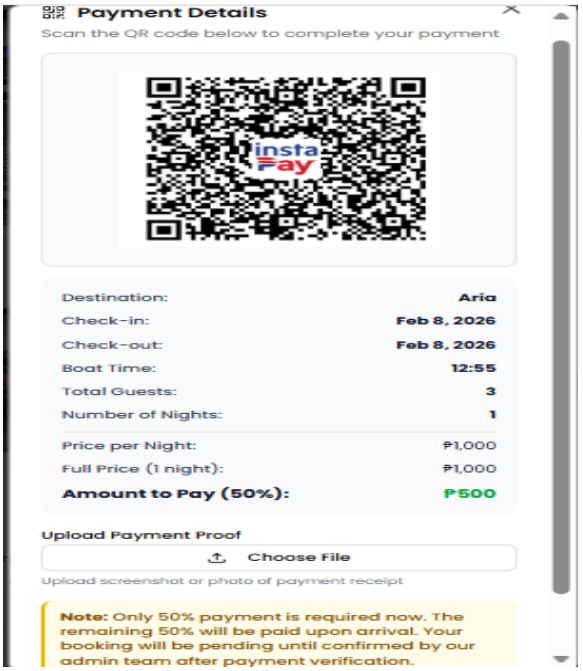


Figure 19. Payment Fees and Details

Figure 19 shows the page of payment options, which helps users conclude their reservations following an articulate and formal process. In this page, the system will provide a detailed overview of the booking, including the descriptive information about the destination, the dates of travelling, and the number of guests. A fifty percent (50%) payment of the total price must be made upfront to the system to obtain the slot, which can be done by scanning the provided QR code. After the transaction is completed, the users have to attach or upload a photo of their payment receipt to ensure that the administrator verifies and confirms the reservation.

The design of the system is such that convenience and security are maintained among the users. It offers immediate confirmation after verification of payment

and records all the transactions completed; therefore, it is easier to trace the booking by both the tourists and the administrators. Moreover, this page gives the user an opportunity to revise all the booking information and make payment in advance, which eliminates the possibility of mistakes and misinterpretations. In making the payment process a part of the package, the system simplifies the process of making reservations, fosters transparency, improves the user experience in general, and facilitates the efficient management of the accommodations and activities by the administrators. Reduces errors associated with manual transactions, ensures timely confirmation of bookings, and provides a secure record of all financial activities for both users and administrators. The reservation processing is also expeditious through this page since all the necessary payment procedures are carried out prior to final confirmation.

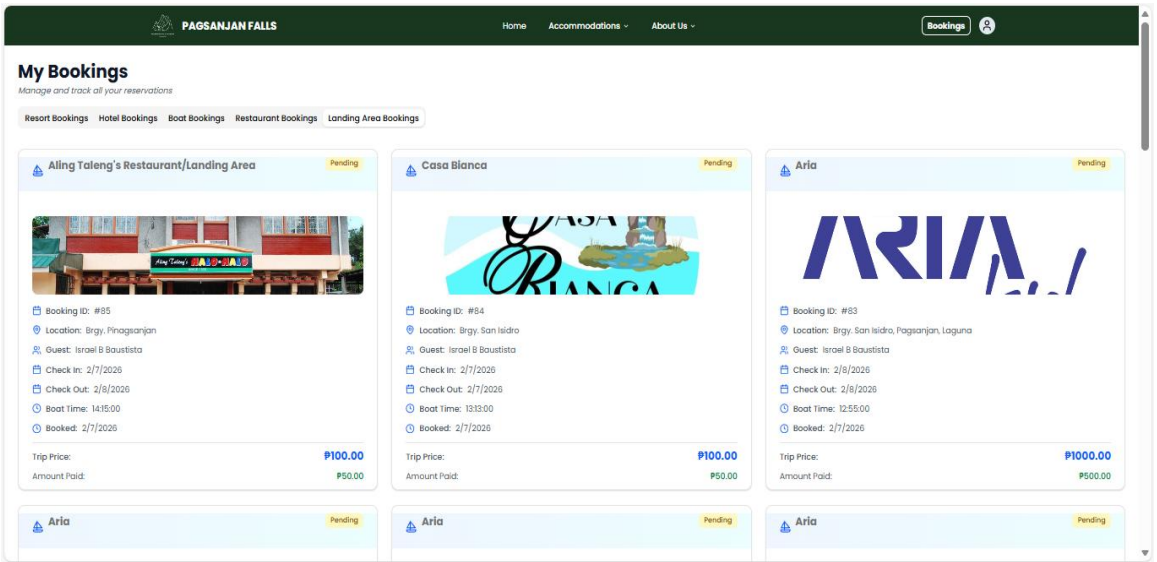


Figure 20. My Bookings Page

Figure 20 illustrates that My Bookings Page serves as an official interface, via which customers can check and control all of their bookings in the Pagsanjan booking system. The page is divided into specific sections such as

resort, hotel, boat, restaurant, and landing area reservations, in order to ensure that the guests can easily retrieve the information when needed. Moreover, users can see the status of their requests, which is usually referred to as pending until being verified by the administrative team regarding payment and finalizing the slot enhances transparency by showing detailed information about each booking, including dates, guest numbers, and selected services, which helps users stay informed and organized. On this page the user can also easily keep a track of what is going on with their reservations in real time. It eliminates the confusion that may arise during the booking process by keeping all the booking information in one place to be reviewed quickly.

Admin Panel

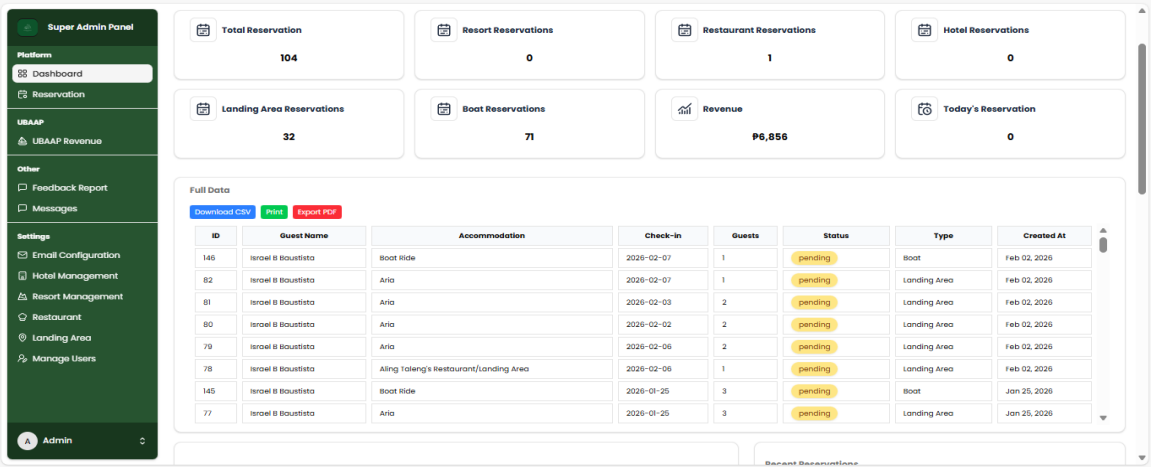


Figure 21. Admin Dashboard

Figure 21 Present the Super Administrator dashboard of the system is being presented, where the administrator would have the ability to view all reservations and activities through one interface. It shows the total landing area, boat, restaurant, and hotel reservations as well as the total revenue and daily bookings. There is also a detailed table where the administrator will be able to monitor and track the transactions by making use

of the guest details, the type of the booking and the schedule as well as the number of guests and the status among others. The side menu also allows user management, feedback, and other tourism establishments, hence aiding centralized and systematic management of the Pagsanjan Ecotourism Web-Based Platform. Centralized system enables administrators to manage operations systematically, identify trends, and respond quickly to issues or requests. It also provides a comprehensive overview of the platform's performance, helping in decision-making and planning to improve overall tourism services.

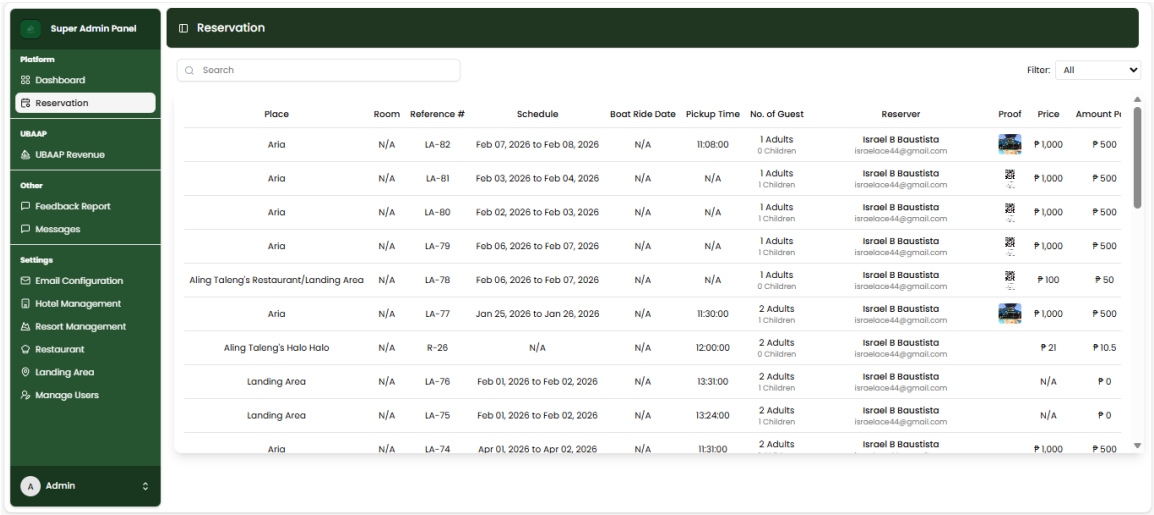


Figure 22. Reservation List

Figure 22 Reservation Management section of Super Administrator panel. It shows all the reservations taken in the system, a location reserved, reference number, schedule, date of the boat excursion, pickup time, number of guests and a person who made the reservation. The evidence of payment, the overall price and amount paid is also presented in the table and this will help the administrator control the transactions and track payments. By using this page, the administrator is able to review, monitor and organize reservations generated by landing sites, restaurants and other services offered by the

tourism industry effortlessly and hence enabling to coordinate the tourist activities appropriately in the Pagsanjan Ecotourism Web-Based Platform. The system will improve administration control, minimize errors, and create a smooth and well-structured running of all tourism activities in the Pagsanjan Ecotourism Web-Based Platform through centralization of reservation and payment information. This ensures an efficient and reliable reservation management process for the entire platform.

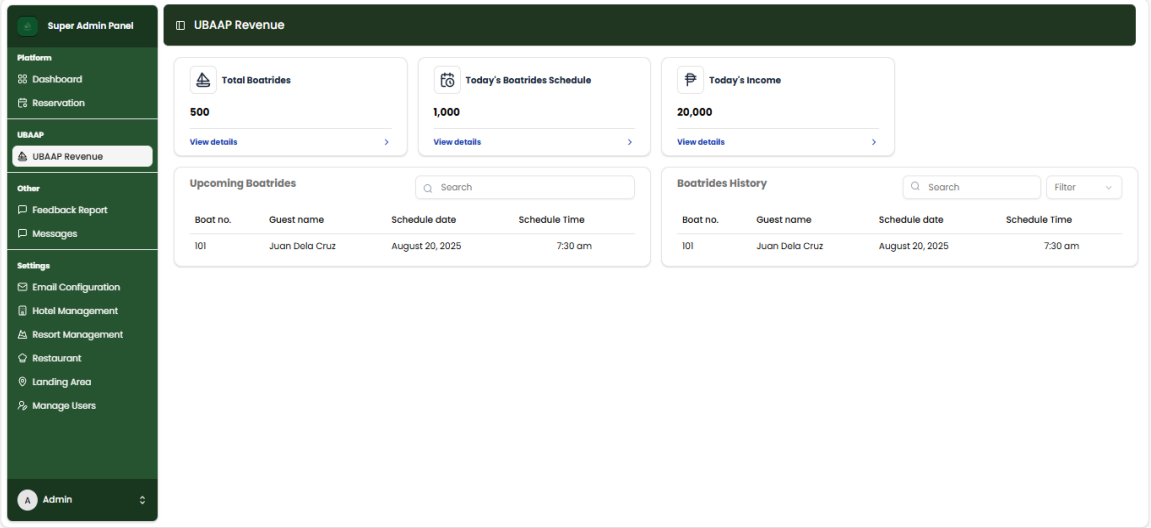


Figure 23. UBAAP Revenue

Figure 23 shows the UBAAP Revenue section of the page under Super Admin. It provides a summative report of the business operations of the boat ride, including the total number of boat rides, the booked boat rides of the day, and the total revenue earned. It also shows the upcoming schedule and the history of the rides already taken on the boat, providing the name and date of the guest, and the time. With the help of this page, the administrator can track daily operations and earnings of the boat tours, thus helping the municipality, keeping an accurate track of the activities, and controlling tourism services related to the Pagsanjan Ecotourism Web-Based Platform. Can manage and control the daily activities in a proper way, so that all the rides and all

transactions are captured. The system enables the administrator to monitor the payment, ensure reservations, and check the completion of the bookings, hence minimizing errors, avoiding duplication of reservations, and enhancing accountability. It also allows the administrator to determine trends, including the high booking periods, high demand periods, and seasonal changes in the activity of tourists, which can prove essential in planning the resources, staffing, and other operational changes.

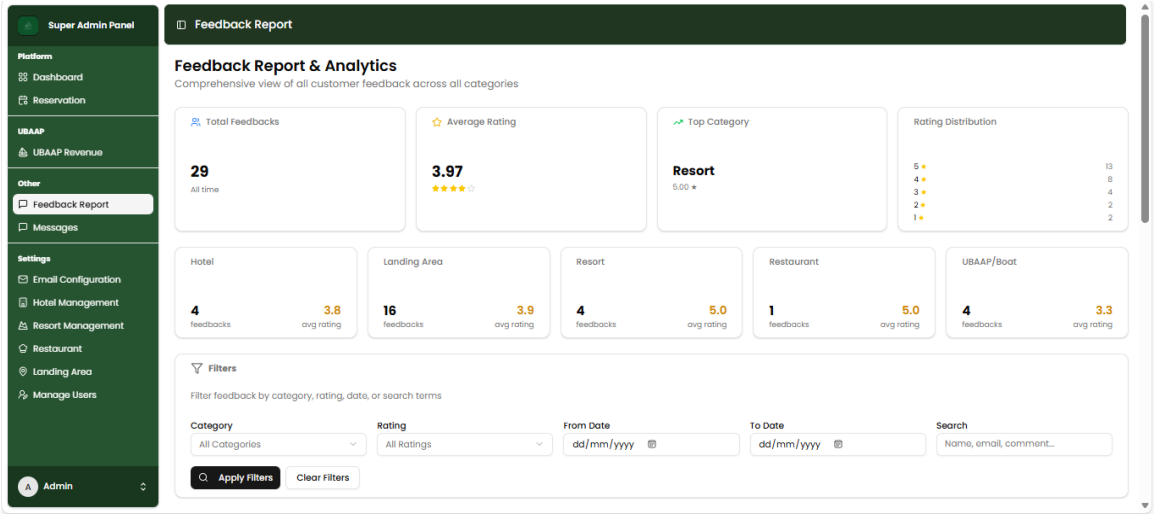


Figure 24. Feedback Report and Analytics

Figure 24 presents the Feedback Report & Analytics, which is used to access and track all customer feedback in the system. It presents the number of feedback overall, the average rating, the best category, and the rating distribution. It also gives results of feedback on each category, like Hotel, Landing Area, Resort, Restaurant, and UBAAP/Boat. This page will help the administrators to evaluate customer satisfaction and go through comments easily. In addition, it also provides filter options to sort feedback by category, rating, date, or keyword search. This module will not only assist administrators in measuring the general level of customer concerns, but it will also be effective in allowing for the review of detailed comments. Additionally, it has the option

of advanced filters with which the feedback may be categorized according to rating, category, submission date, and keyword search. These can be used to make informed decisions to assist the management in discovering the strengths, concerns, and also enhance the quality of services provided to the tourists.

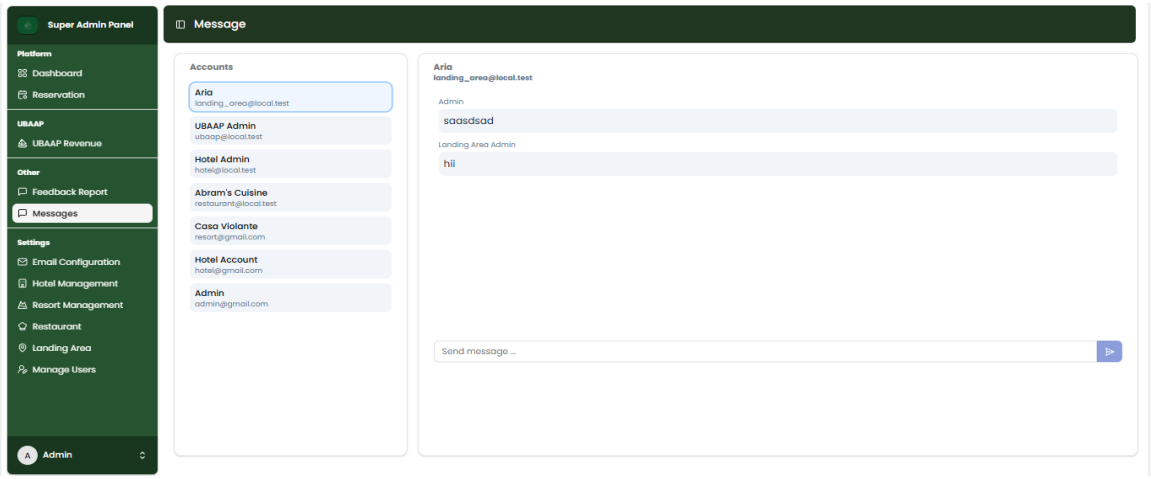


Figure 25. Admin Panel Messages

The Message page will enable the system for the Super Admin and other users of the system to communicate. It shows a list of accounts on the left-hand side and the relevant conversation on the right-hand side. The administrator can choose a user to check past messages and write a new message with the help of the message box, which is located below. This page can help facilitate communication within the system.

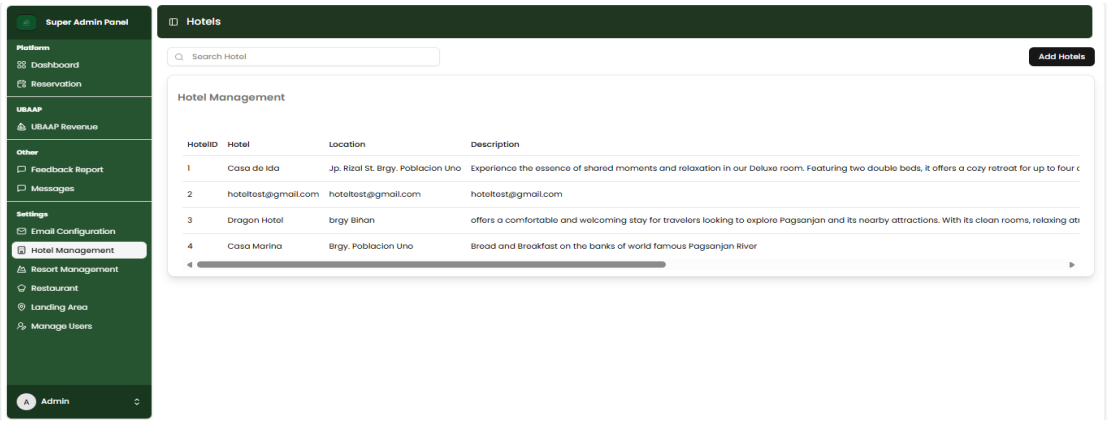


Figure 26. Hotel Management

Figure 26 show the all hotel records within the system are dealt with by using Hotel Management page. It gives a list of hotels with information about the hotel name, location, email and description. The administrator can use the button of adding a hotel to find the hotel and add new hotels. Through this page, the administrator will be able to plan and manage the hotel-based information effectively. System enables administrators to create new hotel entries while they can also update current hotel details and remove hotels that are not operating and maintain accurate records throughout their system. The centralized management system enables tourism operations to plan and make decisions about their activities which include coordinating hotel bookings with boat transportation and landing sites and resort operations.

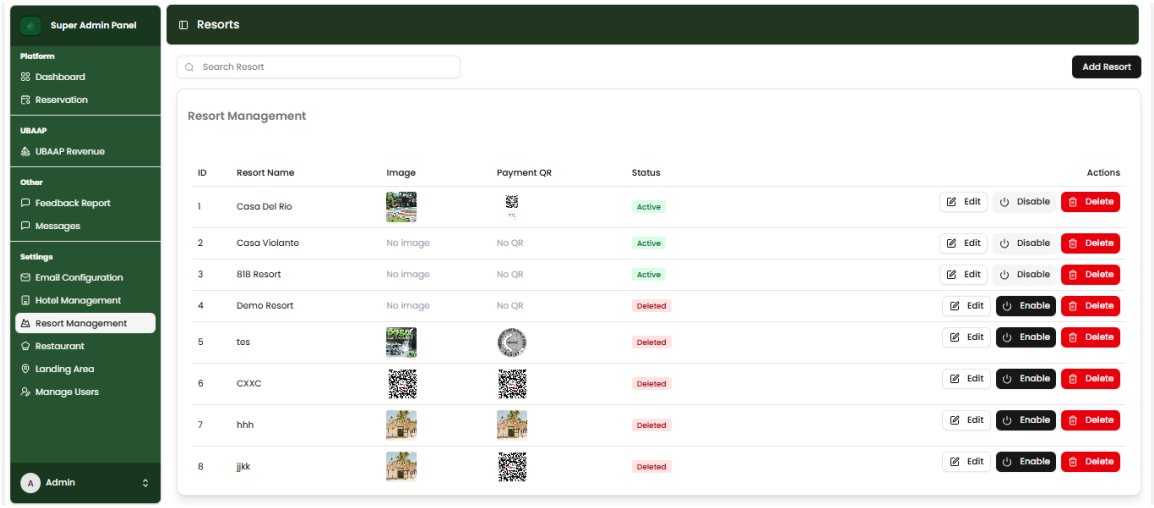


Figure 27. Resort Management

Figure 27 show the Resort Management page allows the administrator to efficiently manage all listed resorts in the system, including popular establishments like Casa Del Rio. The page displays a table summarizing each resort with its name, image, and an active status showing that the resort is available for booking. The administrator can edit information on the resort through controls on the right-hand side, temporarily deactivate a listing of a

resort, or delete a resort if, for any reason, the particular resort is no longer in operation. Furthermore, this page allows administrators to track resort activity and keep up-to-date records and promotional information on the resorts. Centralizing the management of resorts in the system enhances operational efficiency, reduces errors, and ultimately provides a good experience in the booking process for the administrators and tourists.

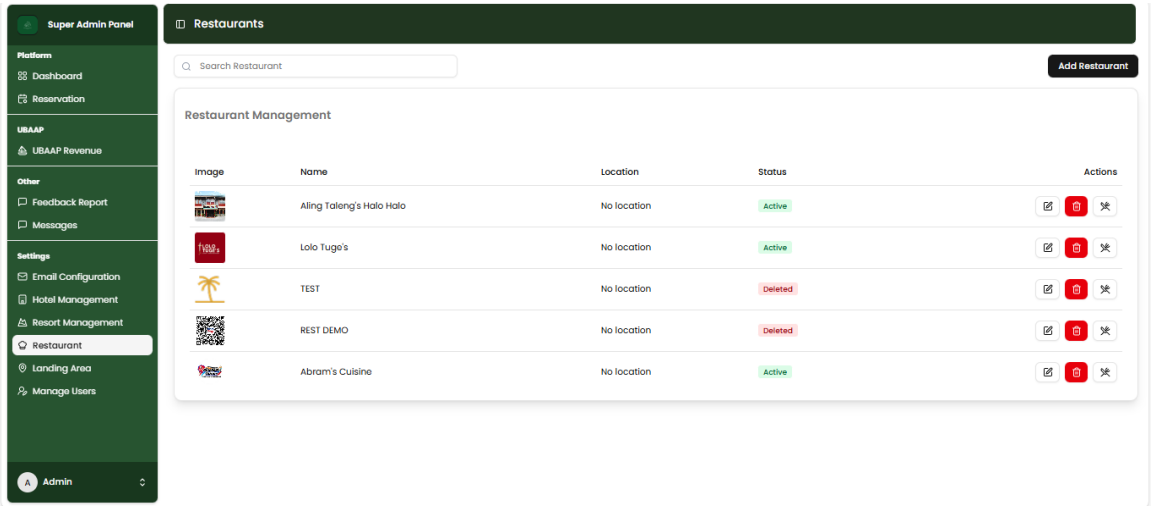


Figure 28. Restaurant Management

Figure 28 the Restaurant Management page offers the list of all dining establishments that are registered in the booking system. It features the highlights of the various venues, including the Aling Taleng Halo Halo, their location, and the status they are currently in. The administrators can easily update the list by clicking the edit and delete icons next to all the restaurant entries to keep the information up to date. To incorporate new restaurants, change the existing details, and delete the ones not in operation, thereby offering authentic operational restaurant options for tourists using the system. By centralizing the information for the restaurants, the system helps in efficient management, making the booking process smoother with reservations, thereby offering a delightful culinary experience for tourists.

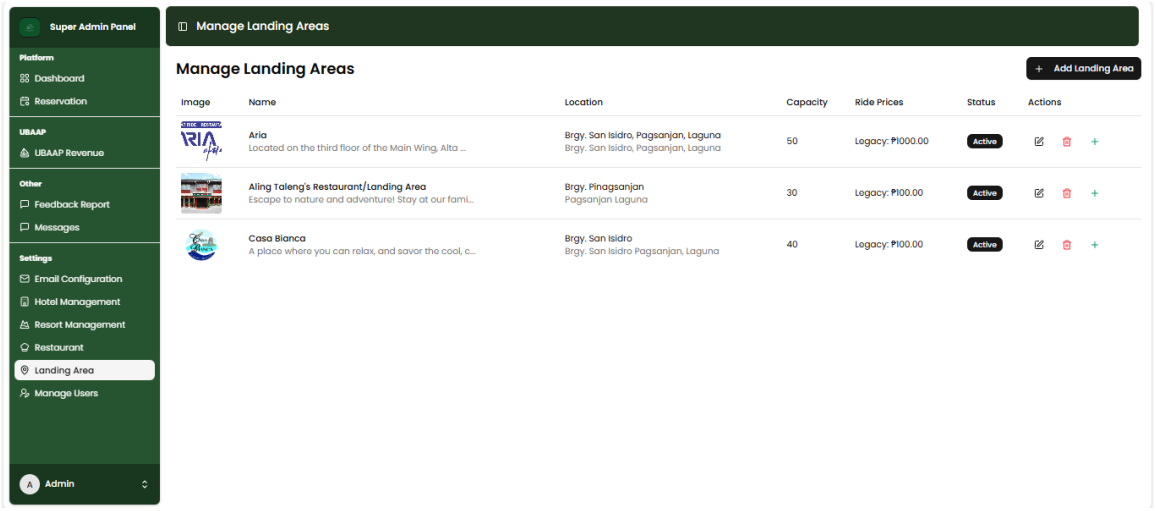


Figure 29. Landing Area Management

Figure 29 illustrate this part of Landing Area Management is supposed to control the individual zones where boat tours and customers are dealt with including Aria and Casa Bianca. These areas (along with the necessary information, such as guest capacity, and typical fare charged on the rides) are listed in the table. Administrators can use the button located on the top of the page to add and update new areas to ensure the right pricing and the availability of the travelers. This page enables zone data to be modified, capacity limits to be set, and fare rate adjustments in line with operational needs.

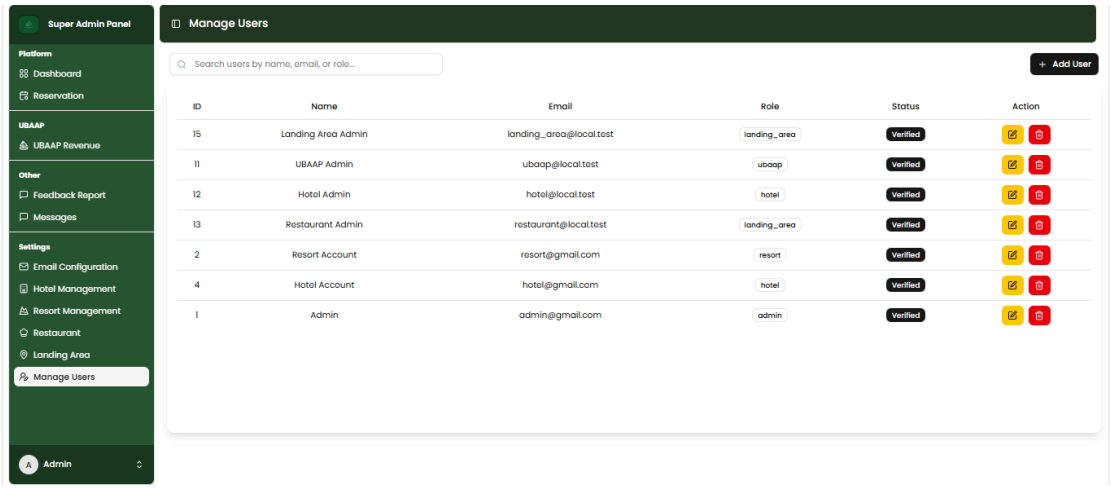


Figure 30. User Management

Figure 30 Show the last function of the Admin Panel, this is a list of all personnel who have administrative access to the system and their names, email addresses, and individual roles are listed here. It helps the chief administrator to trace the people in charge of different areas, including hotel or restaurant accounts, and so, to offer sufficient security. The table also validates the fact that the account of a user is verified and provides a possibility to change his or her permissions or withdraw the access in case of need. This centralized control of user accounts ensures that there is no unauthorized use of the system, protection of critical information, and efficient management of system operations, depending on authorized personnel for these activities. As a result, this structured control of access ensures increased reliability, safety, and integrity of data maintaining organizational control, preventing unwanted access, and making sure that only authorized personnel can access sensitive data and system functions are all made possible by this centralized user management system. Additionally, it enables administrators to track permission changes, keep an exhaustive audit trail of account activity, and produce reports that aid in internal review and security policy compliance.

1. A Unit Testing

The unit testing for the Web-Based Pagsanjan Falls System for Pagsanjan, Laguna was carefully conducted for each important functionality of the system’s modules. This process ensured that all features operate correctly and meet their intended purposes. The testing focused on verifying the reliability, usability, and overall system integrity, providing confidence that administrators, landing area managers, UBAP staff, and tourists can use the system effectively. By examining each module individually, potential errors or

inconsistencies were identified and addressed, ensuring smooth operation across all functions, including payments, reporting, listing management, and analytics and enhance user experience by providing timely, reliable, and secure services. This thorough validation also contributes to the system's long-term sustainability and establishes a solid foundation for future improvements or scaling to accommodate more users and services. It also ensures that each module performs its task accurately under normal system use.

Table 20. Unit Testing for Account Management Module

Tested Case	Expected Result	Actual Output	Response Time (s)	Remarks
Account Registration	User account record created with active status.	Account record saved successfully with status set to "Active".	4.16	PASSED
User Login Success	Valid credentials accepted.	Password verified, session data set.	3.60	PASSED
User Login Failure	Invalid password rejected.	Password mismatch detected.	3.51	PASSED
Role-Based Permissions	User can only access functions allowed for their role.	Role comparison succeeds, access granted.	2.50	PASSED
Session Management	User session is maintained and cleared upon logout.	Session created, stored, and destroyed correctly	2.33	PASSED

Table 20 presents the results of the unit testing conducted on the Account Management Module, demonstrating that all tested functions operated correctly and produced the expected outcomes. During the testing process, the system successfully handled account registration, ensuring that new user

records were accurately created, activated, and stored within the database. The login functionality performed as intended, granting access when valid credentials were entered while properly rejecting invalid passwords with appropriate validation messages. Role-based permissions were thoroughly verified.

Table 21. Unit Testing for Reservation and Scheduling Module

Tested Case	Expected Result	Actual Output	Response Time (s)	Remarks
View Available Landing Areas, Hotels, Resorts, and Restaurants.	System displays all available accommodations with complete details and updated availability.	User successfully browse the available landing area, hotels, resorts and restaurants'.	2.38	PASSED
Submit Reservation Request for Accommodation.	System accepts the reservation, records the request, and sends it to the selected landing area for review.	"Reservation submitted. Waiting for approval."	3.18	PASSED
Landing area approves reservation and confirms check-in.	System updates reservation status to "Approved" and "Check-in Confirmed."	"Your reservation is approved. Check-in confirmed." appears on the user dashboard.	2.0	PASSED
Schedule boat tour after check-in confirmation.	User can select date and time; system saves the schedule.	"Boat tour schedule successfully submitted."	2.30	PASSED
Session Management.	Session values persisted correctly.	Session data stored and retrieved.	2.33	PASSED

Unit testing is a critical tool in the development of any system because it ensures that each module of every function is working properly before it is incorporated into the overall system. It assists in detecting mistakes at initial stages and ensures that the system is functioning as per the expected

functionality. The results of the tests done are provided in table 21 of the unit testing section of the Reservation and Scheduling Module, which confirms that all the key functions of the process of booking and scheduling were in working order and provided the desired results. This proves the fact that the module is already prepared to be used in real-life applications in terms of tourist reservations and boat scheduling. Testing showed that the system is capable of making new reservations making sure it captures all the relevant booking information and makes sure that it is correctly recorded and stored in the database. The users can see schedules and look at the available time slots, and thus provide an accurate and up-to-date view of the availability of the system.

The reservation changes can also be made within the module and the modification recorded in the database instantaneously thus making the reservation accurate and consistent. Furthermore, the cancellation feature functioned as intended, properly removing reservation records and updating the system to reflect the changes, which prevented scheduling conflicts or errors. Performance tests demonstrated that the module processed requests efficiently, with response times ranging from 2.0 to 3.18 seconds, indicating that the system could handle user operations smoothly without delays. Beyond functionality and efficiency, also enhanced user convenience by providing a seamless, user-friendly experience for managing reservations. The results of all the case tests were positive and thus the stability and reliability of the module were successful at the time of working. There was also the capability to handle several reservation actions without experience of any error or loss of data in the system. This means that the module makes it easier to make the right bookings

without any overlaps, as indicated by the results. It also ensures that the users are provided with updated information after every action. Overall, the module meets the necessary requirement of the reservation and scheduling functionality.

Table 22. Unit Testing for Payment and Fees Management Module

Tested Case	Expected Result	Actual Output	Response Time (s)	Remarks
Display of payment breakdown	System should show complete fees: landing area, hotel, resort, restaurant fees, boat tour fee, environmental fee, and subscription fee.	All fees appeared with correct breakdown and accurate total amount.	3.12	PASSED
Processing of 50% down payment	System should compute 50% down payment and accept payment during reservation.	Down payment computed correctly; payment confirmed and saved.	3.28	PASSED
Uploading of proof of gcash/bank transfer	System should allow upload of proof of payment and store it in database.	Upload successful; file stored and linked to reservation record.	4.35	PASSED
Verification of payment by staff	Staff should verify and mark payment as “Paid” or “Pending.”	Verification status updated correctly and reflected on user account.	4.20	PASSED
Automatic charging of ₱100 subscription fee	System should automatically add ₱100 subscription fee upon reservation confirmation.	₱100 fee auto-added and included in final payment summary.	2.33	PASSED

Table 22 displays the results of the unit testing (which includes texted cases, expected results, actual output, response time and remarks) that was performed on the Payment and Fees Management Module to ensure that the system works correctly and all payment-related operations are carried out. The module successfully provides a detailed breakdown of fees according to landing area, hotel, resort, or restaurant fees, boat tour fees, environmental charges, and the fixed ₱100 subscription fee. It accurately processed the mandatory 50% down payment required when making a reservation, while uploaded records of GCash or bank transfers were saved and retrieved without any errors. Payment verification by staff functioned efficiently, with the system updating instantly to reflect completed transactions, demonstrating reliability and accuracy in real-time operations. Response times for the module ranged between 2.33 and 4.35 seconds, indicating good system performance and consistency under various test scenarios. The passing results for all tested cases confirm that the Payment and Fees Management Module is dependable and capable of handling complex financial operations without disruption.

The module also reduces the likelihood of human error by automating calculations, verifying payments, and maintaining a comprehensive record of all transactions. This functionality not only increases efficiency for the staff but also enhances the user experience by providing clear and immediate feedback on payment statuses. Its consistent response times, combined with accurate data handling, indicate that the system can support long-term operations while maintaining efficiency and reliability. The results of the unit testing highlight that this module is a critical component in ensuring smooth financial management and effective operational control within the system.

Table 23. Unit Testing for Data Visualization Module

Tested Case	Expected Result	Actual Output	Response Time (s)	Remarks
Generate tourist arrival trends	System should display a clear graph showing number of tourists per selected date range	Graph displayed correctly with accurate tourist counts	1.12	PASSED
Generate revenue and fees collection report	System should show total income from resorts, hotels, restaurants, landing areas, boat tours, environmental fees, and subscription fees	Full report generated with correct computation of all income sources	1.27	PASSED
Generate tourist demographics report	System should present charts showing gender, age group, and travel type of tourists	Charts appeared correctly with complete demographic breakdown	1.18	PASSED
Generate feedback analysis report	System should summarize common feedback themes and display ratings or comments	Summary report generated with organized feedback insights	1.22	PASSED
Visualization reports	System should allow admin to see visualization results per weeks, months, and year.	Reports will be summarized and visualized in the system	1.31	PASSED

Table 23 show the results of unit testing the Reporting and Analytics Module of the system are provided in Table 30 and indicate that all the major data-visualization and administration issues are properly working and efficient. The trend of tourist's arrival was generated satisfactorily, and there was a clear representation of the number of tourists per date interval with a quick response

of 1.12 seconds. The revenue and fee-collection report was also prepared correctly and it showed the proper calculation of revenue on all sources, namely, resorts and hotels, boats tours, environmental fees, and subscriptions in one and point two seven seconds. Moreover, the system was able to generate reports of tourist-demographics using charts, which depict gender, age group, and type of travel within 1.18 seconds. Lastly, the necessary task to export visualization reports to file formats like PDF or Excel was executed with success and no data loss and took 1.31 seconds.

Table 24. Unit Testing for Landing Areas, Hotel, Restaurant, and Resort Module

Tested Case	Expected Result	Actual Output	Response Time (s)	Remarks
Create landing area, hotel, restaurant, or resort	New listing/item (e.g., room, menu item, ride price) created and saved to the database.	Listing inserted with details, price, and category successfully.	4.49	PASSED
Update listing details (landing area, hotel, restaurant, or resort)	Details of the listing (excluding price) are modified and updated.	Name, description, and category/type updated correctly.	2.25	PASSED
Update listing price (landing area, hotel, restaurant, or resort)	The price field for the specific listing/item is modified.	Price field updated successfully.	3.13	PASSED
Set listing status to hidden (landing area, hotel, restaurant, or resort)	Listing status is set to hidden/inactive, preventing it from appearing in public view.	Status changed to ‘Hidden’ (Inactive).	2.13	PASSED
Get listings by module type	Listings are filtered and retrieved based on their parent module.	Array returned with category/module filter.	2.24	PASSED

Unit testing is also necessary in ensuring that all the modules of the system work properly and hence contribute to the overall functionality of the platform. This is done to confirm that all the significant characteristics are working as anticipated before the entire system is rolled out. Table 24 shows the results of the unit testing of the Landing Area, Hotel, Restaurant and Hotel Resort modules, and the focus on the essential management functions of the given components of the service. The findings reveal that such administrative control over these listings is reliable and efficient. The landing, hotel, restaurant, and resort listing creation function was effectively implemented, and the system was capable of saving the information, like descriptions, prices, and types thereof, within 4.49 seconds, which means that the system had a consistent and reasonably good performance.

Changes to listing information, such as name and description (Price excluded) were successfully communicated, whereas individual price changes were successfully communicated and response times were less than 2.25s and 2.13s respectively. There was also a correct implementation of the functionality to Set listing status to hidden (landing area, hotel, restaurant, or resort) within 2.13 seconds in the module, which provides the ability to temporarily hide items off public view. In addition, filtering of the listings was done efficiently in terms of their type of module (e.g. all Hotel rooms or all Restaurant items), and the results as per expectation took 2.24 seconds. The five cases that were tested were passed and this will prove the reliability of this module in controlling service information and contributing to the effective listing and pricing control that is applicable in all the four sectors of operations. This also serves to aid in making a better decision regarding price, promotions and allocation of

resources and enhances the role of the system in contributing towards the overall operational efficiencies and quality of service provided.

Table 25. Unit Testing for Reports Module

Tested Case	Expected Result	Actual Output	Response Time (s)	Remarks
Generate tourist arrival trends	System displays a clear graph showing the number of visitors over a selected date range.	Graph displayed correctly with accurate counts and trend lines	5.12	PASSED
Generate revenue and fees collection report	System accurately calculates and displays total income from all sources (tours, fees, resort/hotel share).	Full report generated with correct computation of all income sources.	4.27	PASSED
Generate tourist demographics report	System presents charts showing visitor breakdown by gender, addresses group, and travel type.	Charts appeared correctly with complete demographic breakdown.	3.18	PASSED
Generate feedback analysis report	System summarizes common feedback themes and displays corresponding average ratings or comments.	Summary report generated with organized feedback insights and average ratings.	4.22	PASSED

Table 25 presents the unit testing results for the Reports and Analytics Module, where its ability to generate accurate and timely summaries of

ecotourism data was validated. The module successfully produced the tourist arrival trends, clearly displaying visitor counts over time in 5.12 seconds. The essential function to Generate revenue and fees collection report accurately computed and displayed all income from service sources, completing the task in 4.27 seconds. Furthermore, the module effectively generated the tourist demographics report, presenting a clear breakdown of visitors by categories such as addresses and gender in 3.18 seconds. Finally, the system successfully created the feedback analysis report, which organized visitor comments and displayed average service ratings, with a response time of 4.22 seconds. All four remaining tested cases passed, which confirms the module's reliability in generating comprehensive and accurate reports to facilitate effective decision-making and operational monitoring by the Tourism Office.

1. B. Integration Testing

The integration testing of the Pagsanjan Ecotourism Web-Based Platform entailed a critical and methodical test of all the key modules of the system to guarantee a smooth running and interplay of the system. In particular, the testing was performed concerning the Account Management Module, Reservation and Scheduling Module, Payment and Fees Management Module, and Data Visualization Module. The functionality, reliability, and the appropriateness of the operations in each of the modules was tested separately. Subsequently, the modules were tested together to check how they interoperated and the general workflow of the system. This intensive process of integration ensured that all modules are working well independently of each other as well as that they would work well together with other modules in the system, with regard to passing of data, proper processing and uniform human

experience throughout the platform and the said platform is capable of delivering consistent and reliable user experience.

Table 26. Integration Testing

Tested Case	Expected Result	Actual Output	Response Time (s)	Remarks
Account Management to Reservation Flow	Successful Tourist login grants access to the Reservation Module, linking the Tourist's ID to the booking.	Tourist user logs in successfully; reservation form pre-fills user data and logs the Tourist ID in the reservation record.	3.15	PASSED
Reservation to Payment Module Integration	System A valid reservation request (Reservation Module) generates an itemized bill in the Payment Module for the 50% down payment.	Itemized bill correctly generated in Payment Module; payment gateway initializes for 50% of the total cost.	4.38	PASSED
Payment to Reservation/Reporting Flow	Successful 50% payment updates the Reservation status and flows revenue data into the Reports Module.	Reservation status updated to 'Pending Verification'; payment amount and transaction details recorded in Revenue Reports.	5.62	PASSED
Reservation to Account/ Notification Flow	Landing Area Staff (Account Mgt) confirms a check-in, triggering a status update in Reservation and an alert to UBAP.	Staff login is validated; confirmation triggers status change and generates a successful notification/alert to the UBAP account.	4.45	PASSED
Reservation/Payment Data to Reports Module	All finalized booking and payment data flow correctly into the Data Visualization Module.	Reports Module displays accurate counts of Confirmed bookings, tourist demographics, and total collected fees/revenue shares.	4.77	PASSED

Table 26 shows the unit testing of the Special Integration Testing stage where the essential data transfers among the core modules were strictly tested. The Account Management to the Reservation Flow integration succeeded, which proved that the correct authentication of the tourist by the login credentials coupled with matching the identifier with the new booking record took place after 3.15 seconds. This record is then easily passed off to the Reservation to Payment Module Integration whereby a valid booking creates an itemized bill of correct payment to the mandatory 50 percent down payment within 4.38 seconds.

The Payment to Reservation/Reporting Flow was verified after payment and appropriately changed the booking status and registered the revenue information with the Admin in 5.62 seconds. Moreover, the Reservation to Account/Notification Flow was able to validate that the administrative activities (like the confirmation of Landing Area staff) will trigger the required notifications to the other stakeholders (UBAP) in 4.45 seconds. Lastly, the Reservation/Payment Data to Reports Module ensured that all completed booking information and transactions are sent to the Data Visualization Module and can be used to generate accurate revenue and demographic reports within 4.77 seconds. All the five cases that were tested passed, and it ascertained the full integration of the major system modules.

1. C. System Testing

In this phase of the research study system testing is the final and most comprehensive phase in testing the Pagsanjan Ecotourism Web-Based Platform as an integrated whole. This phase is carried out to test a system, its overall functionality, efficiency, and stability, in a real-life scenario or close to

actual use, to ensure that all critical modules and functions of a system, such as booking/reservation, scheduling, payment processing, user management, etc., and all functions are integrated and work seamlessly or function as a unified or integrated system in its entirety to meet specific requirements or standards.

The process involves evaluating the ability of the system to provide precise output, promote seamless transactional systems, and enable the provision of reliable services without compromising its usability, security, and integrity. In addition, the process involves evaluating the ability of the system to process data correctly, to quickly respond to queries, and to maintain its consistency in providing services even during peak usage, including simultaneous interactivities of users on the system. Particularly, the system's usability, responsiveness, tolerance to errors, and compliance with established operational workflows are noteworthy. This phase confirms that the system is fully ready for actual deployment and user operation.

Moreover, this step will ensure that all the units of the system are aligned to each other without conflicts and malfunctions and hence coordination of reservation, scheduling, payment, and user management will be easy. It also looks at the manner in which the system deals with real user interaction such as simultaneous accessibility by more than one user to provide stability and to maintain steady performance. Unidentified problems or system limitations can be found and solved before any actual deployment, which would reduce the risk of failure in the operational use. This also helps in ensuring that the site offers a viable, effective and easy to use experience to the tourists, administrators as well as other stakeholders in the Pagsanjan ecotourism system.

Table 27. Browser Compatibility Testing Results

Tested Case	Chrome Response Time	Firefox Response Time (s)	Microsoft Edge Response Time
1. User Login and Role-Based Authentication	5.15 s	4.20	2.18 s
2. Admin Dashboard and Reporting Access	4.25 s	3.35	3.28 s
3. Tourist Reservation Booking (Form Load)	5.22 s	5.30	2.25 s
4. Landing Area Check-in Confirmation	3.95 s	4.05	3.98 s
5. 50% Down Payment Processing (GCash QR)	3.48 s	3.60	4.55 s
6. Generate Revenue & Fees Report	4.85 s	3.10	4.95 s
7. System Content Management (Admin)	2.10 s	2.25	2.15 s

Table 27 shows the response time of the major functions of the Pagsanjan Ecotourism Web-Based Platform when contacted through Google Chrome, Mozilla Firefox, and Microsoft Edge. The findings show that the system is reliable and works with all browsers and has insignificant differences in loading speed. User login and Role-based Authentication had a response time of 2.18 and 5.15 seconds respectively, and the Tourist Reservation Booking and Landing Area Check in Confirmation took acceptable time indicating effective system processing when carrying out core user transactions. Process-intensive functions, including the 50% Down Payment Processing, through GCash QR, and the creation of Revenue and Fees

Reports, had a bit more response times, with the highest of 4.95 seconds; the numbers are not extremely low considering the needs of payment validation and the data before turning the payment into a report. Among the browsers, the Microsoft Edge tended to have a better response time on some of the functions, whereas Firefox and Chrome remained consistent and practical. All in all, the data prove that the platform is compatible, stable and responsive on major web browsers hence facilitate easy performance by administrators, staff and tourists.

2. Portal Presentation

The Portal Presentation recognizes the key web portals that comprise the Pagsanjan Ecotourism Web-Based Platform and they are the Landing Area, Hotel, Restaurant and Resort portals. All portals are aimed at completing certain tasks in the tourism process, and partners can effectively organize reservations, organize the work with tourist check-ins, and control the availability in a systematic and systematic way. Another point raised in the presentation is that of integration and centralized format of the platform where all the involved establishments and stakeholders could work with a single system. With the synchronization of data in real time, it is possible to update booking information, availability of the services and customer records at any time, reducing the possibility of conflicts and errors in scheduling which can help the municipality assess the tourism activities and make sound decisions to aid the successful and sustainable management of ecotourism. It also explains how the platform supports better communication and data exchange between the stakeholders so that all updates are similar through the system, and help maintain correct records to monitor, report, and make decisions.

2. A Landing Area

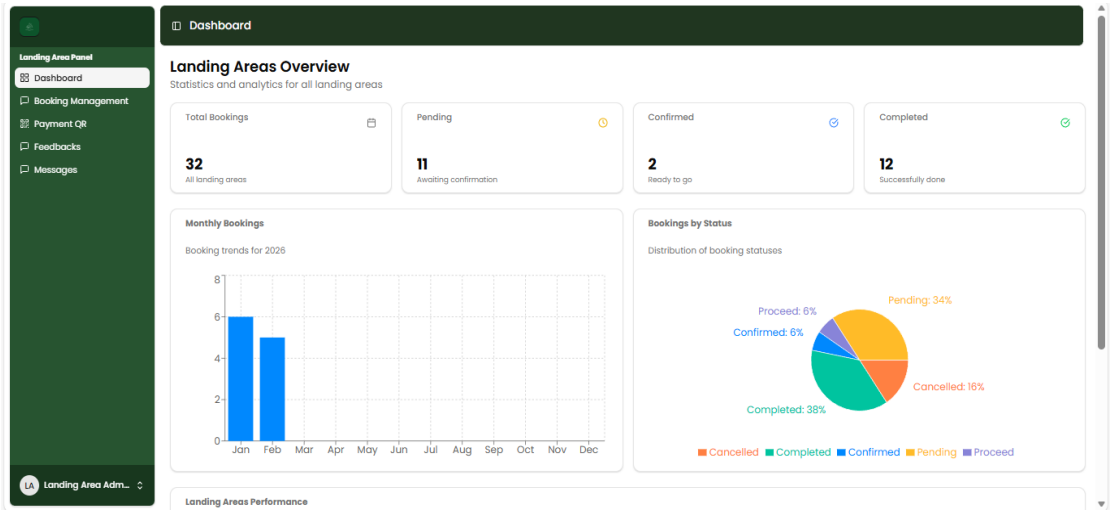


Figure 31. Landing Area Portal

The Landing Area is the primary management platform that helps in overseeing and implementing the tourist booking to the registered landing areas in the Pagsanjan Ecotourism Web-Based Platform. The dashboard will give a quick status of overall booking, with pending, booked, and completed ones so that the landing area personnel can see the status of the real-time arrivals and boat schedules of the tourists. The booking status distribution and the monthly booking chart provide the user with a better overview of the trends and workload in the long run, it becomes easier to arrange operations and collaborate with UBAP and boat operators. Moreover, efficiency and control are emphasized with all booking transactions and operational data being centralized to one interface that is user-friendly. This will streamline faster decision making processes, lessen manual record-keeping and minimize scheduling conflicts. With the access of the correct and appropriate information in time, the system boosts service preparedness, better distribution of resources, and efficient administration of the flow of the tourists, which eventually boosts the overall

dependability and efficiency of the landing area operations. Its real-time monitoring, data visualization, and stakeholder coordination capabilities not only streamline administrative tasks but also enhance service delivery, operational reliability, and visitor satisfaction.

2. B. Hotel

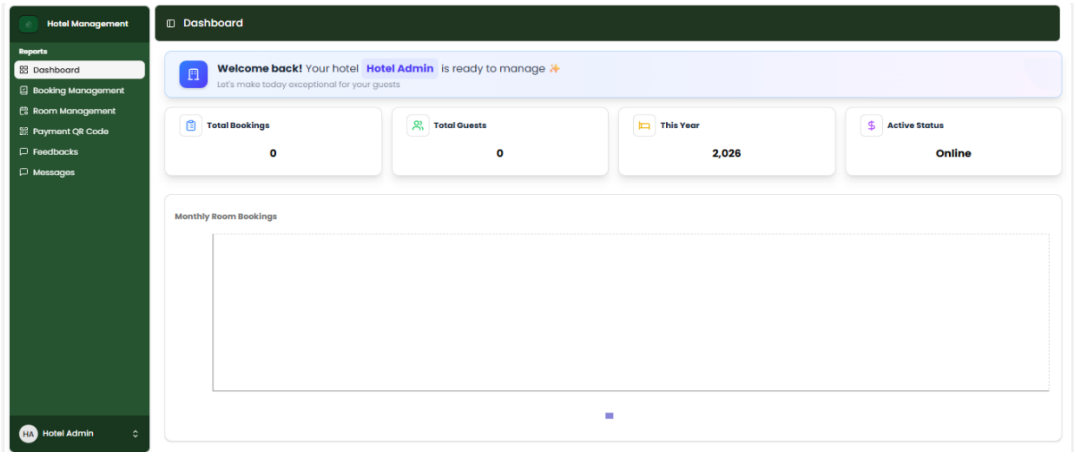


Figure 32. Hotel Portal

The Hotel Portal will help the hotel administrators to handle accommodation reservations in the Pagsanjan Falls Ecotourism Web-Based Platform. The dashboard also gives its users a detailed overview of the details of the booking, the total number of people staying at the hotel, annual events of the hotel, and the general health of the system, allowing hotel employees to effectively manage the daily processes and streamline the business operations. Moreover, the portal includes room management device to track the availability, QR code payment system to complete transactions quicker, feedback assessor to track customer satisfaction and in-built messaging device to provide communication with customers in a seamless manner a higher quality in the service to guests and a well-organized and smooth process of accommodating the guests through centralization of the reservation services, financial

transactions and customer service and interaction. Its timely updates and monitoring features enable the hotel administrators to respond in time to booking modifications, optimization of occupancy rates as well as high service quality standards, which enhances the overall performance of the tourism system.

2. C. Restaurant

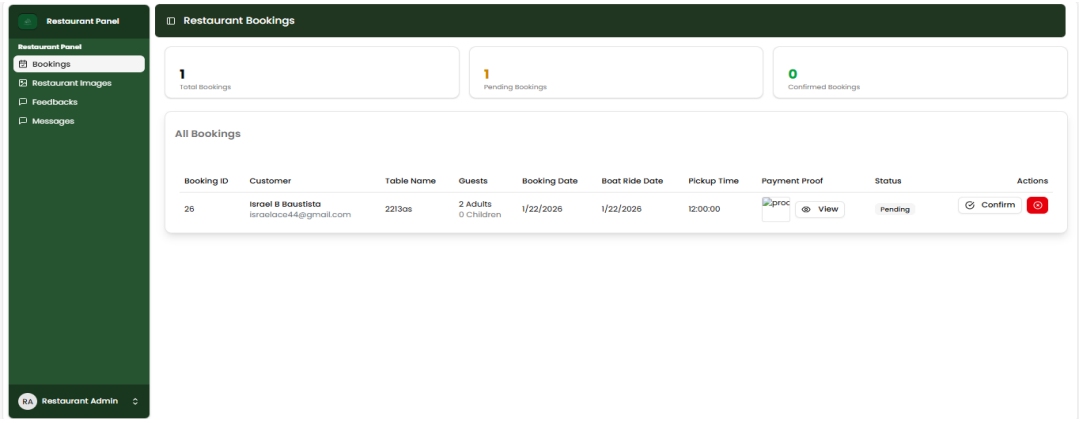


Figure 33. Restaurant Portal

The Restaurant Portal is the centralized information system that is directed towards licensed restaurants in Pagsanjan Falls to help them handle the process of making a table reservation to the tourists in a structured and easy way. With this portal, the staff of restaurants is able to keep track and arrange the bookings instantly, differentiating between the outstanding requests, the requests confirmed, the transactions made, and the cancellations. The reservations also contain finer details of the customer contacts, the number of guests, the date of dining, the schedule of choice, the boat ride schedule and the level of payment. This makes all the dining activities to be aligned to the itinerary of the tourists to avoid conflict and duplication of booking. The portal also enhances efficiency in running the restaurant operations as they are able

to have accurate and updated information and the staffs can offer better services and a smooth and well organized dining experience to the visitors.

.2. D. Resort

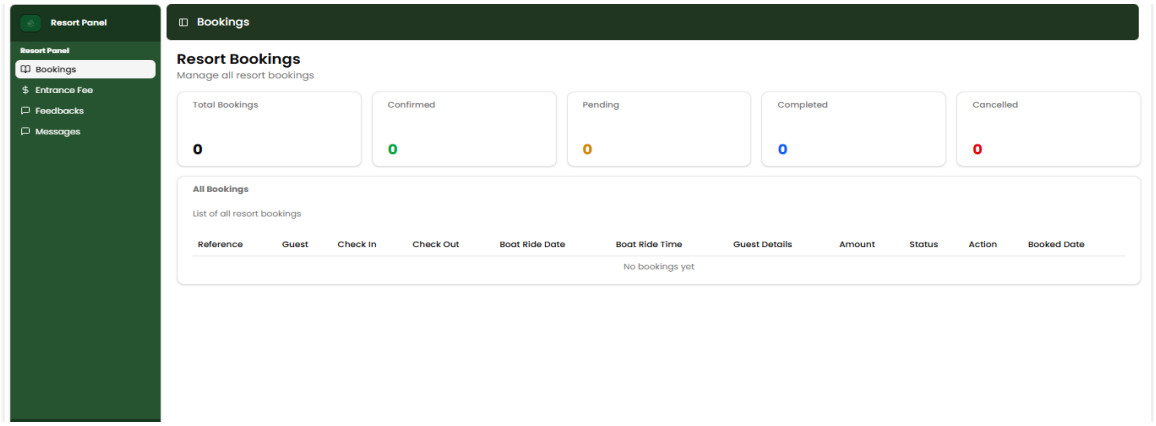


Figure 34. Resort Portal

Resort Portal is a simplified and streamlined process of handling tourist responses and entrance fee databases of the resorts. It provides an overview of total, confirmed, pending, completed and cancelled reservation enabling the employees in the resort to easily monitor the status of the individual reservation. Also, the portal gives detailed information concerning the booking such as the guest information, the check-out and check-in dates, the schedule of the boat rides, and the amount to be paid. The portal is more than a booking system because it has the ability to store all activities of the resort in one user friendly portal where the staff are able to make amendments and view current information in real time. The portal also helps to coordinate with other tourism stakeholders including landing areas, hotels and restaurants so that the itinerary of the guests, boat schedules and accommodation in the resorts are aligned and the visitor has a smooth journey. Its ability to provide real-time information, consolidate all resort activities, and coordinate with tourism partners enhances operational efficiency and visitor satisfaction.

The system not only streamlines the reservation process but also supports long-term planning and sustainable growth for the resort and the surrounding tourism industry. These features indicate that the portal is a valuable tool capable of improving the overall management of tourism services while ensuring a smooth and enjoyable experience for all visitors. It comprehensive functionalities, the portal plays a crucial role in ensuring a smooth, organized, and enjoyable experience for guests, while contributing to the long-term competitiveness and success of the tourism sector.

3. Evaluate the efficiency and effectiveness of the developed system using ISO 25010 and Technology Acceptance Model (TAM).

3. A. System Evaluation Results Based on ISO/IEC 25010 Software Quality Characteristics

Table 28. Results of ISO 25010 Functional Suitability Evaluation

SUBCATEGORY/ITEM	MEAN	SD	INTERPRETATION
Functional Completeness - The system provides all essential features for managing Pagsanjan falls ecotourism.	3.66	±0.57	Strongly Agree
Functional Appropriateness - The system's functions effectively support tourism operations in the municipality.	3.66	±0.57	Strongly Agree
Functional Correctness - The system performs its Pagsanjan Falls ecotourism tasks accurately.	3	±0.00	Agree
Composite Score	**3.44**	**±0.32**	**Strongly Agree**

Legend: *Strongly Agree (3.26 – 4.00), Agree (2.51 – 3.25), Disagree (1.76 – 2.50), and Strongly Disagree (1.0 – 1.75)*

Table 28 is the system evaluation outcomes of the Functional Suitability in terms of ISO/IEC 25010 software quality standard, which is suing the ability of a software system to offer the required functions to suit the needs of users.

The mean score of Functional Completeness was 3.66 out of 4, which shows that the system offers the majority of the core functions needed to handle the Pagsanjan Falls ecotourism activity. This finding indicates that the system can to a large extent support the fundamental functional requirements of tourism management allowing users to undertake the most important activities including tour scheduling, reservations, and supply the tourists with the required information effectively. Likewise, the mean score of Functional Appropriateness also received a score of 3.66 indicating that the system functions are well designed and appropriate to the target tourism activities. This means that the system will assist the users to carry out their activities in the correct and effective way to improve the overall management of the Pagsanjan Falls ecotourism activities and also make the operational processes effective and convenient.

Regarding Functional Correctness, the system scored 3.00 on average, which means that the system does its central job correctly, although there can be a slight improvement in the area concerning the total correctness in all the functions. This implies that majority of the jobs are carried out with little or no errors, but some functions could be streamlined to achieve the highest level of accuracy and depend ability. The total composite score of 3.44, which is on Strongly Agree, indicates that the system is generally functional and can perform reliably to support the ecotourism operations of the Pagsanjan Falls. The standard deviations of between ±0.00 and ±0.57 show there is a moderate to high consistency among the respondents, which shows that there is a high level of agreement when it comes to the functional performance of the system by various groups of users.

Table 29. Results of ISO 25010 Performance Efficiency Evaluation

SUBCATEGORY/ITEM	MEAN	SD	INTERPRETATION
Time Behavior - The system loads and responds quickly for Pagsanjan Falls tourism operations.	3	±0.00	Agree
Resource Utilization-The system uses computing resources efficiently during operation.	3	±0.00	Agree
Capacity - The system remains stable even with many users accessing it.	3.33	±0.57	Strongly Agree
Composite Score	**3.11**	**±0.32**	**Agree**

Legend: *Strongly Agree (3.26 – 4.00), Agree (2.51 – 3.25), Disagree (1.76 – 2.50), and Strongly Disagree (1.0 – 1.75)*

Table 29 is the system evaluation result of the Performance Efficiency per the ISO/IEC 25010 software quality standard, which evaluates the system performance under different circumstances. This indicates that the system can support daily tourism operations with acceptable speed and stability. The Time Behavior criterion scored at 3.00 +0.00 meaning that the system was well responsive enough during the Pagsanjan Falls tourism activities but there is still an opportunity to enhance the load and response time to an even greater degree to allow a smoother operation experience to the user. In the case of Resource Utilization, the system also attained a score of 3.00 ±0.00, which means that it is able to utilize computing resources in an excellent manner, such that the operations do not cripple the hardware and software of the system. This efficiency is necessary to ensure that there is a steady performance through routine tourism management activities.

Capacity on the contrary, had a greater mean of 3.33 0.57 which indicates that the system is capable of supporting a number of users at a time but does not overload or crash and this is paramount during times of high tourist activity. This shows that the system is reliable during high user demand and can maintain stable performance even when many transactions are happening

at once. The total composite score (3.11) that has been rated as Agree indicates that the system carries out its functions effectively in the conditions of normal operation. The standard deviations are relatively low and this shows consistency in the evaluation of the respondents as the respondents agreed that the system was able to remain consistent in its performance. Overall, the findings indicate that the system offers acceptable performance efficiency in running the Pagsanjan Falls ecotourism to balance the use of resources, stability, and responsiveness time. Nonetheless, the user experience and the overall system performance can be improved by specific measures to make it more responsive and optimized, making sure that it would not collapse at the time of the peak load.

Table 30. Results of ISO 25010 System Compatibility Evaluation

SUBCATEGORY/ITEM	MEAN	SD	INTERPRETATION
Co-existence - The system operates reliably on local host across various devices during testing.	3	±0.00	Agree
Interoperability - The system integrates smoothly with other software and hardware used in tourism operations	3.66	±0.00	Agree
Composite Score	**3**	**±0.00**	**Agree**

Legend: *Strongly Agree (3.26 – 4.00), Agree (2.51 – 3.25), Disagree (1.76 – 2.50), and Strongly Disagree (1.0 – 1.75)*

Table 30 show the outcome of the system evaluation with the metrics of compatibility that is in terms of the co-existence and interoperability. Co-existence criterion had a mean score of 3.00 and a standard deviation of 0.00, which is understood to mean Agree. This means that the system is tested to be reliable on a local host on several devices.

It proves that the system can operate on the rest of applications without conflicting with and creating performance problems, the users can access and utilize the system without any problem on any machine. Mean score of

Interoperability criterion is 3.66 with the standard deviation of 0.00 and was also interpreted as Agree. This implies that the system is compatible with other software and hardware utilized in the tourism activities.

Table 31. Results of ISO 25010 Interaction Capability Evaluation

SUBCATEGORY/ITEM	MEAN	SD	INTERPRETATION
Appropriateness Recognizability - The system's functions are easy to understand for all users.	3	±0.00	Agree
Learnability - It is easy to learn how to use the system, even for first- time users.	3.66	±0.57	Strongly Agree
User Error Protection - The system reduces and corrects user mistakes efficiently	3	±0.00	Agree
User Engagement - The system keeps users interested and engaged.	3.66	±0.57	Strongly Agree
Inclusivity - The system is usable by people with varying abilities.	3.33	±0.57	Strongly Agree
User Assistance - The system provides clear guidance when needed.	3.33	±0.57	Strongly Agree
Self-descriptiveness -The system shows clear instructions and labels for easy use.	3	±0.01	Agree
Composite Score	**3.28**	**±0.29**	**Strongly Agree**

Legend: *Strongly Agree (3.26 – 4.00), Agree (2.51 – 3.25), Disagree (1.76 – 2.50), and Strongly Disagree (1.0 – 1.75)*

Table 31 Illustrate the performance of the system in Summarize as per Results of Interaction Capability Evaluation according to ISO 25010 which the mean score of the Appropriateness Recognizability criterion was 3.00, and the standard deviation of the score was 0.00, which was understood to be Agree. This implies that the operations of the system are intuitive and the users can recognize and access the features of the system without any problem. The means of the Learnability criterion were found to be 3.66 with a standard deviation of 0.57, which is understood as Strongly Agree; thus, even first time users can find it easy to learn how to use the system efficiently. In the same

manner, the User Engagement criterion scored 3.66 (Strongly Agree) with a variance of 0.57, which suggests that the system is an engaging one and users are not bored when they carry out their tasks. Under User Error Protection, the system had a score of 3.00 of 0.00 (Agree), which shows that the system is quite effective in minimizing and correcting user errors to achieve a sound interaction experience.

The Self-descriptiveness criterion was as well rated at 3.00 ± 0.01 (Agree) implying that the system has clear guidelines and labels and therefore the users can easily navigate the system without any external help. Other criteria, such as Inclusivity and User Assistance, had scores based on the mean of 3.33 ± 0.57 (Strongly Agree). The result of the composite score of 3.28 with a standard deviation of ± 0.29 that is interpreted as Strongly Agree indicates that the respondents believe that the system is highly user-friendly, interactive, and can support efficient, intuitive, and engaging interactions.

Table 32. Results of ISO 25010 Reliability Evaluation

SUBCATEGORY/ITEM	MEAN	SD	INTERPRETATION
Faultlessness - The system performs all functions consistently without errors.	3	±0.01	Agree
Availability - The system is accessible whenever needed by users.	3	±0.01	Agree
Fault Tolerance - The system continues operating correctly despite minor errors.	3	±0.00	Agree
Recoverability - The system can quickly recover and restore data after failures.	3.33	±0.57	Strongly Agree
Composite Score	**3.08**	**±0.28**	**Agree**

Legend: *Strongly Agree (3.26 – 4.00), Agree (2.51 – 3.25), Disagree (1.76 – 2.50), and Strongly Disagree (1.0 – 1.75)*

Table 32 show the outcome results of ISO25010 which is the Reliability evaluation of the Pagsanjan Ecotourism Web-Based Platform, according to the ISO 25010 standards, reveals that the system works consistently and

dependably in all dimensions. This web-based platform is faultless, as it conducts its intended functions with no single error, hence making sure that operations will be smooth with all users. Besides, it shows strong availability, meaning this system shall be up when needed, which is an important concern for any online reservation and scheduling service that caters to tourists and operators.

These results, on the whole, give a composite reliability rating of 3.08, thus reflecting a general agreement that indeed the system is reliable, stable, and resilient. This is in confirmation that the platform can support tourism activities continuously, maintain information integrity, and give users the assurance of a dependable web-based service experience.

Table 33. Results of ISO 25010 Security Evaluation

SUBCATEGORY/ITEM	MEAN	SD	INTERPRETATION
Confidentiality - The system ensures only authorized users can access sensitive tourism data.	3.33	±0.57	Strongly Agree
Integrity - The system protects information from unauthorized changes or tampering.	3	±0.01	Agree
Non-Repudiation - The system records transactions to prevent users from denying their actions.	3	±0.00	Agree
Accountability - The system tracks user activities to identify responsibility for each action.	3.33	±0.57	Strongly Agree
Authenticity - The system verifies user identities before granting access.	3	±0.01	Agree
Resistance - The system resists unauthorized access and malicious attacks effectively.	3.33	±0.57	Strongly Agree
Composite Score	**3.16**	**±0.30**	**Agree**

Legend: *Strongly Agree (3.26 – 4.00), Agree (2.51 – 3.25), Disagree (1.76 – 2.50), and Strongly Disagree (1.0 – 1.75)*

Table 33 gives the summary of the system's security performance based on ISO 25010 evaluation. The Confidentiality criterion received 3. 33 0. 57 (Strongly Agree) which reveals that the system allows only authorized users to have access to sensitive tourism information such as bookings, payments, and personal data, thus, it is effectively protecting the privacy. Integrity criterion got a mean of 3. 00 0. 01 (Agree) that indicates that the system protects data from unauthorized changes or tampering thus, the stored information accuracy and reliability are kept. The Non Repudiation criterion recorded 3. 00 0. 00 (Agree) which means that all the transactions are recorded correctly so that no user can deny their actions, thus, system operations become more accountable and transparent. The system scored 3. 33 0. 57 (Strongly Agree) for Accountability, which means the system tracks user activity and therefore can assign responsibility for each action. The Authenticity criterion got a rating of 3. 00 0. 01 (Agree), which means that user identities are checked before access is given thereby, application misuse is prevented.

The Resistance criterion got 3. 33 0. 57 (Strongly Agree) which indicates that the system can withstand attacks and prevent unauthorized access thus, data and system functions are always protected. Overall, the combined security score 3. 16 0. 30 (Agree) reveals that users generally find the platform secure. Demonstrates a strong security framework that ensures data protection, user accountability, and safe operation, giving users confidence in the reliability and trustworthiness of the Pagsanjan Ecotourism Web-Based Platform. These findings indicate that the system instills relevant security controls at all major areas. In addition, it

ensures that the user data is safely managed in all transactions. As a result, the platform develops a user trust due to a safe and secured environment.

Table 34. Results of ISO 25010 Maintainability Evaluation

SUBCATEGORY/ITEM	MEAN	SD	INTERPRETATION
Modularity - The system is built with separate modules for easier management and updates.	3.33	±0.57	Strongly Agree
Reusability - The system components can be reused for similar tourism functions or applications	2.66	±0.57	Agree
Analyzability - The system allows easy detection of issues and understanding of its structure.	3.33	±0.57	Strongly Agree
Modifiability - The system can be updated or modified without affecting overall functionality.	3.33	±0.57	Strongly Agree
Testability - The system can be easily tested to ensure all functions work after changes.	3.33	±0.57	Strongly Agree
Composite Score	**3.19**	**±0.00**	**Agree**

Legend: *Strongly Agree (3.26 – 4.00), Agree (2.51 – 3.25), Disagree (1.76 – 2.50), and Strongly Disagree (1.0 – 1.75)*

Table 34 indicates the results of the aspect of maintainability; that is, the system must be able to be conveniently handled, examined, altered, tested and improved without the system encountering any negative effects to its performance. The system achieved a composite mean score of 3.19 with a standard deviation of 0.00, in a verbal form, Agree, which implies that the respondents found the platform to be well structured, flexible, and maintainable in case of making improvements and updates in the future. In particular, the mean score of modularity was 3.33 (SD ±0.57), or Strongly Agree, meaning that the system is divided into distinct and autonomous modules. This architecture enables easy management, troubleshooting and improvement of individual parts without affecting the entire platform. In like manner, the analysability,

modifiability and testability each scored mean scores 3.33 (±0.57) which also reflects as Strongly Agree.

All these results mean that the issues can be easily discovered and interpreted, it is possible to upgrade or better current systems without impacting other functions and to thoroughly test the platform once the changes are realized in order to be sure that it will be reliable and accurate. This means that the platform can be maintained and improved over time without causing major disruptions to its operation. However, reusability was rated a little bit less (2.66 mean score, SD ±0.57) as Agree, indicating that although some of the components can be reused to other similar tourism-related purposes, there is a still a way to enhance the process of creating more general or more generic modules that can be used in other systems or contexts.

Table 35. Results of ISO 25010 Flexibility Evaluation

SUBCATEGORY/ITEM	MEAN	SD	INTERPRETATION
Adaptability - The system adjusts easily to different devices, environments, and user needs.	3.33	±0.57	Agree
Scalability - The system handles more users or bookings without affecting performance.	3	±0.00	Strongly Agree
Installability - The system can be installed and set up easily without technical issues.	3.33	±0.57	Strongly Agree
Replaceability - The system can be upgraded or replaced with minimal disruption to operations	3	±0.00	Agree
Composite Score	**3.16**	**±0.32**	**Agree**

Legend: *Strongly Agree (3.26 – 4.00), Agree (2.51 – 3.25), Disagree (1.76 – 2.50), and Strongly Disagree (1.0 – 1.75)*

Table 35 shows the overall outcomes of the flexibility assessment of the ISO 25010 software quality model. Flexibility is defined as the changes of the system which is able to change to different operation environment, expand as

the number of users increase, and easily installed, upgraded, or changed without the need of having a complex technical facility to install and service the system. This feature is critical to web-based tourism systems, where the changing number of visitors, the changing needs of users, and constant technological changes make it important that the platform should be responsive and adaptable. The system had the composite mean score of 3.16, standard deviation of 0.32, verbally, as Agree, which means that, the respondents tended to agree that the platform was flexible, reliable, and could meet the dynamic operational needs. Regarding the adaptability criterion, the system received an average score of 3.33 (0.57) which implies Agree, meaning that the platform is capable of effectively operating with a variety of devices, browsers, and settings and can cater to the needs of a wide range of users. This makes the system easy to access both by the tourists and administrators irrespective of the device or location hence enhances accessibility and usability. Meanwhile, scalability had a mean score of 3.00(0.00), which in a verbal form, was rated as Strongly Agree, which focused on the capability of the system to facilitate more users, reservations, and running transactions without affecting performance and response time. This is especially important when tourist seasons are high or at other special occasions where there is a high increase in the usage of the system.

The outcome shows that the platform is capable of maintaining operational efficiency even in cases of heavy workloads. In deployment related aspects, the mean score of installability was 3.33 (±0.57) that can be translated as Strongly Agree implying that the system is easily installed, configured and implemented without much technical challenge. This will ease the work of

setting up administrators and lower the downtime during deployment. Moreover, the replaceability was measured at 3.00 (0.00) (Agree), meaning that it is possible to upgrade, change or replace the system without significant disruption of the current services, which contributes to the continuous improvement and sustainability.

Table 36. Results of ISO 25010 Safety Evaluation

SUBCATEGORY/ITEM	MEAN	SD	INTERPRETATION
Operational Constraint - The system operates within safe limits to avoid errors or unsafe conditions.	3	±0.01	Agree
Risk Identification - The system helps identify possible risks or issues during use.	3.33	±0.57	Strongly Agree
Fail Safe – The system automatically activates safety measures when unsafe conditions occur.	3.33	±0.57	Strongly Agree
Hazard Warning - The system provides clear alerts when it detects unsafe conditions.	3.33	±0.57	Strongly Agree
Safe Integration - The system integrates safely with other tools or components without causing problems.	2.66	±0.57	Agree
Composite Score	**3.13**	**±0.25**	**Agree**

Legend: *Strongly Agree (3.26 – 4.00), Agree (2.51 – 3.25), Disagree (1.76 – 2.50), and Strongly Disagree (1.0 – 1.75)*

Table 36 shows the overall outcomes of the Safety assessment of the ISO 25010 software quality model. Operational Constraint criterion (Agree): The mean of the Operational Constraint criterion was 3.00 (SD ± 0.01 Agree), which means that the system operates within the limits of safety to avoid the dangerous situations or mistakes. Risk Identification got an average of 3.33 (SD ±0.57, Strongly Agree), indicating that the platform is capable of identifying potential risks or problems in the course of use and users can take the relevant measures. Both the Fail Safe and Hazard Warning criteria achieved 3.33 (SD ±0.57, Strongly Agree), proving that the system automatically takes on safety

measures in unsafe circumstances and gives clear warning to alert users of possible potential hazards, which is a proactive measure of safety management. The Safe Integration criterion was a bit less with a score of 2.66 (SD ±0.57, Agree) meaning that the system can (and will) integrate safely with other tools or components, though there is still room to improve in order to guarantee a perfect compatibility in every case.

Table 37. Results of System Quality Evaluation (ISO 25010)

CATEGORY	MEAN	SD	INTERPRETATION
Functional Suitability	3.44	±0.32	Strongly Agree
Performance Efficiency	3.11	±0.32	Agree
Compatibility	3	±0.00	Agree
Interaction Capability	3.28	±0.00	Strongly Agree
Reliability	3.08	±0.28	Agree
Security	3.16	±0.00	Agree
Maintainability	3.19	±0.00	Agree
Flexibility	3.16	±0.32	Agree
Safety	3.13	±0.25	Agree
Composite Score	**3.17**	**±0.16**	**Agree**

Table 37 presents a summary of the System Quality Evaluation done on the Pagsanjan Ecotourism Web, Based Platform as per the ISO 25010 standard that defines the quality of software in several attributes. The assessment reveals that the system continues a high degree of compliance with the functions it is intended to carry out. Functional Suitability (3.44) and Interaction Capability (3.28) scored as Strongly Agree, which is in line with a user, friendly interface and efficient performance in satisfying the users' needs.

Other quality characteristics such as Performance Efficiency, Compatibility, Reliability, Security, Maintainability, Flexibility, and Safety scored a mean between 3.00 and 3.19 (Agree), meaning that the platform is dependable, flexible to user needs, data security are guaranteed, and it maintains its performance at a steady level even when changing the device or browser. The

overall Composite Score of 3.17 (Agree) indicates that the system offers a reliable, secure, and efficient platform for users to make reservations, schedule boat rides, and get information about the ecotourism at Pagsanjan Falls.

3. B. System Acceptability Evaluation Results Based on the Technology Acceptance Model (TAM)

Table 38. Results of Perceived Usefulness

ITEM	MEAN	SD	INTERPRETATION
1. The system makes reserving accommodations and boat tours in Pagsanjan more efficient than the manual process.	3.98	±0.24	Excellent
2. Using the platform improves the organization of my travel plan and schedule to Pagsanjan Falls.	3.73	±0.44	Excellent
3. The online payment feature makes the transaction process easier and more secure.	3.73	±0.44	Excellent
4. The system helps the Tourism Office and local partners manage and monitor tourist services better.	3.55	±0.54	Excellent
Composite	**3.74**	**±0.12**	**Excellent**

n=4, Legend: Excellent (3.25–4.00), Good (3.24–2.50), Fair (2.49–1.75), (1.81–2.60), Poor (1.74–1.00

Table 38 shows the review of the system by the user in terms of its usefulness and advantages to the management of tourism services in Pagsanjan. Everything was rated as Excellent, which means that the respondents agree that the platform can be useful and effective. The overall mean score (3.98 ±0.24) indicates that the customers are highly convinced. The following, i.e., the travel planning improvement as well as the online payment option (both 3.73 ±0.44), prove that the platform helps users to plan their trip

and provide transactions in a convenient and secure way. The system also helps the Tourism Office and the local partners to manage and monitor the services more efficiently (3.55 ±0.54); such is the lowest mean score, but it is regarded as Excellent.

The overall mean of 3.74 ±0.12 and is acceptable by the consumers to handle the reservations and the activities of tourism. It shows a high possibility of simplifying travel preparations, making transactions safe and convenient, and service monitoring and service delivery. With the combination of these features, the platform will help make the tourism environment in Pagsanjan more organized, responsive, and sustainable. This is based on the findings that the further use of the system and its subsequent advancement may considerably contribute to the overall tourism experience, the cooperation between stakeholders, and the subsequent growth and competitiveness of the local tourism industry.

Table 39. Results of Perceived Ease of Use

ITEM	MEAN	SD	INTERPRETATION
1. The overall interface is clear and easy to understand.	3.61	±0.57	Excellent
2. I can complete a reservation and schedule a boat ride quickly without confusion.	3.61	±0.53	Excellent
3. The system is easy to use even without specific training or instructions.	3.53	±0.54	Excellent
4. Interacting with the system does not require a lot of mental effort.	3.51	±0.54	Excellent
Composite	**3.56**	**±0.01**	**Excellent**

n=4, Legend: Excellent (3.25–4.00), Good (3.24–2.50), Fair (2.49–1.75), (1.81–2.60), Poor (1.74–1.00).

Table 39 shows the comparison of the system on its usability and ease of use. The rankings of all the items were Excellent, which means that users found the platform to be simple to understand and use. The largest mean scores (3.61 ± 0.57 and 3.61 ± 0.53) prove that the interface is understandable and users can quickly and without complications complete reservations and book boat rides. The system is also easy to use without previous training (3.53 ± 0.54), and the users do not feel that they have to put much of their brainpower into using the system (3.51 ± 0.54). The composite mean of 3.56 ± 0.01 validates the fact that Pagsanjan Ecotourism Web-Based Platform is convenient and easy to use by the tourists, the Tourism Office, and the partner establishments, when it comes to handling reservations and services.

Table 40. Attitude Towards Usage

ITEM	MEAN	SD	INTERPRETATION
1. I have a positive attitude toward using the Pagsanjan Falls Eco-Tourism Web-Based System for accessing tourism information and services.	3.83	±0.37	Excellent
2. Using the system is a good idea because it improves convenience and efficiency in planning visits to Pagsanjan Falls.	3.55	±0.54	Excellent
3. I am willing to continue using the Pagsanjan Falls Eco-Tourism Web-Based System for future trips and transactions.	3.77	±0.42	Excellent
4 I would recommend the Pagsanjan Falls Eco-Tourism Web-Based System to other tourists, visitors, and travelers.	3.77	±0.42	Excellent
Composite	**3.61**	**±0.05**	**Excellent**

n=4, Legend: Excellent (3.25–4.00), Good (3.24–2.50), Fair (2.49–1.75), (1.81–2.60), Poor (1.74–1.00).

Table 40 outlines the evaluation of the attitudes of users on the use of the system. This was an excellent rating since all of the statements were rated as excellent, which means that the respondents have a positive attitude towards the platform. The largest mean score (3.83 ± 0.37) indicates that the users are also positive with the usage of the Pagsanjan Falls Eco-Tourism Web-Based System to access tourism information and services. It is also viewed by the respondents as a prudent decision making the system convenient and efficient towards planning the visit (3.55 ± 0.54). Moreover, the users indicate their willingness to use the system on their next visits and in the future transactions and tend to share it with other tourists (both 3.77 ±0.42). The composite mean value of 3.61 with a SD of 0.05 confirms that users highly accept the system and they consider the system as one of the useful tools in the management of their Pagsanjan traveling activities.

Table 41. Behavioral Intention to Use (BI)

ITEM	MEAN	SD	INTERPRETATION
1. I intend to use this platform for future trips to Pagsanjan.	3.83	±0.37	Excellent
2. I would recommend the Pagsanjan Ecotourism Web-Based Platform to other people	3.55	±0.54	Excellent
3. I believe the platform should be the standard way to book Pagsanjan tours and accommodations.	3.77	±0.42	Excellent
4. Overall, I am satisfied with the functionalities and services provided by the web-based system.	3.77	±0.42	Excellent
Composite	**3.73**	**±0.07**	**Excellent**

n=4, Legend: Excellent (3.25–4.00), Good (3.24–2.50), Fair (2.49–1.75), (1.81–2.60), Poor (1.74–1.00)

Table 41 demonstrates the behavioral intention of the users to use the system. Everything was perceived to be Excellent meaning that the respondents are willing to use the Pagsanjan Ecotourism Web-Based Platform further. The mean of the highest number (3.83 ±0.37) means that the site will be used by the users in future visits to Pagsanjan. They also do not hesitate to suggest the system to other people (3.55 ±0.54) and think that it needs to become the common way of making tours and accommodation arrangements (3.77 ±0.42). In addition, the users were satisfied with the platform functions and services (3.77 ±0.42). The composite mean of 3.73 ±0.07 confirms that the users are highly agreeing to the system and intending to use the system in their activities of travelling.

Table 42. Results of the Technology Acceptance Model (TAM) Evaluation

ITEM	MEAN	SD	INTERPRETATION
1. Perceived Usefulness	3.74	±0.12	Excellent
2. Perceive Ease of Use	3.56	±0.01	Excellent
3. Attitudes Towards Usage	3.61	±0.05	Excellent
4. Behavioral Intention to Use (BI)	3.73	±0.07	Excellent
Composite Score	**3.66**	**±0.04**	**Excellent**

n=4, Legend: Excellent (3.25–4.00), Good (3.24–2.50), Fair (2.49–1.75), (1.81–2.60), Poor (1.74–1.00)

Table 42 expresses the rating of the respondents to the Pagsanjan Falls Eco-Tourism Web-Based System according to the Technology Acceptance Model (TAM) constructs. The ratings of all aspects were Excellent, demonstrating high satisfaction and high levels of accepting the system. The score of Perceived Usefulness was 3.74 ±0.12, meaning that users believe the system enables them to have a better eco-tourism experience that includes such activities as trip planning, schedule check, and information access. The

score of Perceived Ease of Use was 3.56 +0.01, which indicates that the site is user-friendly, straightforward to use, and needs the least amount of effort to work with. The Attitudes Towards Usage of the users was 3.61 ±0.05, which indicates that users have a positive opinion and favorable ideas about the features and services of the system. In the same way, Behavioral Intention to Use was 3.73 ±0.07, which establishes that the users would adopt the system in future visits, recommend the same to other users and to use it in making arrangements related to eco-tourism. The total effective, usable, and acceptable score of the platform of 3.66 with the standard deviation of 0.04 also indicates the overall effectiveness, usability and acceptance of the platform and implies that it fulfills the expectations of the users and that they are willing to use the system continuously to plan and improve the eco-tourism activities in Pagsanjan falls.

CHAPTER V

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

This chapter presents the summary of the study's results and key findings. It provides an overview of the system's development, functionality, and user acceptance, highlights the main outcomes, draws conclusions, and offers recommendations for further system improvement. "This study contributes to local government digital tourism management by demonstrating how ISO 25010 and TAM can jointly validate operational readiness and user acceptance of LGU-led ecotourism platforms."

Summary

The idea behind the capstone project: Pagsanjan Ecotourism Web- Based Platform was to address the acute problems associated with ineffective operations that were in the form of manual check-in, inconsistency of policies, and inability to have a central coordination among the key stakeholders in the Pagsanjan Falls ecotourism sector. Previously, the work was anchored on the manual system of checking-in and unstructured booking leading to the occurrence of errors, delays, and unauthorized transactions which decreased the satisfaction of the tourists. To curb these gaps, the researchers were ready to develop, prepare, and roll out a web based system that was destined to centralize the process of tourist check-in, and control the boat-ride schedule, and even exhibit transparency in financial reporting. The development process of the project was constituted by Agile Scrum methodology that was designed to meet the continuous input of the user who included the representatives of the Tourism Office, the Landing Area staff, and the tourists. This system was modelled on four major integrated modules, which are Account Management,

Reservation and Scheduling, Payment and Fees Management, and Data Visualization.

The technologies and frameworks involved were PHP-Laravel and MySQL database and this has provided a system that is highly scalable, efficient and easy to use, providing this system with capabilities of processing real time data. The built-in system involves a tourist landing at a certified landing field and subsequently the system facilitates the notification of a boat tour by the tourist. The Payment module is utilized to pay the transparent 50% down- payment process and the Data Visualization module will show the real-time report of the tourist demographics, revenues, and trends to the Tourism Office. The unit testing and the integration testing that were performed during system testing showed that the modules were performing satisfactorily.

Moreover, the effectiveness of the system was measured with the help of 50 respondents using the Technology Acceptance Model (TAM). The mean composite mean score of 3.69 with a SD of (pm) 0.04 which was seen as Excellent indicated that users believed that the system was so useful, easy to use, and willing to adopt the application of the technology.

Conclusion

According to the analysis and the examination of the Web-based system of the Pagsanjan Falls Eco-Tourism, it can be stated that the created system has managed to deliver a structured Internet-based service to regulate the reservations and boat tour booking. The system promotes the municipal policy as all the tourists have to book the services via certified Landing Areas and UBAP boat assignment system thus contributing to the safety, order and adequate distribution of income among the boatmen. The system will also serve

as a one-stop shop where tourists can see the open landing sites, hotels, resorts, and restaurants, make reservations, confirm check-in, and even boat scheduling hence less confusion and greater convenience. The results of Technology Acceptance Model (TAM) showed that the acceptance by the user was very strong and the average score of Perceived Usefulness was 3.74, Perceived Ease of Use 3.56, Attitude Towards Usage 3.73 and Behavioral Intention to Use 3.73, gave an overall composition mean of 3.69 (SD ±0.04), which is considered to be Excellent.

These findings express that tourists, staff and administrators are ready to keep on using the system and they feel that it can be useful in organizing the visit and the management of the tourism services. Nevertheless, despite the high ratings of the majority of the features, the services to provide full information about safety warnings, local rules, and guidelines on tourism still need to be improved, allowing the users to learn about the key messages more thoroughly before visiting Pagsanjan Falls.

Recommendation

Based on the principal findings, data-driven conclusions, and opportunities identified during the development and testing phases, the following recommendations are hereby proposed to further strengthen the functionality and effectiveness of the Pagsanjan Ecotourism Web-Based Platform:

1. Safety and Information Awareness Improvement.

Even though the system received favorable reviews, some of the users have noted that tourism information, and safety guidelines could still be better. One of the recommendations is to provide a specific information portal containing safety warnings, visitor regulations, local rules, weather notices, and a step-by-

step manual on how to book and schedule a boat ride. This addition will allow the tourists to understand the process better and to visit Pagsanjan Falls with less danger.

2. Enhance System Performance and Server Implementation.

Since the system has been tested in a local host system, and the ISO 25010 test on performance efficiency produced an agree rating, it is suggested to install the application on a trusted web hosting system. Proper hosting and database optimization will reduce the load time, support concurrent users and enhance real-time operation of reservation and scheduling.

3. UBAP Boat Scheduling Management Direct.

At the present time, the process informs UBAP of the reservation confirmation. It is also recommended to introduce a special UBAP scheduling interface, according to which an UBAP administrator is able to allocate available boatmen directly, through a queue or availability list. This will facilitate boat dispatching, minimize waiting time as well as equitable allocation of trips among boatmen.

4. Increased Online Payment.

To make it more convenient and to facilitate transactions, the system must integrate other payment options such as credit/ debit cards, and locally available e-wallet services in addition to the currently existing payment option. This would be an improvement to both local and foreign tourists hence simplifying and making payment processing faster.

5. Feedback Rating and Feedback Feature of Boatman.

The suggestion to increase the feedback module will involve allowing tourists to rate and comment on the boatperson they are assigned and their respective services. This improvement would give the Tourism Office and UBAP a

powerful tool on how to track the quality of services, motivate high performance and encourage the constant development of ecotourism services.

6. Enhancing Data backup and Security.

To make the system reliable and compliant with ISO 25010, it is recommended that the system be configured to perform automated database backups, secure authentication systems, and periodically monitor user actions. These will protect reservation information, payment information and tourist information against loss and illegal access of data.

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APPENDICES

APPENDIX A

Letter of Approval

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the capstone project entitled:

" **PAGSANJAN FALLS ECO-TOURISM: A WEB BASED SYSTEM FOR THE MUNICIPALITY OF LAGUNA"**

ISRAEL BAUTISTA, ANDREA JADE BORJA, and ROSE MALYN SAN ISIDRO have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the proposal presentation.

GENE MARCK B. CATEDRILLA
Capstone Project Adviser

APPENDIX B

Consent Letter

Republic of the Philippines
Laguna State Polytechnic University-Los Baños Campus
Province of Laguna

To Office of the Mayor

Dear Sir

We are fourth-year college students from Laguna State Polytechnic University – Los Baños Campus, currently working on a proposal for a web-based system intended to support the tourism industry of Pagsanjan, Laguna. As part of our study and system development, we would like to respectfully conduct a survey for our Capstone project. The purpose of this study is to evaluate how the system can improve the efficiency, convenience, and sustainability of tourism management in Pagsanjan. It also aims to determine how the system's features—such as ease of access, usefulness, and user satisfaction—affect its adoption by tourists, local businesses, and tourism staff. The participations and feedbacks of correspondents are essential to understanding the system's effectiveness and ensuring that it supports the Municipality of Pagsanjan in promoting organized, secure, and eco-friendly tourism operations. This information will serve as a reference for us to better understand the current state of tourism and to propose effective features that will benefit the landing area, hotels, resorts, restaurants, bangkero, and the overall Pagsanjan Ecotourism sector.

Rest assured that the data will be used strictly for academic purposes and treated with confidentiality and respect. We hope for your kind approval of our request. Thank you very much for your time and support. We look forward to your positive response.

Sincerely,

Israel B. Bautista

Andrea Jade D. Borja

Rose Malyn L. San Isidro

BSIT 4th Year Students at Laguna State Polytechnic University

Noted by:

Gene Marck Catedrilla

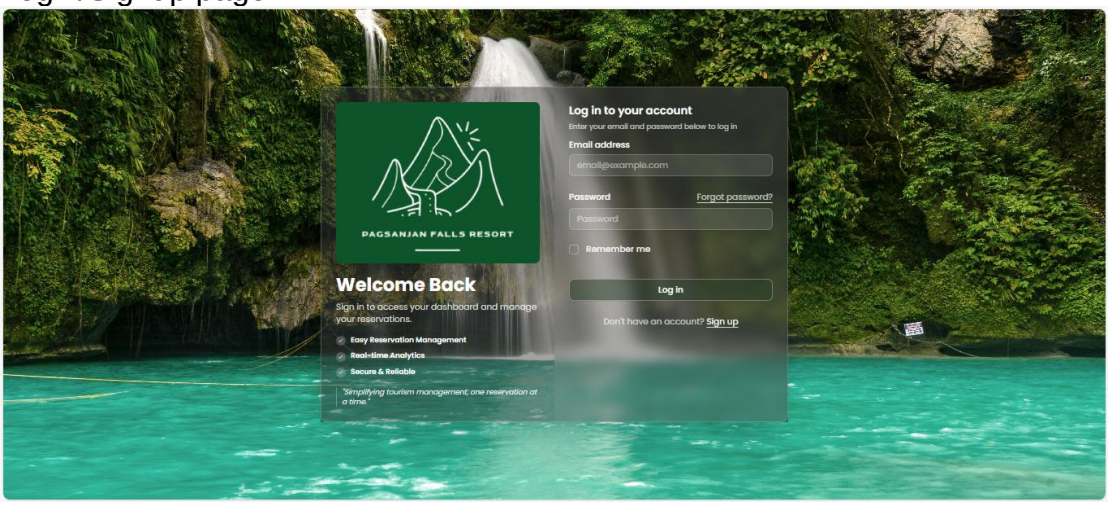
Thesis Advisor

MUNICIPALITY OF PAGSANJAN
Office of the Mayor
RECEIVED
By:
Date: 11/18/2024 Time:

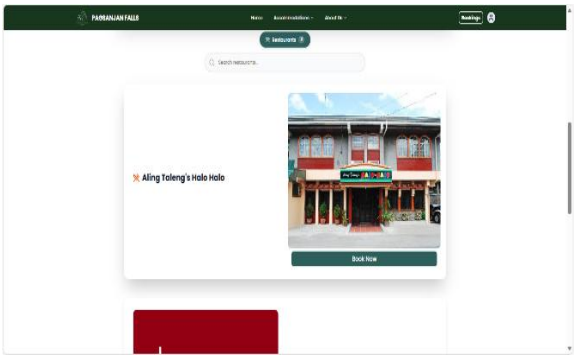
APPENDIX C

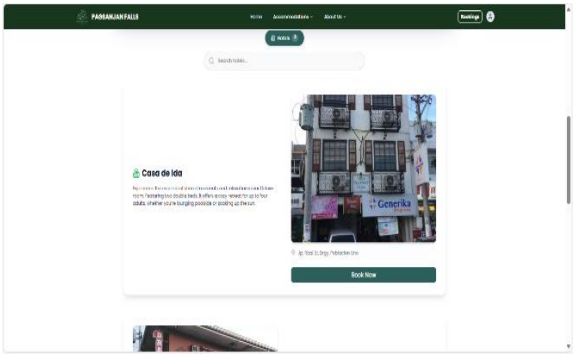
Output Screenshots

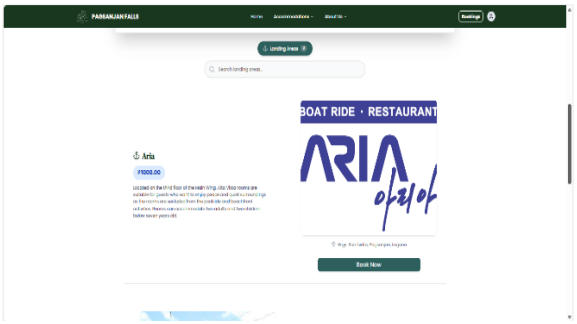
Login/Signup page

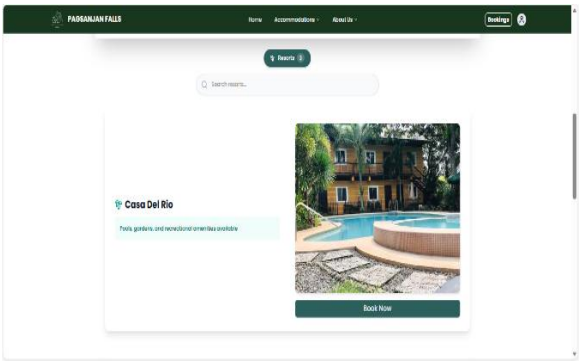


Landing Area, Hotel, Resort and Restaurant Accommodation

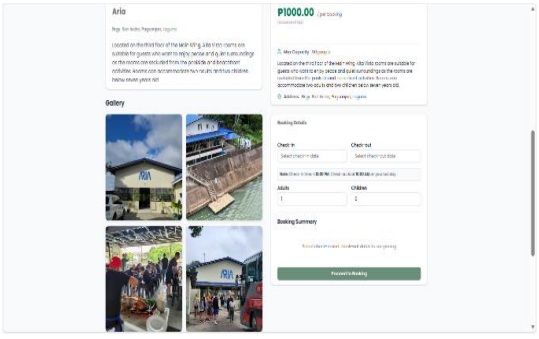


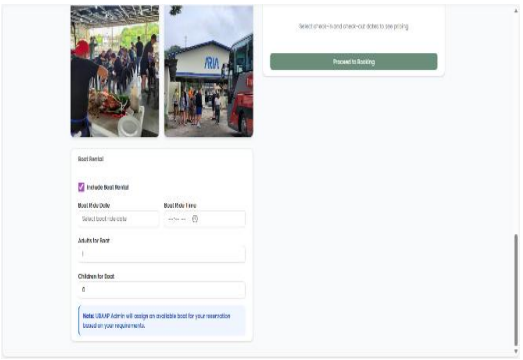




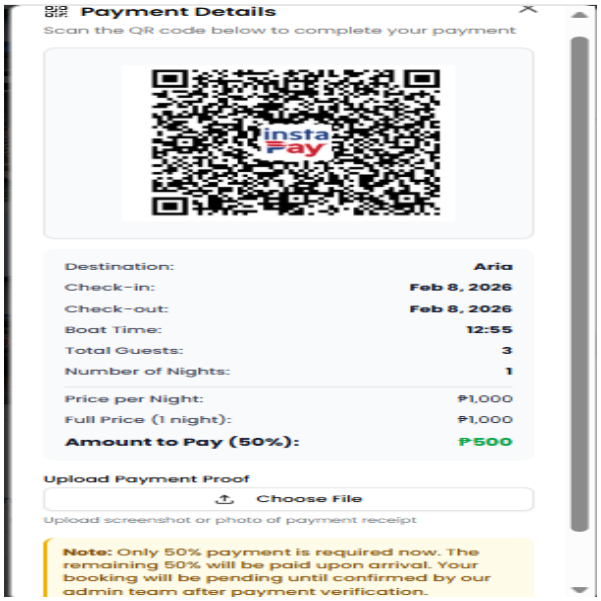


Booking Details for Accommodation

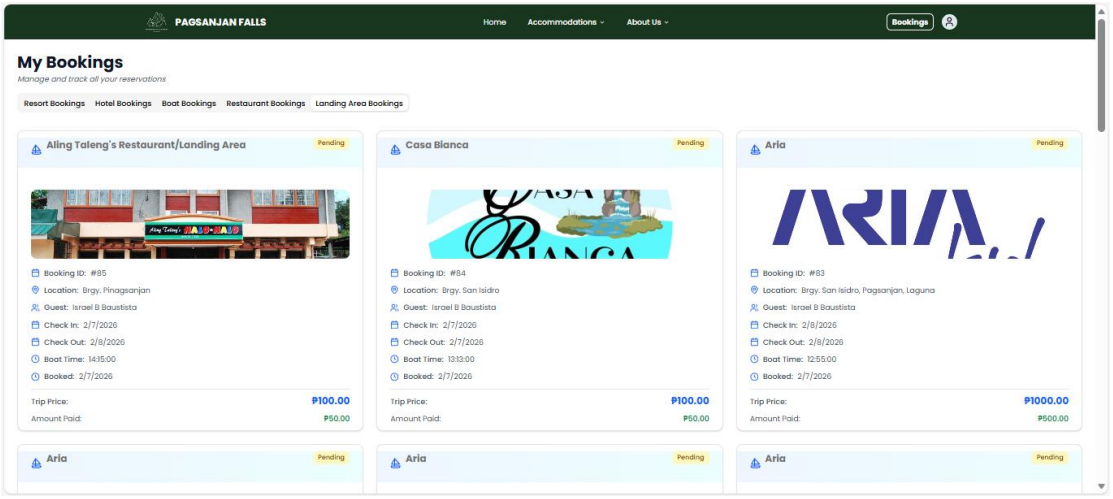




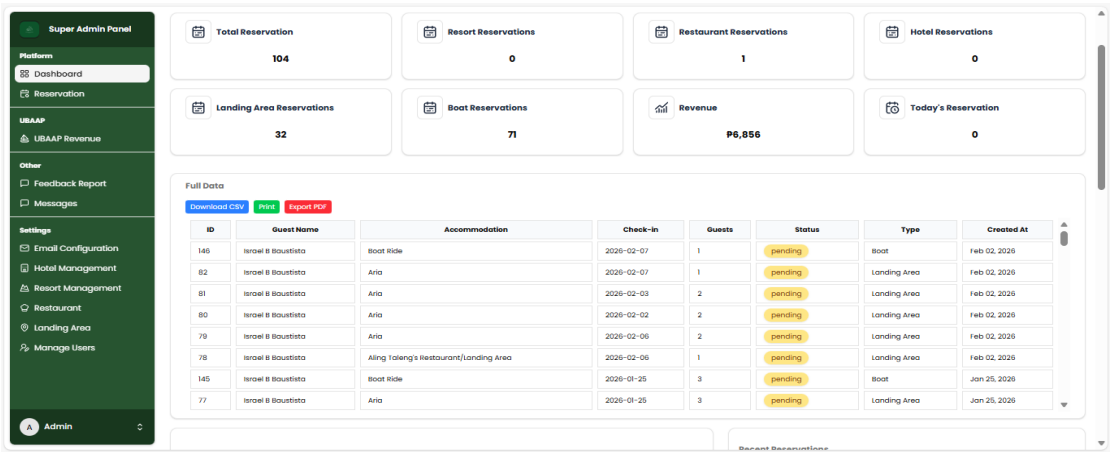
Payment Fees and Details



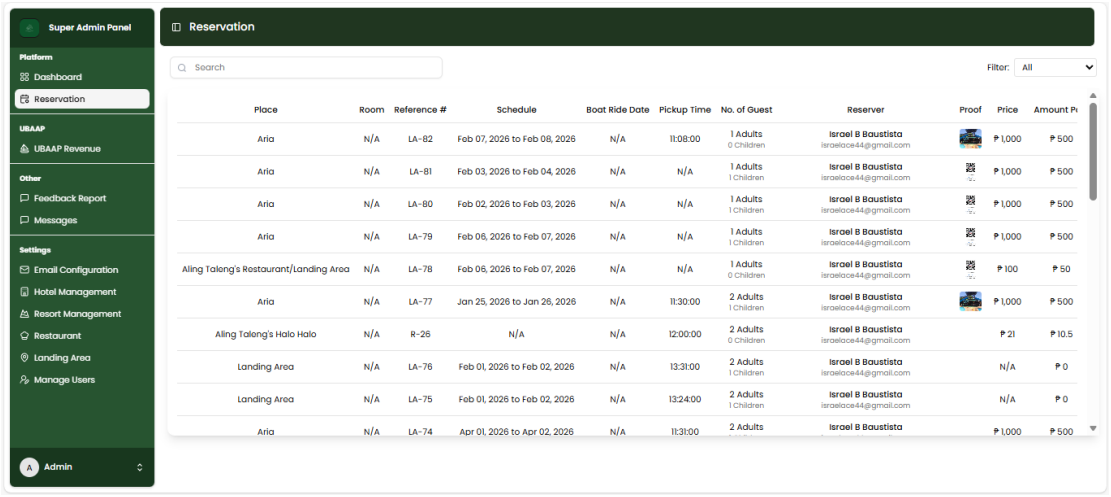
My Bookings Page



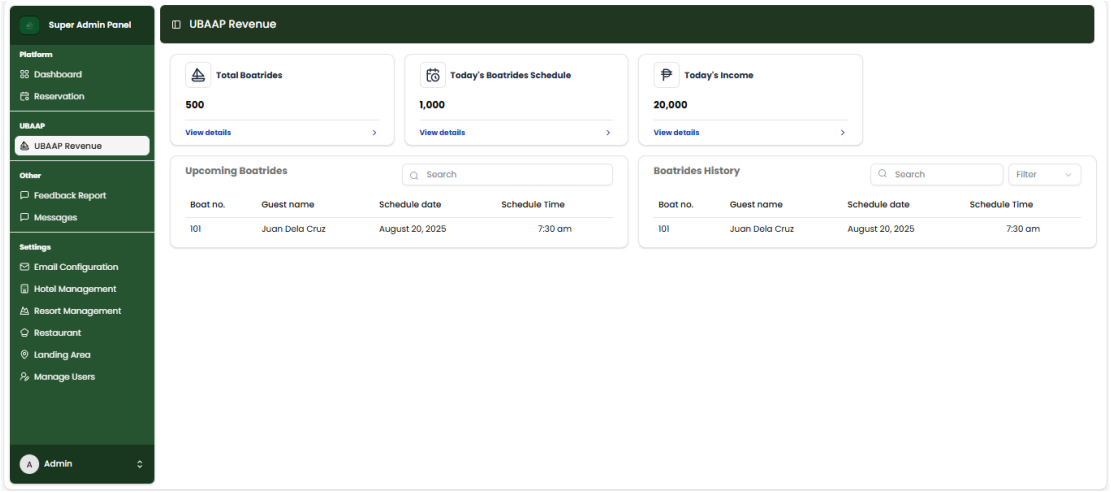
Admin Dashboard



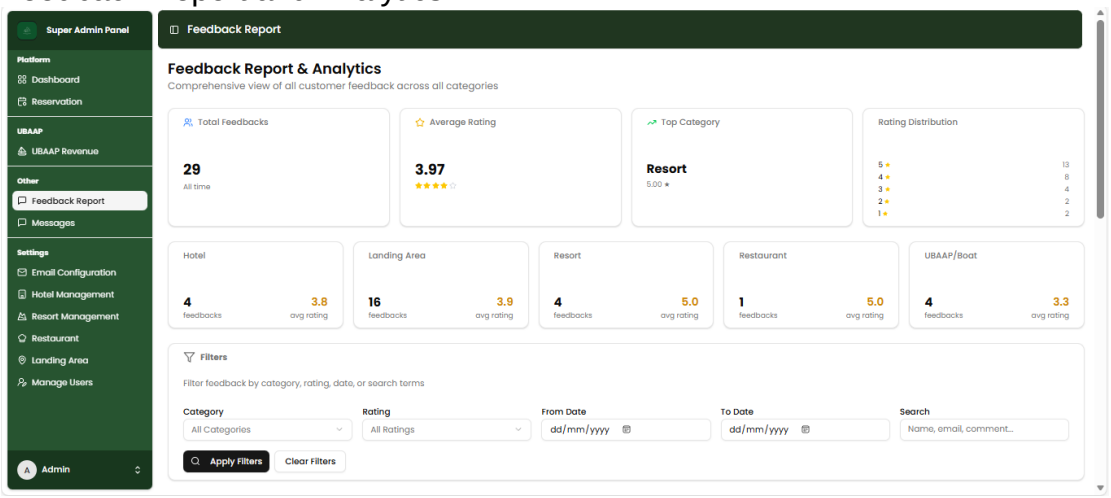
Reservation List



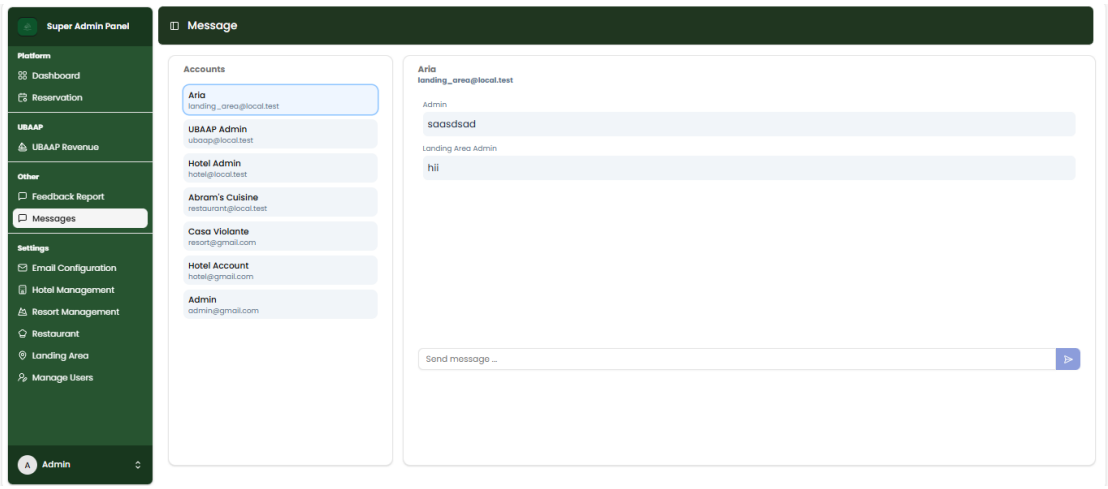
UBAAP Revenue



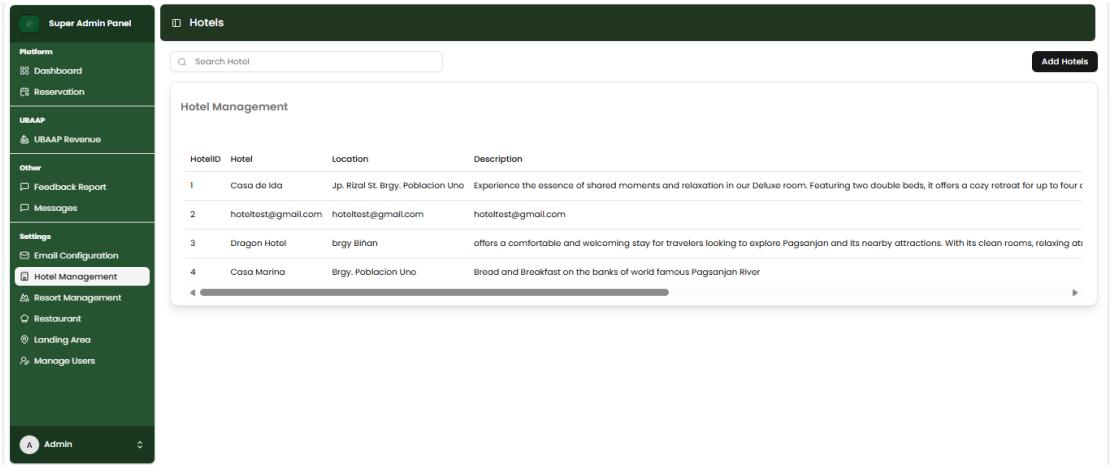
Feedback Report and Analytics



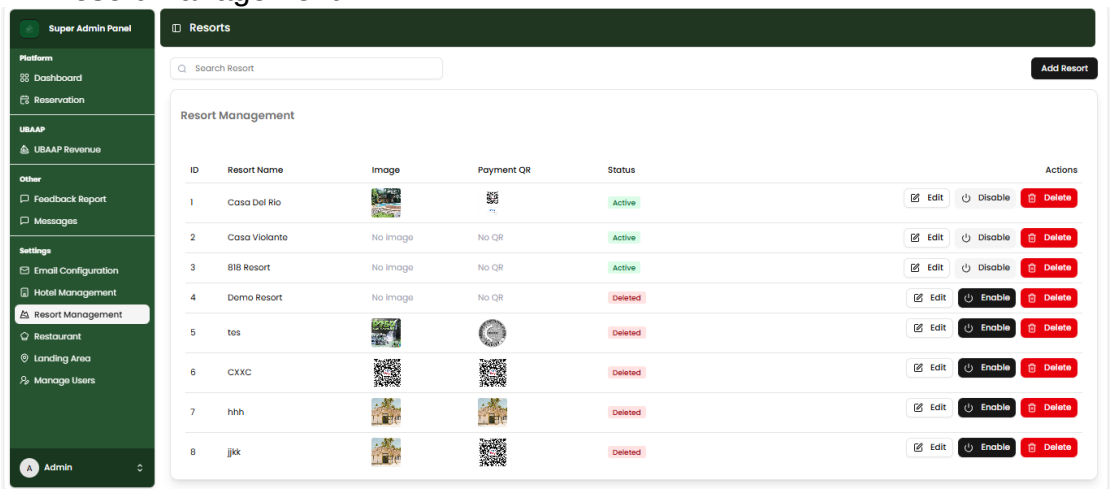
Admin Panel Messages



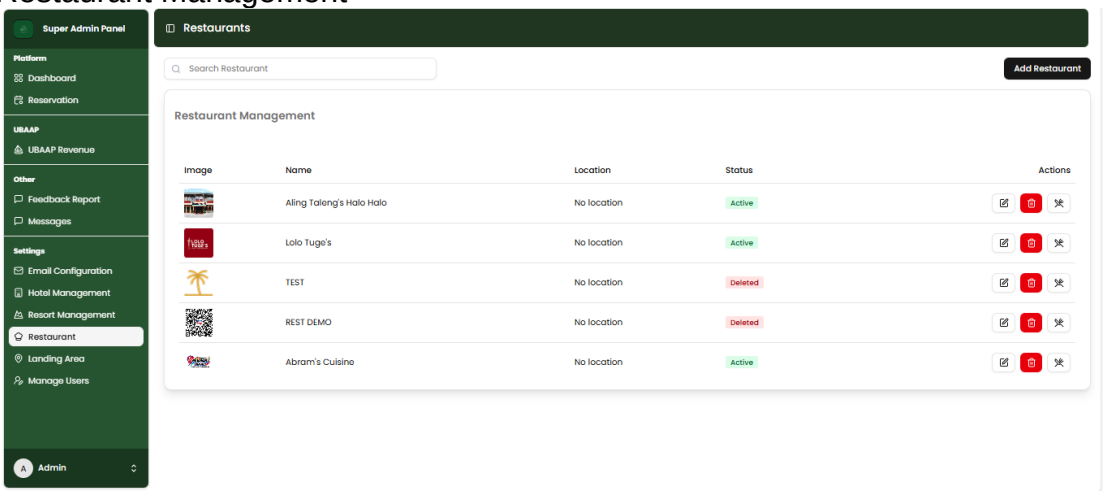
Hotel Management



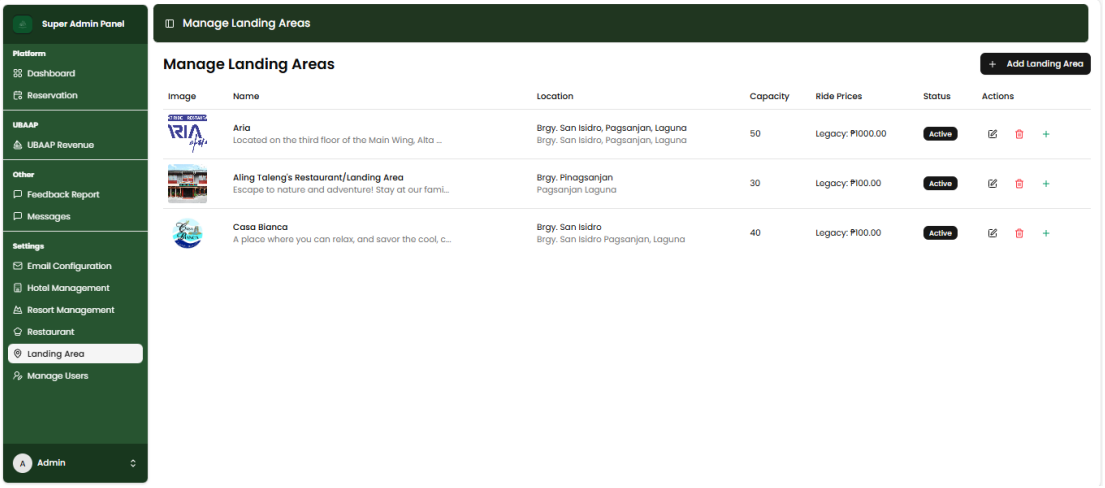
Resort Management



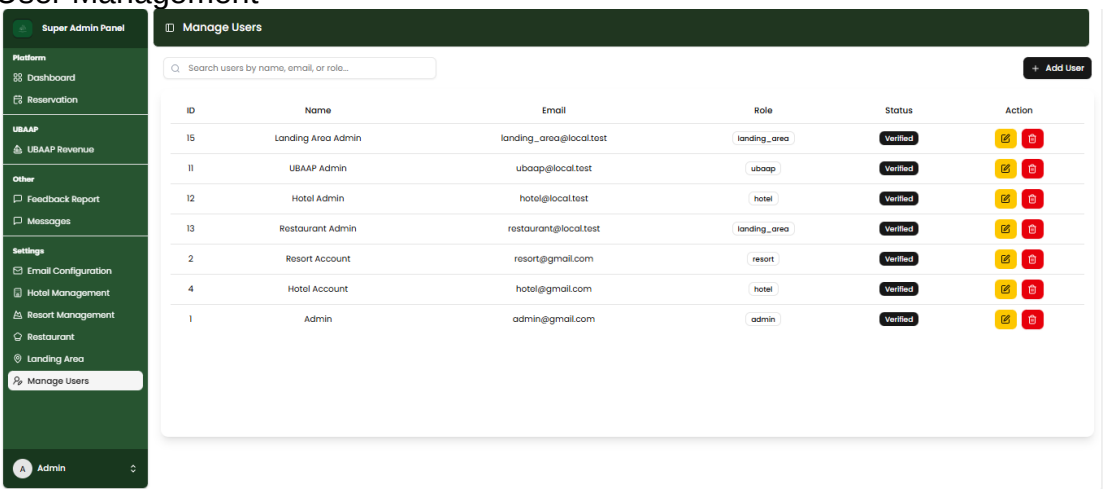
Restaurant Management



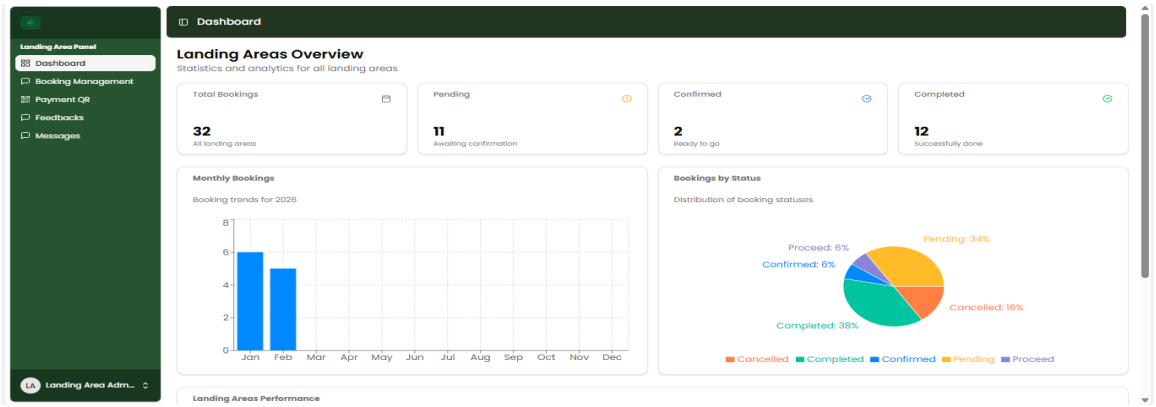
Landing Area Management



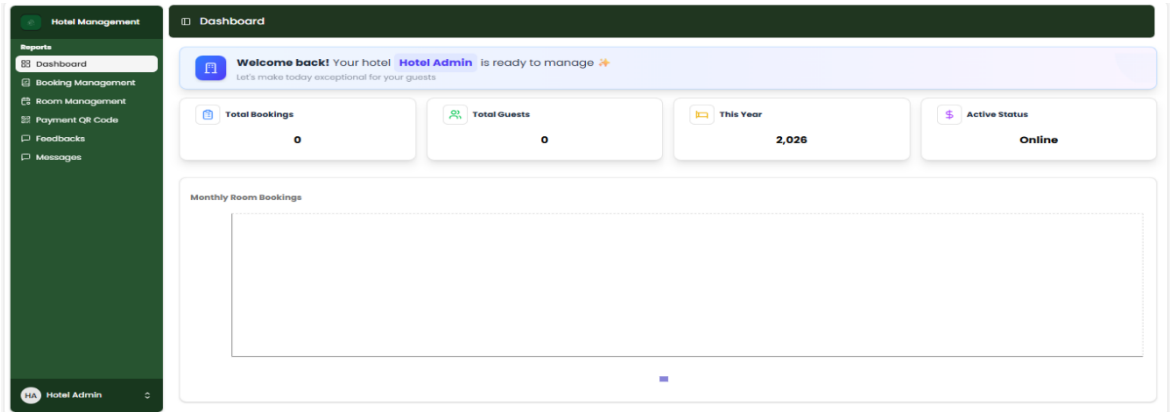
User Management



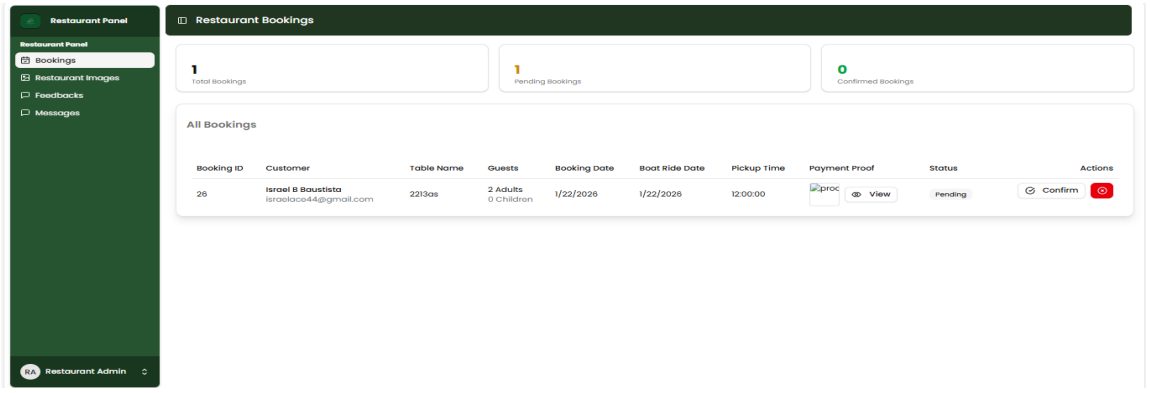
Landing Area Portal



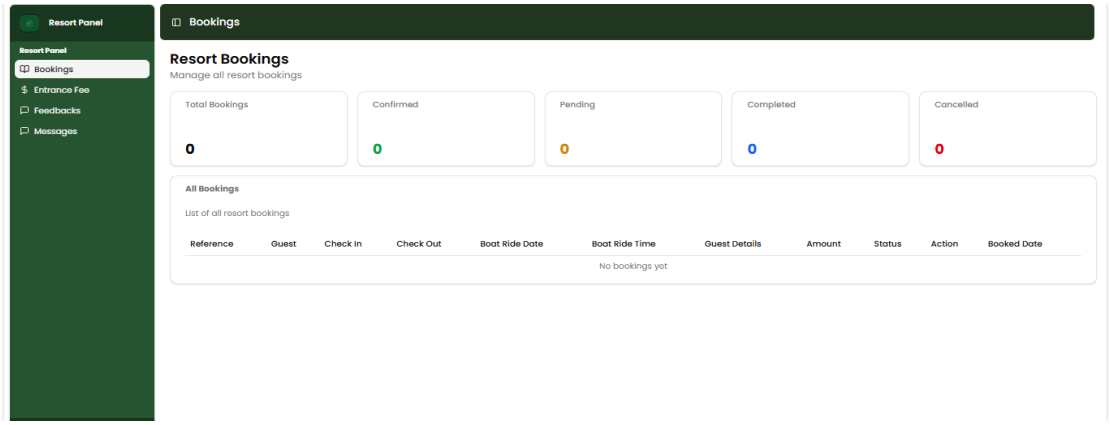
Hotel Portal



Restaurant Portal

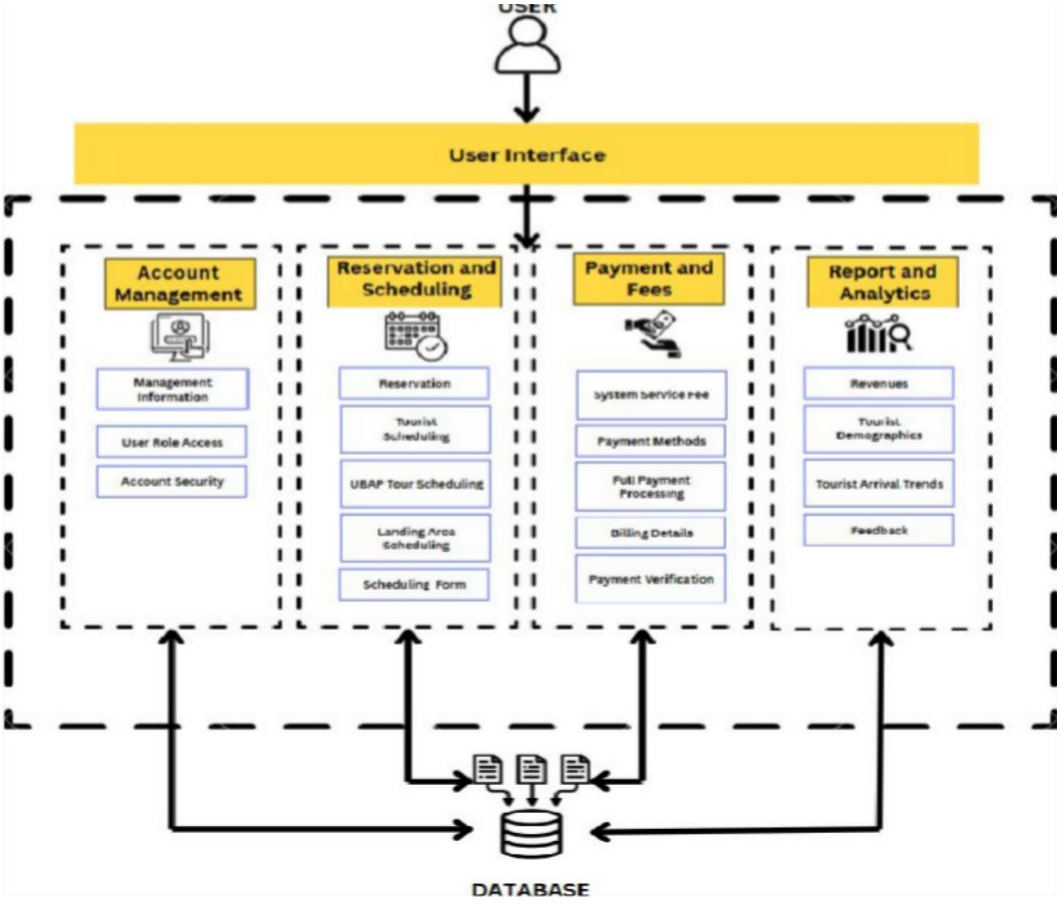


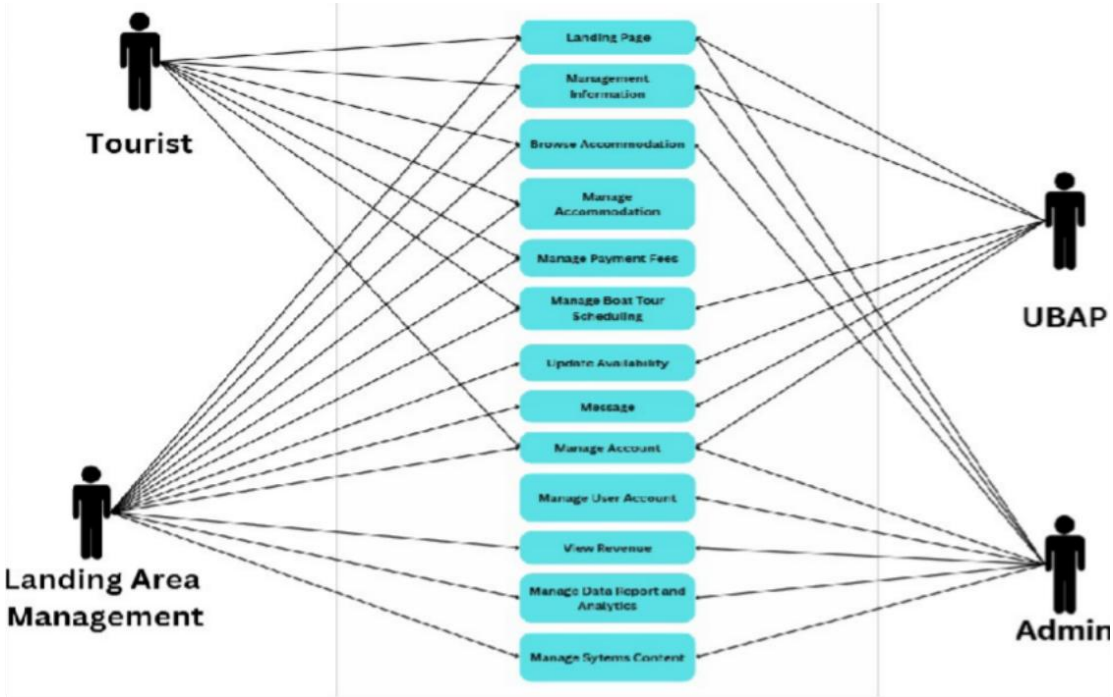
Resort Portal

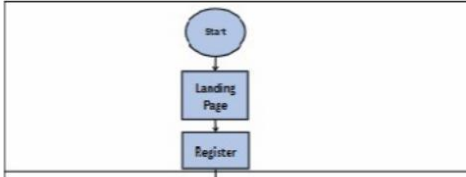


APPENDIX D

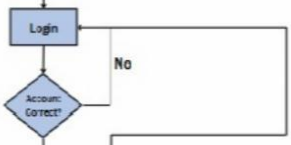
Diagrams


USER
User Interface
Account Management
Management Information
User Role Access
Account Security
Reservation and Scheduling
Reservation
Tourist Scheduling
UBAP Tour Scheduling
Landing Area Scheduling
Scheduling Form
Payment and Fees
system service fee
Payment Methods
Full Payment Processing
Billing Details
Payment Verification
Report and Analytics
Revenues
Tourist Demographics
Tourist Arrival Trends
Feedback
DATABASE

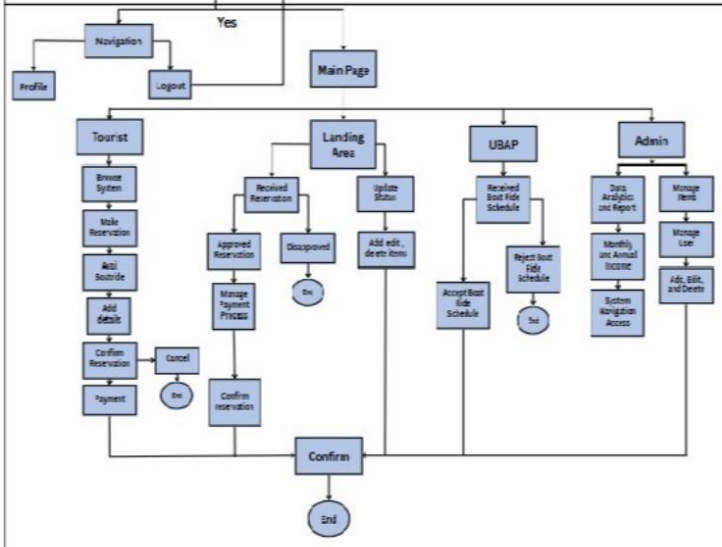

Tourist
Landing Area Management
UBAP
Admin
Landing Page
Management Information
Browse Accommodation
Manage Accommodation
Manage Payment Fees
Manage Boat Tour Scheduling
Update Availability
Message
Manage Account
Manage User Account
View Revenue
Manage Data Report and Analytics
Manage Sytems Content



When users first open the system, they can view the entire system except that they can't make a reservation to the system, they must first need to register an account if they do not already have one. During registration, users can select their role from four types: Tourist, Landing Area (resort), UBAP (United Boatmen Association of Pagsanjan), or Admin (Tourism Office).



Once registered, the user can proceed to log in using their account credentials. If the login credentials are incorrect, the system will ask them to try again until successful. After a successful login, the user will be directed to their corresponding main page, where they will access features based on their role. The system will determine users account which is which. Users may also choose to visit their profile or log out at any time.



Once registered, the user can proceed to log in using their account credentials. If the login credentials are incorrect, the system will ask them to try again until successful. After a successful login, the user will be directed to the application's main page, where they will access features based on their role. Users may also choose to visit their profile or log out at any time. If the login is successful, users are directed to their designated main page:

- Tourist: Tourists can browse the system, make a reservation and schedule a boat ride. Before they can confirm their reservation and payment they are requires by the system to input their details for the system to confirm the reservation that they make.
- Landing Area: Landing Area staff can accept, manage payment transactions done with the guests and confirm the reservation the tourist's reservation request. They also have access to update the status of their management (lancing area, hotel, restaurant and resort), add, edit, or delete certain information.
- UBAP (United Boatmen Association of Pagsanjan): UBAP receives boat ride schedule upon the successful confirmation of reservation of the tourist from the Landing Area, hotel, restaurant and resort to provide the boat service. They assign a boat and ensure the service is delivered accordingly.
- Admin (Tourism Office): The Admin is responsible for managing data analytics and report, as well as viewing monthly revenue reports and a system navigation access. Admin also manage the content of the systems items and users in the system and by adding, editing, or deleting entries when needed.

Once all tasks for each role are completed and confirmed, the system process reaches its end.

APPENDIX E

Survey Questionnaire/Evaluation Tool

PROJECT TITLE: **"PAGSANJAN FALLS ECO-TOURISM: A WEB BASED SYSTEM FOR THE MUNICIPALITY OF LAGUNA"**

THE RESEARCHERS: Israel B. Bautista, Andrea Jade DC. Borja, Rose Malyn L. San Isidro

Good Day! We are 4th year students, currently pursuing a Bachelor of Science in Information Technology specializing in Service Management Program. As part of our academic endeavors, we have developed a comprehensive web-based system adhering to the ISO 25010 standard.

To assess the effectiveness and user satisfaction of our system, we have devised a set of questions. Your responses, rated on a scale from 1 to 4, will help us evaluate various aspects of the system, from usability, functionality, security and overall user experience.

4 = Strongly Agree 3= Agree 2 =Disagree 1= Strongly Disagree

Functional Suitability	**4**	**3**	**2**	**1**
Functional Completeness:	All necessary features for booking tours, payments, and notifications are fully implemented.	The system vides most features for booking tours but may need improvements.	The system provides limited features for booking tours and lacks essential functionality.	The system does not provide features for booking tours or has no clear purpose.
Functional Correctness:	The system accurately processes payments and confirms all transactions successfully.	Operations generally run accurately with few minor issues that do not affect overall reliability	System has frequent errors affecting bookings or payments.	Many functions fail consistently system produces incorrect results and is unreliable.
Functional Appropriateness:	Users can easily check tour availability and make bookings without guidance.	Most system functions are Relevant and assist in achieving booking-related goals.	Some functions are mismatched or not helpful for bookings.	The system does not support integration with any external platforms.
Performance Efficiency				
Time Behaviour:	The system performs actions instantly with no noticeable delay in browsing, booking, or payments.	System maintains good speed with few instances of lag.	The system shows moderate delays in booking, payments, or schedule updates, causing minor inconvenience or repeated user actions.	The system often loads slowly, causing interruptions in booking.

Resource Utilization:	System uses resources efficiently; tasks complete without lag.	The system manages resources effectively with minor slowdowns when processing several tasks.	The system consumes noticeable resources, causing minor lags or delays on some devices.	The system performs poorly due to high resource consumption, resulting in frequent lag or system crashes.
Capacity:	The system can handle multiple users and large amounts of data smoothly without performance issues.	The system can accommodate several users at once with slight performance change	The system experiences lag when many users access it at once.	System cannot handle multiple users; performance fails under normal usage.
Compatibility				
Co-Existence:	Users can book tours, access schedules, and complete payments smoothly because the system works perfectly with external services.	The system can accommodate several users at once with slight performance change	The system experiences lag when many users access it at once.	System cannot handle multiple users; performance fails under normal usage.
Interoperability:	The booking system fully supports external connections like emails, online payment, and notifications.	The system integrates with most external platforms. Minor issues may occur in data synchronization (e.g., slight delay in booking confirmation), but core functionality is unaffected.	The system Only partially interoperates with external services. Frequent minor errors, delays, or data mismatches occur, affecting tasks like booking confirmation or schedule updates.	The system lacks any working external integration.
Interaction Capability				
Appropriateness Recognizability:	All important information (dates, pricing, and availability) is clearly visible and easy to identify.	Most features are easily recognizable; minor features may require brief explanation but users can complete tasks with minimal guidance	Some information is unclear or difficult to locate	Users often fail to understand the system's functions. Booking, payment, or schedule tasks are confusing, and users cannot complete tasks effectively even with some assistance.
Learnability:	The system is extremely easy to learn and navigate. Users can instantly understand how to book tours, view schedules, and explore ecotourism details without assistance.	Users can learn how to use the system with minimal effort; most features are understandable after a few interactions.	The system requires time and user effort to learn; some features or navigation paths are unclear and may confuse new users.	The system is difficult to learn; users frequently require assistance to understand and complete basic operations.
Operability:	The system provides a seamless experience; all features, such as tour booking, feedback	The system operates well with only minor delays or usability issues that don't affect performance.	The system's operation is occasionally disrupted by slow responses or unclear buttons that affect the	The system frequently encounters operational issues, such as unresponsive buttons or loading times.

	submission, and schedule viewing, work efficiently and respond quickly to user actions.		user experience.	
User Error Protection: .	The system efficiently minimizes user errors through form confirmation and clear feedback messages, ensuring a smooth booking process.	The system effectively minimizes user mistakes and Provides clear Guidance when errors occur.	The system occasionally allows errors without proper Guidance for correction.	The system does not prevent errors. Users frequently make mistakes that lead to failed bookings, incorrect payments, or scheduling problems, requiring significant external intervention.
User Engagement:	Interface encourages users to explore features and packages actively with interactive prompts and clear options.	The system maintains moderate engagement; users find it functional but not visually or interactively appealing.	The system is moderately engaging; design and interaction are functional but lack attractiveness or interactivity.	The system is unengaging; users rarely explore or use additional features
Inclusivity:	Text, fonts, colors, and layout are fully readable and accessible to users of all abilities	Mostly accessible; minor usability issues may affect a small number of users, but most tasks can be completed independently.	The system has noticeable inclusivity limitations; certain users experience difficulty accessing or understanding the platform.	System is not inclusive; many users cannot access or use core features effectively.
User Assistance:	Provides clear, interactive guidance and feedback at every step of booking, payment, and schedule viewing. Users can complete all tasks confidently external help.	The system offers assistance for most features, though some areas could benefit from clearer guidance or feedback.	Limited support; users often struggle to complete viewing without external help.	The system lacks assistance and clear instructions, making it difficult for users to complete tasks independently.
Self-Descriptiveness:	Icons and interface clearly indicate functions without confusion.	Most functions are understandable; some may require explanation.	Some elements are unclear; users need instructions to understand functions.	Features are poorly labeled and unintuitive; users cannot understand how to perform bookings, payments, or access schedules.
Reliability				

Faultlessness:	The system operates without errors or crashes during all functions	The system works reliably for most tasks. Minor glitches may occur occasionally, but users can still complete	The system experiences errors that affect some functions. bookings, payments, or schedule access successfully.	Frequent system errors prevent users from completing essential tasks. Bookings may fail, payments are often rejected, and schedules are inaccessible, making the system practically unusable
Availability:	The system is consistently accessible with no downtime, ensuring continuous service to all users.	The system is mostly available brief downtime or maintenance may occur but does not significantly affect usability.	Noticeable downtime occurs periodically, limiting users' ability to book tours, view schedules, or make payments reliably.	System is frequently unavailable; users cannot reliably book tours or view schedules.
Fault Tolerance:	The system continues to function correctly even during minor server, network, or database errors, ensuring all bookings, payments, and schedule updates remain accurate.	Handles most errors with minimal disruption; minor delays may appear.	The system becomes unstable when faults occur but recovers after some delay.	The system crashes or becomes unusable when faults occur.
Recoverability:	The system quickly restores all data and functionality after errors or crashes, ensuring no loss of booking, payment, or user information.	The system recovers effectively after minor faults with minimal data loss and short recovery time.	The system takes a long time to recover from faults, occasionally losing user or transaction data.	The system fails to recover from errors or crashes, resulting in major data loss or requiring technical intervention.
Security				
Confidentiality:	All the sensitive information, including user data and payment, is safely stored in the system and cannot be accessed by an unauthorized person.	The system maintains go od confidentiality with minor vulnerabilities that do not significantly expose user data.	The system has noticeable weaknesses that could allow limited unauthorized access to user information.	The system provides insufficient data protection, making user and payment information vulnerable to unau acc
Integrity:	All transactions and stored information are correct and consistent and tampering or corruption is highly guarded.	There is typically no loss or damage of data, the inconsistency of data is not very great, but only in the subsidiary processes.	Data is sometimes modified or lost due to weak integrity controls, affecting reliability of bookings or payments.	This system often causes loss or distortion of vital data undermining both accuracy of transactions and user confidence.

Nonrepudiation:	All user actions (booking, payment, or update) are logged in securely to avoid the denial of transactions performed.	Most actions are traceable; minor gaps exist in transaction logging	Some user actions lack proper verification, occasionally allowing disputes over completed transactions.	The system provides no proof of user transactions; actions can be denied or unverified.
Accountability:	The system keeps detailed logs of all user and admin activities, allowing full tracking of actions.	System records most activities, though minor logs or user actions may be missing.	Only some user or admin activities are recorded, making it difficult to trace issues.	There is no logging or tracking, no user or administrator actions can be tracked.
Authenticity:	The system verifies user identities accurately through secure login and authentication methods.	User authentication works effectively with minor issues that rarely affect access.	Authentication sometimes fails, allowing unauthorized access or blocking legitimate users.	The system lacks prop er authentication, allowing unauthorized users to access accounts or data.
Resistance:	System components can be reused efficiently for other applications or modules	System resists most threats but may show minor vulnerabilities under certain attack conditions.	The system has wea k resistance, occasionally affected by unauthorized attempts or malicious activity.	The system is highly vulnerable to attacks, frequently compromised or disrupted.
Maintainability				
Modularity:	The system is divided into clear, independent modules that can be easily modified or updated without affecting others.	Modules are mostly independent; minor dependencies exist that slightly affect updates	Modules are interdependent, making maintenance or updates moderately complex.	The system lacks modular structure; changes in one area cause failures in others.
Reusability:	System components can be reused efficiently for other applications or modules.	Most components are reusable with minimal modification.	Some parts are reusable but require significant adjustments.	Component s are not reusable; each part must be rebuilt from s[illegible]
Analyzability:	System structure is well- documented and easy to analyze for performance issues or errors.	System is generally analyzable; minor parts require extra effort to review or debug.	The system is difficult to analyze due to poor documentation or complex code structure.	System is not analyzable; identifying and fixing issues or complex code structure.
Modifiability:	Changes can be made easily without affecting system stability or other modules.	Modifications are possible with minimal system impact.	Modifications are complex and may introduce errors.	Systems is rigid; any modification causes major issues or requires full redevelopment.

Testability:	The system is easy to test; functions can be verified accurately with minimal effort.	System testing is manageable, though some components require extra testing time.	Testing is difficult due to unclear processes or incomplete documentation.	System cannot be reliably tested; functions are unpredictable or lack test access points.
Flexibility				
Adaptability:	The system adapts seamlessly to different devices, browsers, and user environments.	The system adjusts well with minor display or function issues across platforms.	Adaptability is limited; users may face issues on certain devices or browsers.	System is not adaptable; only functions properly in one specific environment.
Scalability:	The system scales efficiently as user demand increases without performance loss.	System scales moderately; performance slightly decreases under heavy load.	System struggles to scale, showing noticeable slowdowns as users increase.	The system cannot handle additional users; performance drops significantly.
Install ability:	The system installs easily on any device or server with clear instructions and no errors.	Installation is generally smooth; minor configuration issues may arise.	Installation requires technical assistance or additional steps.	The system is difficult to install or fails during installation.
Replace ability:	System components can be easily replaced or upgraded without affecting other functions.	Component s are replaceable with minor adjustments needed.	Replacement s are difficult and may require partial reconfiguration.	Component s cannot be replaced without complete system overhaul.
Safety				
Operational Constraint:	The system operates safely within all defined limits, preventing misuse or unsafe operations.	System follows safety limits with minor exceptions that do not impact user security.	System occasionally operates beyond safe limits, causing potential risk.	System often violates safety constraints, posing serious risks to users or data.
Risk Identification:	The system effectively identifies and mitigates potential risks before they cause issues.	System identifies most risks; some may require manual monitoring.	Risk detection is limited; several issues arise without early warning.	System lacks risk identification; problem occur without detection or prevention.
Fail Safe:	The system maintains data integrity and user safety during failures, automatically reverting to a safe state.	The system safely handles most failures with minimal disruption.	System sometimes fails to protect data during unexpected errors.	The system crashes completely during failures, losing and user progress.

Hazard Warnings:	The system clearly alerts users to potential risks, errors, or unsafe operations.	System provides adequate warnings, though some alerts may be unclear or delayed.	Warnings are inconsistent; users may not be informed of certain issues.	The system lacks any warning mechanism; users are unaware of hazards or failures.
Safe Integration:	The system integrates securely with external platforms (e.g., payment gateways, booking APIs) without compromising safety or data integrity.	Integration is generally safe; minor vulnerabilities or compatibility issues exist.	System integration sometimes introduces security or reliability risks.	Integrations are unsafe; external connections often lead to failures or data breaches.

Signature of Respondent_____ Date_____

TECHNOLOGY ACCEPTANCE MODEL QUESTIONNAIRE

Good Day, We are students from Laguna State Polytechnic University - Los Baños Campus, currently conducting a study on the deployment and user acceptance of the Pagsanjan Ecotourism Web-Based Platform. As part of our project, we have designed and developed this online system to modernize the booking and scheduling process for Pagsanjan falls tours and local accommodations. Your responses, rated on a scale from 1 (Strongly Disagree) to 4 (Strongly Agree), will help us evaluate the system's effectiveness and its value to tourists and local stakeholders.

Instruction: Evaluate the web-based system by selecting a rating from 1 (Strongly Disagree) to 4 (Strongly Agree) for each statement provided

Perceived Usefulness Assesses how relevant and impactful the film is in promoting awareness about women's participation in governance.	**4**	**3**	**2**	**1**
1. The system makes reserving accommodations and boat tours in Pagsanjan more efficient than the manual process.				
2. The platform improves the organization of my travel plan and schedule to Pagsanjan Falls. Using the platform improves the organization of my travel plan and schedule to Pagsanjan Falls.				
3. The online payment feature makes the transaction process easier and more secure.				
4. The system helps the Tourism Office and local partners manage and monitor tourist services better.				
Perceived Ease of Use (PEOU) How easy, smooth, and clear it is to navigate and operate the web-based system.	4	3	2	1
1. The overall interface is clear and easy to understand.				
2. I can complete a reservation and schedule a boat ride quickly without confusion.				
3. The system is easy to use even without specific training or instructions.				
4. Interacting with the system does not require a lot of mental effort.				

Attitudes Towards Using the System (ATU) The users' overall evaluation of their feelings, acceptance, and willingness to use and recommend the Pagsanjan Falls Eco-Tourism Web-Based System for tourism-related activities.	**4**	**3**	**2**	**1**
1. I have a positive attitude toward using the Pagsanjan Falls Eco-Tourism Web-Based System for accessing tourism information and services.				
2. Using the system is a good idea because it improves convenience and efficiency in planning visits to Pagsanjan Falls.				
3. I am willing to continue using the Pagsanjan Falls Eco-Tourism Web-Based System for future trips and transactions.				
4. I would recommend the Pagsanjan Falls Eco-Tourism Web-Based System to other tourists, visitors, and travelers.				
Behavioral Intention to Use (BI) The user's willingness to adopt and recommend the platform for future use.	4	3	2	1
1. I intend to use this platform for future trips to Pagsanjan				
2. I would recommend the Pagsanjan Ecotourism Web-Based Platform to other people.				
3. I believe the platform should be the standard way to book Pagsanjan tours and accommodations				
4. Overall, I am satisfied with the functionalities and services provided by the web-based system.				

APPENDIX F

ISO 25010 and TAM Results

ISO RESULTS:

CATEGORY	MEAN	SD	INTERPRETATION
FUNCTIONAL SUITAILITY	3.44	±0.32	STRONGLY AGREE
PERFORMANCE EFFICIENCY	3.11	±0.32	AGREE
COMPATIBILITY	3	±0.00	AGREE
INTREACTION CAPABILITY	3.28	±0.00	STRONGLY AGREE
REALIABILITY	3.08	±0.28	AGREE
SECURITY	3.16	±0.00	AGREE
MAINTAINABILITY	3.19	±0.00	AGREE
FLEXIBILITY	3.16	±0.32	AGREE
SAFETY	3.13	±0.25	AGREE
	3.17	**±0.16**	**AGREE**

TAM RESULTS:

ITEM	MEAN	SD	INTERPRETATION
PERCIEVED USEFULNESS	3.74	0.12	EXCELLENT
PERCEIVE EASE OF USE	3.56	0.01	EXCELLENT
ATTITUDES TOWARDS USAGE	3.61	0.05	EXCELLENT
BEHAVIORAL INTENTION TO USE(BI)	3.73	0.07	EXCELLENT
	3.66	0.04573474245	EXCELLENT

APPENDIX G

Source Code

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;
use Illuminate\Database\Eloquent\SoftDeletes;
use App\Models\User;
use App\Models\LandingArea;
use App\Models\Boat;
use App\Models\BoatBooking;
use App\Models\LandingAreaImage;

class LandingAreaRequest extends Model
{
    use HasFactory, SoftDeletes;

    protected $fillable = [
        'user_id',
        'landing_area_id',
        'landing_area_image_id',
        'customer_name',
        'customer_email',
        'customer_phone',
        'pickup_date',
        'check_out_date',
        'pickup_time',
        'number_of_adults',
        'number_of_children',
        'boat_id',
        'total_amount',
        'landing_area_fee',
        'status',
        'special_requirements',
        'admin_notes',
        'is_paid',
        'payment_proof',
        'payment_qr',
    ];

    protected $casts = [
        'pickup_date' => 'date',
        'check_out_date' => 'date',
        // keep pickup_time as string so we don't coerce incorrectly in forms
        'pickup_time' => 'string',
        'total_amount' => 'decimal:2',
        'landing_area_fee' => 'decimal:2',
        'is_paid' => 'boolean',
    ];
```

```
        // Relationships
        public function user()
        {
            return $this->belongsTo(User::class);
        }

        public function landingArea()
        {
            return $this->belongsTo(LandingArea::class);
        }

        public function image()
        {
            return $this->belongsTo(LandingAreaImage::class,
'landing_area_image_id');
        }

        public function boat()
        {
            return $this->belongsTo(Boat::class);
        }

        public function boatBooking()
        {
            return $this->hasOne(BoatBooking::class, 'landing_area_request_id');
        }

        // Scopes
        public function scopePending($query)
        {
            return $query->where('status', 'pending');
        }

        public function scopeConfirmed($query)
        {
            return $query->where('status', 'confirmed');
        }

        public function scopeForLandingArea($query, $landingAreaId)
        {
            return $query->where('landing_area_id', $landingAreaId);
        }

        // Accessors
        public function getTotalPeopleAttribute()
        {
            return ($this->number_of_adults ?? 0) + ($this->number_of_children
?? 0);
        }
```

```
    public function getStatusBadgeColorAttribute()
    {
        return match ($this->status) {
            'pending' => 'yellow',
            'confirmed' => 'blue',
            'assigned' => 'purple',
            'completed' => 'green',
            'cancelled' => 'red',
            default => 'gray',
        };
    }
}
```

APPENDIX H

Grammarly Results

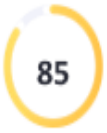
GRAMMARLY CHAPTER 1-2

by PARAGUYA NICA ELA D.

General metrics

41,075	5,952	287	23 min 48 sec	45 min 47 sec
characters	words	sentences	reading time	speaking time

Score



This text scores better than 85% of all texts checked by Grammarly

Writing Issues

162	19	143
Issues left	Critical	Advanced

Writing Issues

2 **Clarity**

1	Paragraph can be improved	
1	Wordy sentences	

1 **Correctness**

2	Improper formatting	
2	Confused words	
1	Incorrect noun number	

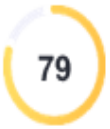
FINAL -3

by PARAGUYA NICA ELA D.

General metrics

29,321	4,150	239	16 min 35 sec	31 min 55 sec
characters	words	sentences	reading time	speaking time

Score



This text scores better than 79% of all texts checked by Grammarly

Writing Issues

151	8	143
Issues left	Critical	Advanced

Writing Issues

8	**Correctness**	
2	Determiner use (a/an/the/this, etc.)	
3	Incorrect punctuation	
1	Comma misuse within clauses	
2	Misplaced words or phrases	
4	**Clarity**	
4	Wordy sentences	

FINAL -3 (1)

by PARAGUYA NICA ELA D.

General metrics

19,924	2,950	163	11 min 48 sec	22 min 41 sec
characters	words	sentences	reading time	speaking time

Score



This text scores better than 77% of all texts checked by Grammarly

Writing Issues

112	15	97
Issues left	Critical	Advanced

Writing Issues

6 Clarity

- 4 Wordy sentences
- 2 Paragraph can be improved

15 Correctness

- 3 Determiner use (a/an/the/this, etc.)
- 7 Incorrect punctuation
- 1 Misplaced words or phrases

4-5

by Borja Andrea Jade De Claro

General metrics

66,393	10,051	369	40 min 12 sec	1 hr 17 min
characters	words	sentences	reading time	speaking time

Score



This text scores better than 79% of all texts checked by Grammarly

Writing Issues

337	✓	337
Issues left	Critical	Advanced

Unique Words

Measures vocabulary diversity by calculating the percentage of words used only once in your document

15%

unique words

Rare Words

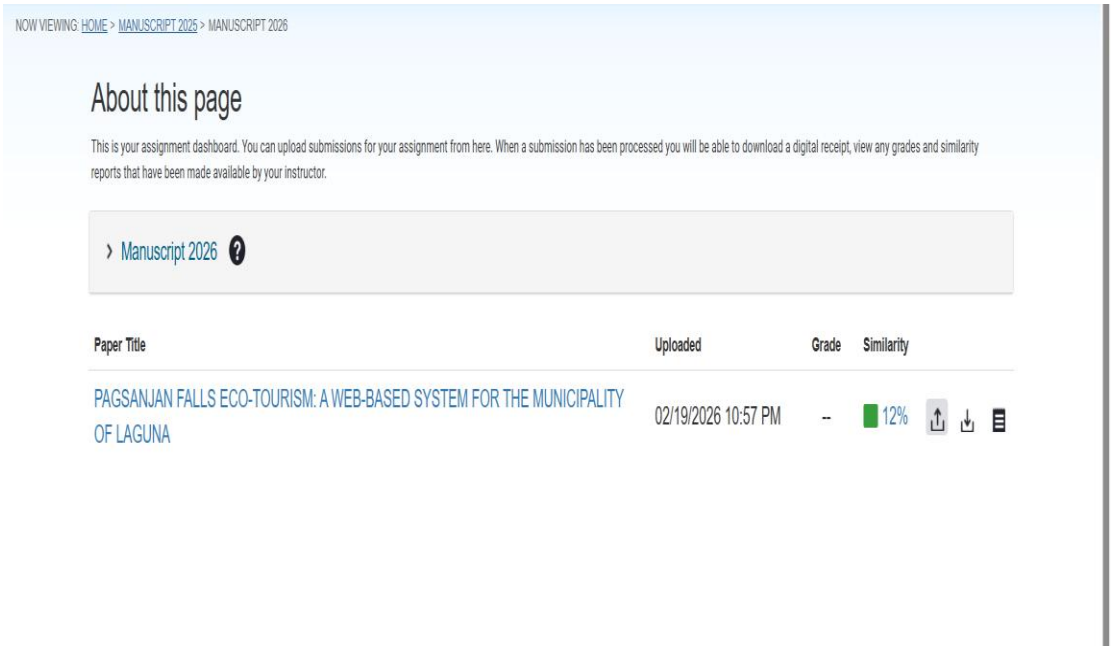
Measures depth of vocabulary by identifying words that are not among the 5,000 most common English words.

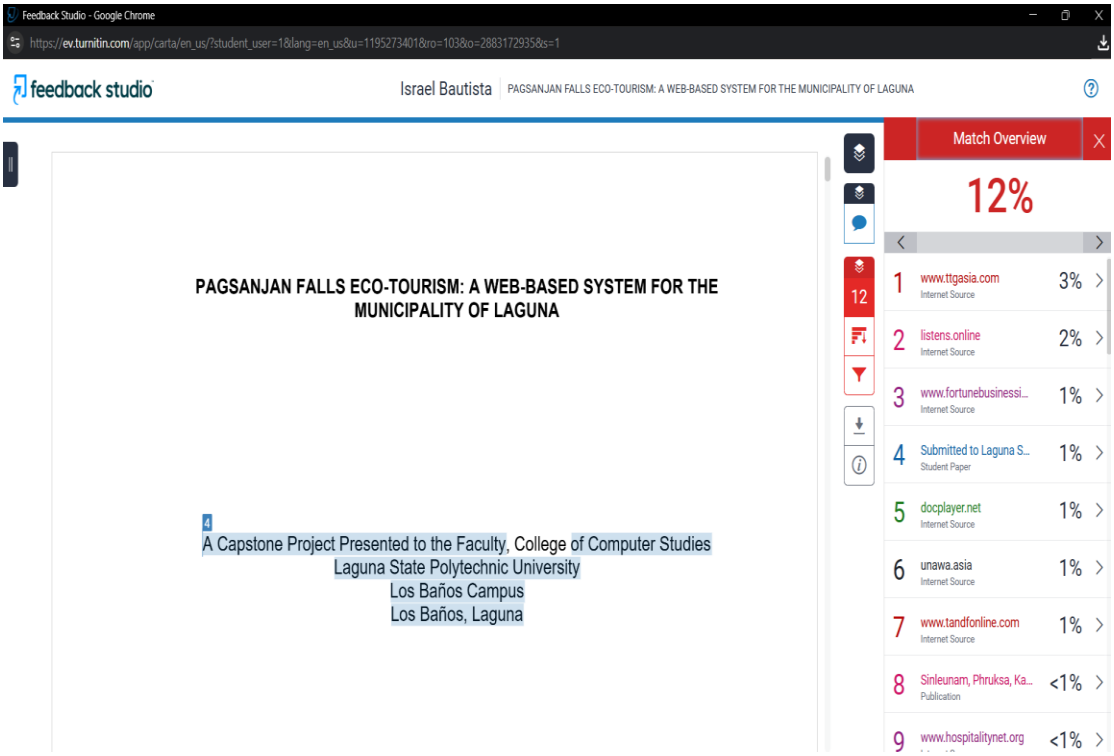
49%

rare words

APPENDIX I

Turnitin Results


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PAGSANJAN FALLS ECO-TOURISM: A WEB-BASED SYSTEM FOR THE MUNICIPALITY OF LAGUNA
A Capstone Project Presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna
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9 www.hospitalitynet.org <1%

APPENDIX J

Publishable Paper (IEE Format)

PAGSANJAN FALLS ECO-TOURISM: A WEB-BASED SYSTEM FOR THE MUNICIPALITY OF LAGUNA.

Andrea Jade DC. Borja, Rose Malyn L. San Isidro, Israel B. Bautista
College of Computer Studies (CCS), Laguna State Polytechnic University, Los Baños, Laguna, Philippines

***Abstract* —** Manual and uncoordinated tourism processes in Pagsanjan Falls have caused difficulties in managing reservations, boat tour schedules, fee collection, and access to tourism information, resulting in delays, booking errors, and inconvenience for tourists and service providers. To address these issues, this study developed a web-based ecotourism system for the Municipality of Laguna that integrates online reservations, boat tour scheduling, payment and fees management, and tourism data reporting under the supervision of the Municipal Tourism Office. The system aims to improve service efficiency, enhance the tourist experience, and support organized and sustainable tourism by allowing better coordination among landing areas, hotels, resorts, restaurants, and the United Boatmen Association of Pagsanjan (UBAP). The system was evaluated using ISO 25010 software quality standards and the Technology Acceptance Model (TAM), and the results showed that it met the required quality attributes and was positively accepted by users. Overall, the study confirms that the developed web-based system is an effective tool for improving tourism operations, strengthening stakeholder coordination, and supporting sustainable ecotourism management in Pagsanjan Falls.

Keywords — :Ecotourism, web-based system, tourism management, reservation system, ISO 25010, Technology Acceptance Model (TAM), Pagsanjan Falls

I. INTRODUCTION

The Ecotourism has its rise in the scholarly literature in the late 1980s as a kind of nature tourism, which focuses on protecting the environment, community well-being, and valuable experiences of tourists. The initial research by different scholars understood ecotourism as a responsible tourism to natural environments whose operations conserve the environment and benefit the local communities (Ceballos-Lascurain, 1996; Orams, 1995). In the course of time, ecotourism has become an aspect of interest of the study of tourism, which resulted in the declaration of the year 2002 as the International Year of Ecotourism by the United Nations, and the Journal of Ecotourism was launched.

With the elaboration of ecotourism as a field of study, scholars put forward different definitions and guidelines that were based on sustainability and long-term growth. Preservation of natural resources, economic gains, participation of the locals, education, and social development are the primary objectives of eco tourism (Weaver and Lawton, 2007; Whitelaw et al., 2014). It is also found that ecotourism may lead to rural growth and alleviation of poverty especially in developing nations, especially through the provision of livelihoods and sustenance of local economies (Snyman, 2017).

The ecotourism is a type of tourism that is one way of traveling to nature and culture with responsibility without disturb or destroy natural resources and environment but having the main goal to see, learn and enjoy the scenario, forest, wild animals and also the culture appeared in that location. The national park is just one of the ecotourism sites where the tourist pay attention to due with high competitive in tourist industry, and high number of TV commercials and other medias. The last 10 years visitor statistics in national park, there is declining of tourists and less realize on ecotourism. Based on the above problems, the researcher raises the issue of creating the web-based decision support system for ecotourism planning by choosing the national park in Chiangmai province where the majority of national parks are situated. Additionally, national parks in Chiangmai province have biological diversity and natural diversity appropriate to be promoted as an ecotourism destination. This study illustrates the concept and two models developing approach of web-based decision support system for ecotourism planning to enable the user selects, the outcome is the users are in a good satisfaction on the system (Tritilanunt et.al. 2021).

In short, the easier and more beneficial a system is perceived to be, the more likely users are to adopt it. (Marikyan and Papagiannidis, 2024), the formulation of the model and technology acceptance measures has made major theoretical contributions and has been of immense practical utility. The use of the model for testing usability in IS has facilitated the assessment of users' motivation to embrace a variety of technologies an activity that had not occurred previously as a consequence of the unavailability of validated subjective measures. The creation of constructs that strongly and significantly correlated with use behavior allowed for understanding cognitive and affective factors that mediate the effect of system characteristics on technology acceptance

III. METHODOLOGY

The analysis of this chapter focuses on evaluating the web-based reservation system development methods systematically. A structured approach defined the study process which combined project design with development processes and testing and evaluation procedures for assuring system functionality and usability effectiveness.

3.1 Project Design

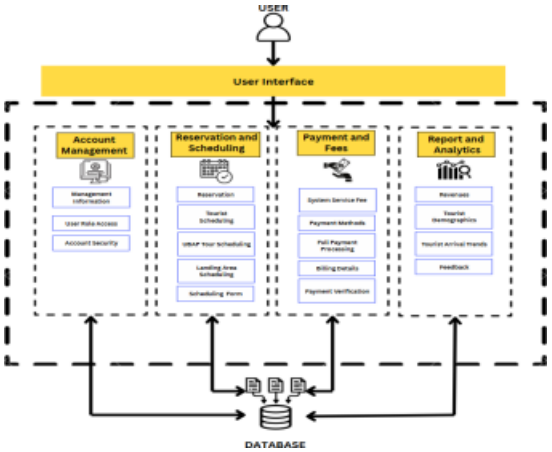


Figure 1. System Architecture

Figure 1 shows the system's architecture. An illustration presents the system layout by showing all modules available in the platform. The Account Management Module handles user profiles, roles, and security, making sure only registered users can access the system. The Reservation and Scheduling Module allows tourists to book resorts and boat tours, while the system helps manage tour schedules, landing areas, and coordination with the United Boatmen Association of Pagsanjan (UBAP). The Payment and Fees Module includes all payment processes such as system service fees, online or manual payments, and payment verification, ensuring all transactions are smooth and secure. Lastly, the Report and Analytics Module collects important data such as tourist arrivals, revenues, and feedback, which helps the admin make better decisions. All modules are connected to a central database to store and manage information efficiently and support the operations of the entire system.

3.2 Materials

Software

The following software was utilized during the development of the web- based booking system: The structure and scalability of backend features.

& Laravel Framework to process reservation and payments and control user administration.

MySQL Database – Serves as the primary database for securely storing user information, reservations, and transaction records.

Hardware

Name	Specification
RAM (Random Access Memory	8.00 Gigabyte
CPU (Central Processing Unit)	AMD Ryzen™ 3
OS (Operating System)	Microsoft Windows 11
Storage	SSD 237GB (Recommended: 256, 512GB SSD or higher)

Table 1. Hardware Minimum Requirement

Table 1 shows the basic equipment requirements for the project to be started. The system should have at least 8GB of RAM but 16GB RAM will provide enhanced multitasking capabilities. The necessary CPU for this project should be a quad-core model starting from an AMD Ryzen™ 3 or equivalent because of its programming complexity. Windows 11 Operating System will be heavily recommended for the latest and better version of windows. The computer requires a minimum SSD storage capacity of 256GB but 512GB SSD provides the fastest data access together with better storage performance. Furthermore, creating this web-based system uses two or more laptop.

3.3 Project Development

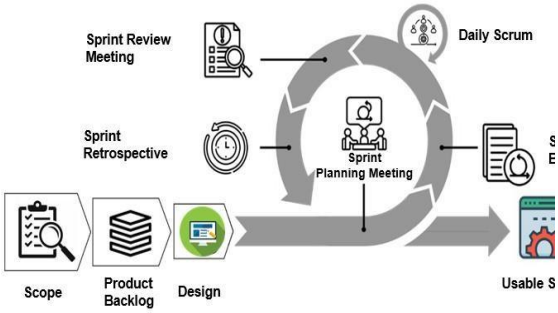


Figure 2. Agile Scrum Methodology

Figure 2 illustrates Agile Scrum, a project design and build the web-based booking system, the developers used the Agile Software Development Life Cycle (SDLC) technique. A Scrum methodology of Agile development divided project work into separate phases for the purpose of flexible user-responsive development. The development method used continuous user interactions for continuous improvement during every phase of development. Figure 2 presents Project management framework Agile Scrum which enables development through continuous stages with periods between two weeks and four weeks. A typical cycle within Agile Scrum spans two to four weeks and groups most important features as priorities to achieve fast functionality alongside usability.

IV. RESULTS AND DISCUSSION

This chapter presents the results and discussion derived from the methods and procedures stated in the previous chapter. The results are organized and discussed in relation to the research objectives and insights drawn from the collected data.

1. Presentation of the System

Information management for tourists and access relevant ecotourism information efficiently.

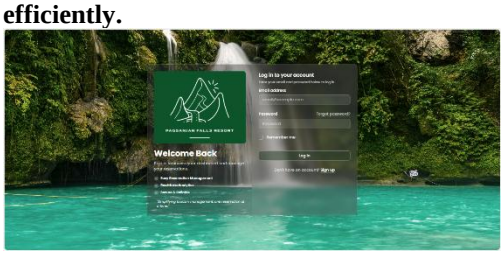


Figure 13. Login/Signup page

User Login Page will enable registered users to access the Pagsanjan Ecotourism Web-Based Platform with their email address and password. It provides an easy and straightforward log-in form with the following features like the options of Remember me and Forgot password to help the user with ease of use and account recovery.

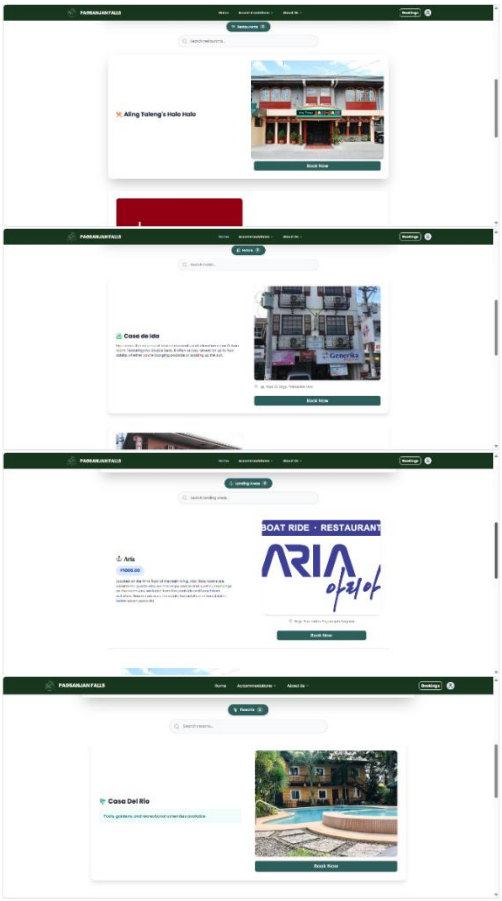


Figure 14, 15, 16, 17. Landing Area, Hotel, Resort and Restaurant Accommodation

Figure 14, 15, 16, 17. Present these four pictures depict the accommodation part that is displayed in the Pagsanjan web-based booking system designed to help the guests find the right accommodation. The pages list some of the different types of accommodation such as landing areas, restaurants, resorts, and hotels with clear photos and relevant information related to the services offered and prices. The available options can be easily viewed by the users to evaluate what each of the establishments can provide. the system provides a seamless booking experience where guests can check availability in real-time and make reservations instantly. The platform also includes detailed descriptions of facilities, nearby attractions, and special offers, allowing users to make informed decisions. This feature not only improves convenience for tourists but also assists the accommodation providers in managing bookings more efficiently and increasing visibility to potential guests.

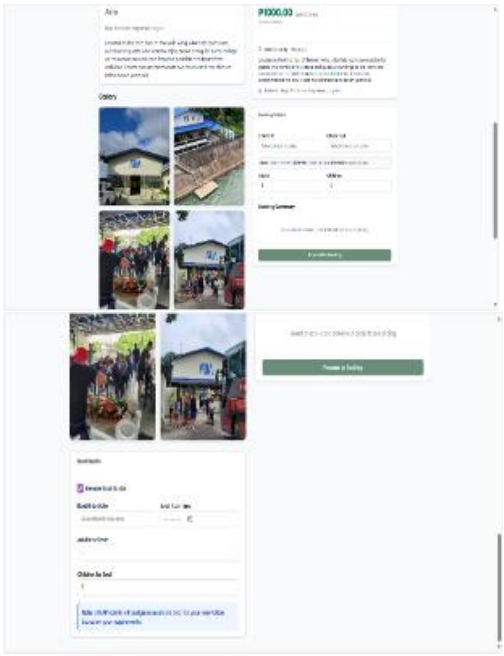


Figure 18. Booking Details for Accommodation

Figure 18 includes the booking information of accommodation including its location, description, per-booking price, and number of guests. It will also provide a photo gallery to help tourists to visualize the premises with clear booking fields to check-in and check-out date and the number of adults and children. It also gives a summary of booking to direct the details to users before their reservation is confirmed and thus make it easier to do and easily understand by the tourist using the system. The website supports the introduction of new services, like boat tours to the famous Pagsanjan Falls or rapid-shooting trips by enabling the user to choose the preferred time and the quantity of guests. This platform does not only support the visitors but it also supports the accommodation providers and tour operators to manage the reservations efficiently. The system streamlines the administration by enabling information and booking to be centralized in a single location thus reducing the number of administrative work done, double booking of rooms and gives a clear track record of all transactions made. This combination of accommodation and activity booking under a single platform would bring customers a seamless experience, and tourism in Pagsanjan will be promoted, which will make the services both more convenient and satisfied and more efficient, in general, to the users and service providers.



Figure 19. Payment Fees and Details

Figure 19 show the page of payment options helps users conclude their reservations following an articulate and formal process. In this page the system will provide a detailed overview of the booking, including the descriptive information about the destination, the dates of travelling, and the number of guests. A fifty percent (50%) payment of the total price must be made upfront to the system to obtain the slot which can be done by scanning the provided QR code. After the transaction is completed, then the users have to attach or upload a photo of their payment receipt to ensure that the administrator verifies and confirms the reservation. The design of the system is such that convenience and security is maintained among the users. It offers immediate confirmation after verification of payment and records all the transactions completed; therefore, it is easier to trace the booking by both the tourists and the administrators. Moreover, this page gives the user an opportunity to revise all the booking information and make payment in advance which eliminates the possibility of mistakes and misinterpretations. In making the payment process a part of the package, the system simplifies the process of making reservations, fosters transparency, and improves the user experience in general and facilitates the efficient management of the accommodations and activities by the administrators.

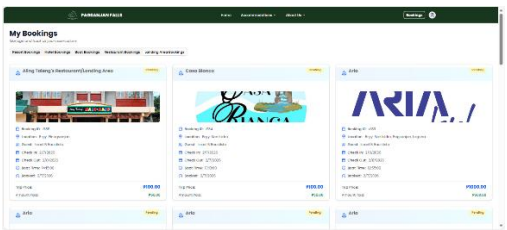


Figure 20. My Bookings Page

Figure 20 illustrates My Bookings Page serves as an official interface, via which customers can be able to check and control all of their bookings in the Pagsanjan booking system. The page is divided into specific sections such as resort, hotel, boat, restaurant,

and landing area reservations in order to ensure that the guests can easily retrieve the information when needed. Moreover, users can see the status of their requests, which is usually referred to as pending until being verified by the administrative team regarding payment and finalizing the slot enhances transparency by showing detailed information about each booking, including dates, guest numbers, and selected services, which helps users stay informed and organized.

Admin Panel

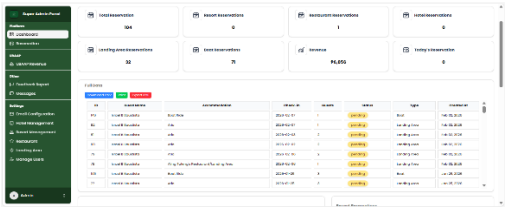


Figure 21. Admin Dashboard

Figure 21 Present the Super Administrator dashboard of the system is being presented, where the administrator would have the ability to view all reservations and activities through one interface. It shows the total landing area, boat, restaurant, and hotel reservations as well as the total revenue and daily bookings. There is also a detailed table where the administrator will be able to monitor and track the transactions by making use of the guest details, the type of the booking and the schedule as well as the number of guests and the status among others. The side menu also allows user management, feedback, and other tourism establishments, hence aiding centralized and systematic management of the Pagsanjan Ecotourism Web-Based Platform. centralized system enables administrators to manage operations systematically, identify trends, and respond quickly to issues or requests. It also provides a comprehensive overview of the platform's performance, helping in decision-making and planning to improve overall tourism services.

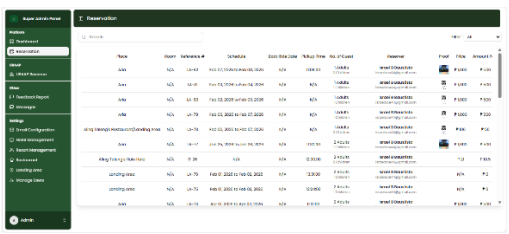


Figure 22. Reservation List

Figure 22 Reservation Management section of Super Administrator panel. It shows all the reservations taken in the system, a location reserved, reference number, schedule, date of the boat excursion, pickup time, number of guests and a person who made the reservation. The evidence of payment, the overall price and amount paid is also presented in the table and this will help the administrator control the transactions and track payments. By using this page, the administrator is able to review, monitor and organize reservations generated by landing sites, restaurants and other services offered by the tourism industry effortlessly and hence enabling to coordinate the tourist activities appropriately in the Pagsanjan Ecotourism Web-Based Platform. The system will improve administration control, minimize errors, and create a smooth and well-structured running of all tourism activities in the Pagsanjan Ecotourism Web-Based Platform through centralization of reservation and payment information.

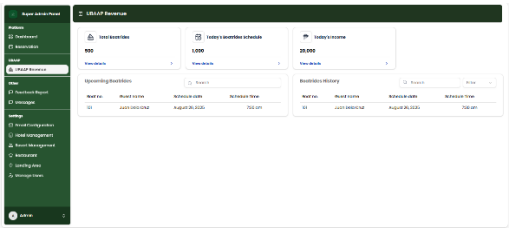


Figure 23. UBAAP Revenue

Figure 23 show the UBAAP Revenue section of the page under Super Admin. It provides a summative report of the business operations of the boat ride including the total number of boat rides, the booking boat rides of the day and the total revenue earned. It also shows the upcoming schedule and the history of the rides already taken on the boat providing the name and date of the guest and time. With the help of this page, the administrator can track daily operations and earnings of the boat tours, thus helping the municipality, keeping an accurate track of the activities and control tourism services related to the Pagsanjan Ecotourism Web-Based Platform. Can manage and control the daily activities in a proper way, so that all the rides and all transactions are captured. The system enables the administrator to monitor the payment, ensure reservations, and check the completion of the bookings hence, minimizes errors, avoidance of duplication of reservations and enhances accountability. It also allows the administrator to determine trends including the high booking periods, high demand periods, season changes in the activity of the tourist which can prove essential in planning the resources, staffing and other operational changes.

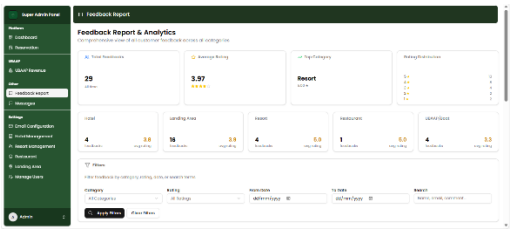


Figure 24. Feedback Report and Analytics

Figure 24 Present the Feedback Report & Analytics is used to access and track all feedbacks of customers in the system. It presents the number of feedbacks overall, average rating, the best category and the rating distribution. It also gives results of feedback on each category like Hotel, Landing Area, Resort, Restaurant, and UBAAP/Boat. This page will help the administrators to evaluate customer satisfaction and go through comments easily. In addition, it also provides filter options to sort feedback by category, rating, date or keywords search. This module will not only assist administrators to measure the general level of customer concerns, but it is also effective in allowing reviewing detailed comments. Additionally, it has the option of advanced filters with which the feedback may be categorized according to rating, category, submission date and keyword search. These can be used to make informed decisions to assist the management to discover the strengths, concerns and also enhance the quality of services provided to the tourists.

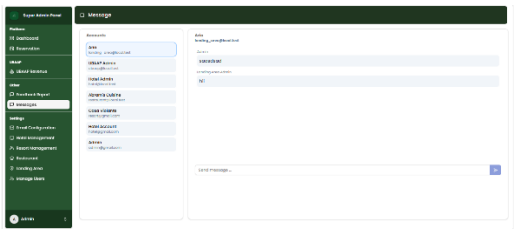


Figure 25. Admin Panel Messages

The Message page will enable the system between the Super Admin and other users of the system to communicate. It shows a list of accounts on the left-hand side and the relevant conversation on the right hand side. The administrator can choose a user to check past messages and write a new message with help of message box which is located below. This page can help in facilitating the communications within the system. By offering an organized and centralized messaging interface, this page can also aid in improving the efficiency of system communications by enabling administrators to keep a clear record of all interactions, monitor conversations, and respond quickly.

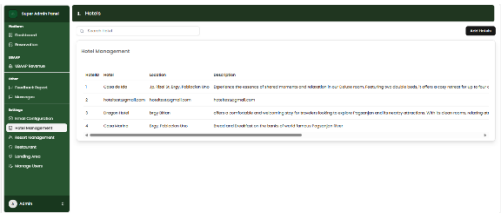


Figure 26. Hotel Management

Figure 26 show the all hotel records within the system are dealt with by using Hotel Management page. It gives a list of hotels with information about the hotel name, location, email and description. The administrator can use the button of adding a hotel to find the hotel and add new hotels. Through this page, the administrator will be able to plan and manage the hotel-based information effectively. System enables administrators to create new hotel entries while they can also update current hotel details and remove hotels that are not operating and maintain accurate records throughout their system. The centralized management system enables tourism operations to plan and make decisions about their activities which include coordinating hotel bookings with boat transportation and landing sites and resort operations.

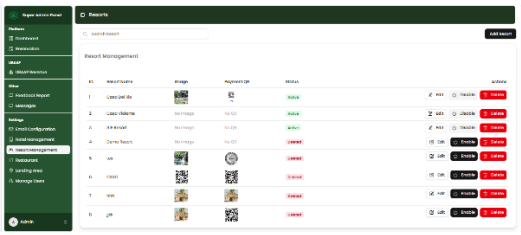


Figure 27. Resort Management

Figure 27 show the Resort Management page allows the administrator to efficiently manage all listed resorts in the system, including popular establishments like Casa Del Rio. The page displays a table summarizing each resort with its name, image, and an active status showing that the resort is available for booking. The administrator can edit information on the resort through controls on the right-hand side, temporarily deactivate a listing of a resort, or delete a resort if, for any reason, the particular resort is no longer in operation. Furthermore, this page allows administrators to track resort activity and keep up-to-date records and promotional information on the resorts. Centralizing the management of resorts in the system enhances operational efficiency, reduces errors, and ultimately provides a good experience in the booking process for the administrators and tourists alike.

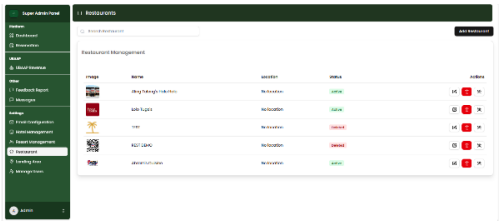


Figure 28. Restaurant Management

Figure 28 the Restaurant Management page offers the list of all dining establishments that are registered in the booking system. It features the highlights of the various venues, including the Aling Taleng Halo Halo, their location, and the status they are currently in. The administrators can easily update the list by clicking the edit and delete icons next to all the restaurant entries to keep the information up to date. To incorporate new restaurants, change the existing details, and delete the ones not in operation, thereby offering authentic operational restaurant options for tourists using the system. By centralizing the information for the restaurants, the system helps in efficient management, making the booking process smoother with reservations, thereby offering a delightful culinary experience for tourists.

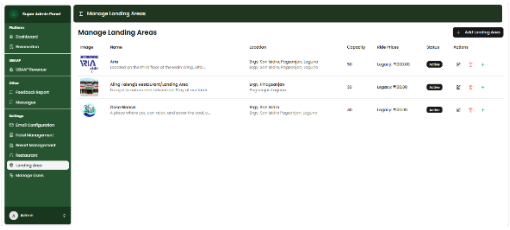


Figure 29. Landing Area Management

Figure 29 illustrate this part of Landing Area Management is supposed to control the individual zones where boat tours and customers are dealt with including Aria and Casa Bianca. These areas (along with the necessary information, such as guest capacity, and typical fare charged on the rides) are listed in the table. Administrators can use the button located on the top of the page to add and update new areas to ensure the right pricing and the availability of the travelers. This page enables zone data to be modified, capacity limits to be set, and fare rate adjustments in line with operational needs. Such functionality ensures order is maintained in scheduling operations, as well as ensuring that no overcrowding occurs. This gives a clear indication of how a centralized database for landing area data would enable better operation and coordination among players, leading to a much better tourism experience.

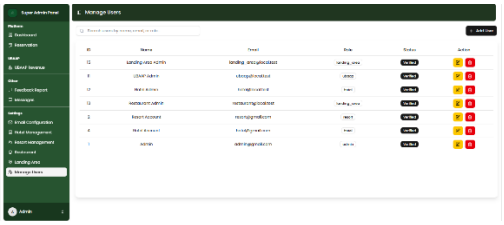


Figure 30. User Management

Figure 30 shows the last function of the Admin Panel, this is a list of all personnel who have administrative access to the system and their names, email addresses, and individual roles are listed here. It helps the chief administrator to trace the people in charge of different areas, including hotel or restaurant accounts, and so, to offer sufficient security. The table also validates the fact that the account of a user is verified and provides a possibility to change his or her permissions or withdraw the access in case of need. this centralized control of user accounts ensures that there is no unauthorized use of the system, protection of critical information, and efficient management of system operations, depending on authorized personnel for these activities. As a result, this structured control of access ensures increased reliability, safety, and integrity of data. Maintaining organizational control, preventing unwanted access, and making sure that only authorized personnel can access sensitive data and system functions are all made possible by this centralized user management system. Additionally, it enables administrators to track permission changes, keep an exhaustive audit trail of account activity, and produce reports that aid in internal review and security policy compliance.

V. 5.SUMMARY, CONCLUSION, AND RECOMMENDATIONS

Summary

The idea behind the capstone project: Pagsanjan Ecotourism Web-Based Platform was to address the acute problems associated with ineffective operations that were in the form of manual check-in, inconsistency of policies, and inability to have a central coordination among the key stakeholders in the Pagsanjan Falls ecotourism sector. Previously, the work was anchored on the manual system of checking-in and unstructured booking leading to the occurrence of errors, delays, and unauthorized transactions which decreased the satisfaction of the tourists. To curb these gaps, the researchers were ready to develop, prepare, and roll out a web based system that was destined to centralize the process of tourist check-in, and control the boat-ride schedule,

and even exhibit transparency in financial reporting. The development process of the project was constituted by Agile Scrum methodology that was designed to meet the continuous input of the user who included the representatives of the Tourism Office, the Landing Area staff, and the tourists. This system was modelled on four major integrated modules, which are Account Management, Reservation and Scheduling, Payment and Fees Management, and Data Visualization. The technologies and frameworks involved were PHP-Laravel and MySQL database and this has provided a system that is highly scalable, efficient and easy to use, providing this system with capabilities of processing real time data. The built-in system involves a tourist landing at a certified landing field and subsequently the system facilitates the notification of a boat tour by the tourist. The Payment module is utilized to pay the transparent 50% down-payment process and the Data Visualization module will show the real-time report of the tourist demographics, revenues, and trends to the Tourism Office. The unit testing and the integration testing that were performed during system testing showed that the modules were performing satisfactorily. Moreover, the effectiveness of the system was measured with the help of 50 respondents using the Technology Acceptance Model (TAM). The mean composite mean score of 3.62 with a SD of (pm) 0.20 which was seen as Excellent indicated that users believed that the system was so useful, easy to use, and willing to adopt the application of the technology. Overall, the created platform was capable of centralizing ecotourism services and addressing the problem of manual policy implementation and reap considerable advantages in terms of organizing the booking process and safety, which benefits the interests of the municipality in terms of sustainable and effective tourism management.

Conclusion

Based on the study and assessment of the Pagsanjan Ecotourism Web-Based Platform, the following conclusions can be identified:

System Utility and Policy Enforcement: In designing and developing the platform, it successfully met the objective of putting in place an orderly online service platform. The system directly enforces municipal policy, as all tours pass through regulated channels (Landing Area and UBAP), thus ensuring equal distribution of income for the boatmen (Bangkero) and improving the general safety and orderliness of the ecotourism service. The high Perceived Usefulness (Mean = 3.95) strongly indicates that users consider the system a significant enhancement over the manual process, especially in terms of ensuring online payments and service management.

It acts as a one-stop portal, and indeed, the objective of providing a single entry point is achieved. It allows the tourists to check on the availability of hotels or accredited Landing Areas such as hotels, resorts, and restaurants, makes requests for reservations, and then creates boat tours after the check-in has been confirmed. This single access reduces confusion and provides convenience, as reflected by the high scores in Perceived Ease of Use (Mean = 3.56) and Aesthetic Quality (Mean = 3.50).

High Acceptance and Adoption: The result of the TAM evaluation showed a very high level of acceptance for the system, with an overall composite Excellent rating (Mean = 3.62). Further, the strong Behavioral Intention to Use (Mean = 3.73) affirms that all stakeholders involved, such as the Tourism Office, staff, and tourists themselves, are highly willing to adopt the system for long-term use. This intention is predominantly driven because the platform is perceived as very enjoyable to use (Mean = 3.61) and economically viable (Mean = 3.52).

Content Enhancement Need: While the features and core functionality of the platform are rated very highly, there is a particular area of content enhancement that needs attention. The item related to the provision of relevant information with regards to Pagsanjan Falls, safety precautions, and local policies obtained the lowest mean (Mean = 3.30, a Good interpretation). This signifies that while the users believe in the accuracy of the system, the completeness or prominence of vital operational and safety information needs attention.

Recommendation

Based on the principal findings, data-driven conclusions, and opportunities identified during the development and testing phases, the following recommendations are hereby proposed to further strengthen the functionality and effectiveness of the Pagsanjan Ecotourism Web-Based Platform:

1.Safety and Policy Information Enhancement (Data-Driven Priority): Since providing safety guidelines and local policies ranked as the only item below “Excellent”, with a Mean of 3.30, it is strongly recommended that enhancement be provided to the Information Awareness module. Establishment of a dedicated, highly visible section on safety, local ordinances, and weather advisories will raise

this score into the Excellent range and increase compliance with basic visitor safety.

2.Improve Mobile Responsiveness: It is recommended to optimize further the responsiveness of the system to ensure consistent performance and full usability across all devices such as smartphones and tablets. This will ensure that tourists can easily reach booking and check-in features on-site to support the high Behavioral Intention to Use.

3.Integrate the direct management of UBAP Scheduling: Although there are notifications to UBAP, consider implementing a dedicated module within which the UBAP administrator will directly be able to manage the boatmen using some kind of queue system to further streamline the dispatch of boat rides and increase the efficiency of the operation.

4.Expand payment gateway options: To maximize high ratings for Perceived Usefulness of the payment feature (Mean = 3.96), add more major credit/debit cards, such as Visa and Mastercard, and other local e-wallet options in addition to the current selection. This will make it easier for both domestic and international tourists to complete the final 50% payment process with less hassle.

5.Implement a Rating and Review System for Boatmen: Further develop the Tourist Feedback module by allowing specific ratings for the assigned boatman. This not only collects data on quality control but also encourages better delivery of services and aids in service improvement, building on the strength of the high Perceived Enjoyment score.

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APPENDIX K

Curriculum Vitae



Andrea Jade DC. Borja

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Bambang Los Baños Laguna

SKILLS

- Microsoft Office
- Basic Programming (c++, c#, python)
- Time Management
- Team Coordination & Communication
- Problem Solving
- Flexibility

Career Objective

To gain experience as an Information Technology student by applying my basic knowledge in computer systems, software, and technology in a professional work environment. I aim to learn new skills and improve my understanding of IT processes while working with experienced professionals. I also want to develop my problem-solving and communication skills to become more confident in handling technical tasks. Through hard work and continuous learning, I hope to contribute positively to the organization and grow as a future IT professional.

EDUCATIONAL BACKGROUND

Primary

- 2010-2016
- Los Baños Central ES - Laguna

Secondary

- 2017-2022
- LBNHS Poblacion - Laguna
- Bambang Senior High School-Laguna

College

- 2022-2026
- Laguna State Polytechnic University
- Bachelor of Science in Information Technology
- Major in Service Management and Program
- Malinta, Los Baños, Laguna

PROJECTS

Purrfect Care Web System — April-May 2024
https://purrfectcarewebsystem.netlify.app/

Developed a fully responsive pet care web application designed to showcase veterinary services and support pet owners in managing their pet's health and wellness. The system features user-friendly navigation, service overviews, team profiles, photo galleries, and an integrated contact/reservation interface. Built with modern web technologies and deployed on Netlify, the project emphasizes clean UI/UX, accessibility, and ease of interaction for both pet owners and clinic staff. It highlights foundational skills in front-end development, responsive design, content structuring, and deployment of live web applications.

TilaMseek – Web-Based E-Commerce System -Dec 2025

Developed a web-based multi-vendor e-commerce system that connects tilapia suppliers with businesses and consumers. The platform includes role-based access for customers, sellers, and administrators, product listing, cart and order management, real-time messaging, seller verification, email notifications, and sales analytics dashboards. Built using HTML, CSS, JavaScript, PHP, and MySQL, following the Agile Scrum methodology with comprehensive system testing.

Israel B. Bautista



My Contact

1148 Brgy. Bambang, Los Baños, Laguna

09303471414
09926611878

israelace44@gmail.com

My Skills

- Basic Coding with Database
- Problem-Solving
- Creativity
- Leadership

Education Background

- **Elementary**
  2009-2015 — Los Baños Central Elem School
- **High School**
  2015-2019 — Los Baños National High School-Poblacion
- **Senior High School**
  2019-2021 — Los Baños Lalakay National Senior High School-Laguna

Work Experience

- **Production Operator**
  2021-2022 — Samsung

Projects

- **Develop a Brgy. Lamot II Monthly Committee Reports**
  2024 — Calauan Laguna
- **Develop a Web Based Flower Shop in Calamba, Laguna**
  2025 — Calamba Laguna



Rose Malyn L. San Isidro

Contact

09564253125

rosellorensanisidro@gmail.com

Purok 1A Tadlac Los Baños Laguna

Skills

- Communication Skills
- Adaptability
- Attention to details
- Problem Solving

Education

ELEMENTARY (2009 -2015)
TADLAC ELEMENTARY SCHOOL

HIGH SCHOOL (2015-2019)
LOS BAÑOS NATIONAL HIGH SCHOOL POBLACION

SENIOR HIGH SCHOOL (2019-2021)
LOS BAÑOS LALAKAY NATIONAL SENIOR HIGH SCHOOL

Experience

PRODUCTION OPERATOR
(2021 -2022)

Experience

- Develop a KK Profiling system
- Develop a Web based Flower Shop In Calama Laguna